MAX9295A
1 x 4 CSI-2 GMSL2 Serializer

ES3 Register Table Rev C1

Reserved and Unused Register Bits

Not all register bits in the register space is shown on the register table. All bits not explicitly defined in the register table should be treated as reserved and should not be written to. If write is required to a register with both defined and undefined register bits, read out the register's contents, change the defined bits, and write the new values back to the register.

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Register Map

MAX9295A

1 x 4 CSI-2 GMSL2 Serializer

Address	Name	MSB							LSB
DEV									
0x00	REG0[7:0]	DEV_ADDR[6:0]							CFG_BLOCK
0x01	REG1[7:0]	IIC_2_EN	IIC_1_EN	DIS_LOCAL_CC	DIS_REM_CC	TX_RATE[1:0]		RX_RATE[1:0]	
0x02	REG2[7:0]	VID_TX_EN_U	VID_TX_EN_Z	VID_TX_EN_Y	VID_TX_EN_X	AUD_TX_EN_Y	AUD_TX_EN_X	–	–
0x03	REG3[7:0]	–	–	UART_2_EN	UART_1_EN	AUD_RX_SRC	RCLK_ALT	RCLKSEL[1:0]	
0x05	REG5[7:0]	LOCK_EN	ERRB_EN	–	–	PU_LF3	PU_LF2	PU_LF1	PU_LF0
0x06	REG6[7:0]	–	–	RCLKEN	–	–	–	–	–
0x07	REG7[7:0]	PAR_DIN_EN[3:0]				PAR_INV_PCLK	PAR_VS_EN	PAR_HS_EN	PAR_VID_EN
0x0D	REG13[7:0]	DEV_ID[7:0]							
0x0E	REG14[7:0]	–	–	–	–	DEV_REV[3:0]			
0x26	REG26[7:0]	–	LF_1[2:0]			–	LF_0[2:0]		
0x27	REG27[7:0]	–	LF_3[2:0]			–	LF_2[2:0]		
0x38	IO_CHK0[7:0]	PIN_DRV_EN_0[7:0]							
0x39	IO_CHK1[7:0]	PIN_DRV_SEL	PIN_DRV_EN_1[6:0]						
0x3B	IO_CHK3[7:0]	PIN_IN_0[7:0]							
0x3C	IO_CHK4[7:0]	–	PIN_IN_1[6:0]						
OVERLAP									
TCTRL									
0x0C	PWR4[7:0]	–	DIS_LOCAL_WAKE	WAKE_EN_B	WAKE_EN_A	–	–	–	–
0x10	CTRL0[7:0]	RESET_ALL	RESET_LINK	RESET_ONESHOT	AUTO_LINK	SLEEP	REG_ENABLE	LINK_CFG[1:0]	
0x11	CTRL1[7:0]	–	–	–	BACK_COMP_SPLTR	–	CXTP_B	–	CXTP_A
0x12	CTRL2[7:0]	IBLEED_OFF	–	–	–	–	–	–	–
0x13	CTRL3[7:0]	–	–	LINK_MODE[1:0]		LOCKED	ERROR	CMU_LOCKED	–
0x18	INTR0[7:0]	–	–	–	–	AUTO_ER	DEC_ERR_THR[2:0]		

Address	Name	MSB							LSB
						R_RST_EN			
0x19	INTR1[7:0]	PKT_CNT_EXP[3:0]				AUTO_CNT_RST_EN	–	–	–
0x1A	INTR2[7:0]	PHY_INT_OEN_B	PHY_INT_OEN_A	REM_ERR_OEN	–	LFLT_INT_OEN	IDLE_ERR_OEN	DEC_ERR_OEN_B	DEC_ERR_OEN_A
0x1B	INTR3[7:0]	PHY_INT_B	PHY_INT_A	REM_ERR_FLAG	–	LFLT_INT	IDLE_ERR_FLAG	DEC_ERR_FLAG_B	DEC_ERR_FLAG_A
0x1C	INTR4[7:0]	EOM_ERR_OEN_B	EOM_ERR_OEN_A	–	HDCP_INT_OEN	MAX_RT_OEN	RT_CNT_OEN	PKT_CNT_OEN	WM_ERR_OEN
0x1D	INTR5[7:0]	EOM_ERR_FLAG_B	EOM_ERR_FLAG_A	–	HDCP_INT_FLAG	MAX_RT_FLAG	RT_CNT_FLAG	PKT_CNT_FLAG	WM_ERR_FLAG
0x1E	INTR6[7:0]	VDDCMP_INT_OEN	PORZ_INT_OEN	VDDBAD_INT_OEN	–	–	ADC_INT_OEN	APRBS_ERR_OEN	MIPI_ERR_OEN
0x1F	INTR7[7:0]	VDDCMP_INT_FLAG	PORZ_INT_FLAG	VDDBAD_INT_FLAG	–	–	ADC_INT_FLAG	APRBS_ERR_FLAG	MIPI_ERR_FLAG
0x20	INTR8[7:0]	ERR_TX_EN	–	–	ERR_TX_ID[4:0]				
0x21	INTR9[7:0]	ERR_RX_EN	–	–	ERR_RX_ID[4:0]				
0x22	CNT0[7:0]	DEC_ERR_A[7:0]							
0x23	CNT1[7:0]	DEC_ERR_B[7:0]							
0x24	CNT2[7:0]	IDLE_ERR[7:0]							
0x25	CNT3[7:0]	PKT_CNT[7:0]							
GMSL									
0x29	TX1[7:0]	–	–	ERRG_EN_B	ERRG_EN_A	–	–	–	–
0x2A	TX2[7:0]	ERRG_CNT[1:0]		ERRG_RATE[1:0]		ERRG_BURST[2:0]			ERRG_PER
0x2B	TX3[7:0]	–	–	–	–	–	TIMEOUT[2:0]		
0x2C	RX0[7:0]	PKT_CNT_LBW[1:0]		–	–	PKT_CNT_SEL[3:0]			
0x30	GPIOA[7:0]	GPIO_RX_FAST_BIDIR_EN	GPIO_TX_CASC	GPIO_FWD_CDLY[5:0]					
0x31	GPIOB[7:0]	GPIO_TX_WNDW[1:0]		GPIO_REV_CDLY[5:0]					
CC									
0x40	I2C_0[7:0]	–	–	SLV_SH[1:0]		–	SLV_TO[2:0]		
0x41	I2C_1[7:0]	–	MST_BT[2:0]			–	MST_TO[2:0]		
0x42	I2C_2[7:0]	SRC_A[6:0]							–
0x43	I2C_3[7:0]	DST_A[6:0]							–

Address	**Name**	**MSB**							**LSB**
0x44	I2C_4[7:0]	SRC_B[6:0]							–
0x45	I2C_5[7:0]	DST_B[6:0]							–
0x46	I2C_6[7:0]	–	–	–	–	I2C_AUTO_CFG	I2C_SRC_CNT[2:0]		
0x48	UART_0[7:0]	–	–	REM_MS_EN	LOC_MS_EN	BYPASS_DIS_PAR	BYPASS_TO[1:0]		BYPASS_EN
0x4C	I2C_PT_0[7:0]	–	–	SLV_SH_PT[1:0]		–	SLV_TO_PT[2:0]		
0x4D	I2C_PT_1[7:0]	–	MST_BT_PT[2:0]			–	MST_TO_PT[2:0]		
0x4F	UART_PT_0[7:0]	BITLEN_MAN_CFG_2	DIS_PAR_2	–	–	BITLEN_MAN_CFG_1	DIS_PAR_1	–	–
CFGV VIDEO_X									
0x50	TX0[7:0]	TX_CRC_EN	–	–	–	PRIO_VAL[1:0]		PRIO_CFG[1:0]	
0x51	TX1[7:0]	BW_MULT[1:0]		BW_VAL[5:0]					
0x53	TX3[7:0]	–	–	TX_SPLT_MASK_B	TX_SPLT_MASK_A	–	–	TX_STR_SEL[1:0]	
CFGV VIDEO_Y									
0x54	TX0[7:0]	TX_CRC_EN	–	–	–	PRIO_VAL[1:0]		PRIO_CFG[1:0]	
0x55	TX1[7:0]	BW_MULT[1:0]		BW_VAL[5:0]					
0x57	TX3[7:0]	–	–	TX_SPLT_MASK_B	TX_SPLT_MASK_A	–	–	TX_STR_SEL[1:0]	
CFGV VIDEO_Z									
0x58	TX0[7:0]	TX_CRC_EN	–	–	–	PRIO_VAL[1:0]		PRIO_CFG[1:0]	
0x59	TX1[7:0]	BW_MULT[1:0]		BW_VAL[5:0]					
0x5B	TX3[7:0]	–	–	TX_SPLT_MASK_B	TX_SPLT_MASK_A	–	–	TX_STR_SEL[1:0]	
CFGV VIDEO_U									
0x5C	TX0[7:0]	TX_CRC_EN	–	–	–	PRIO_VAL[1:0]		PRIO_CFG[1:0]	
0x5D	TX1[7:0]	BW_MULT[1:0]		BW_VAL[5:0]					
0x5F	TX3[7:0]	–	–	TX_SPLT_MASK_B	TX_SPLT_MASK_A	–	–	TX_STR_SEL[1:0]	
CFGI INFOFR									
0x78	TR0[7:0]	TX_CRC_EN	RX_CRC_EN	–	–	PRIO_VAL[1:0]		PRIO_CFG[1:0]	
0x79	TR1[7:0]	BW_MULT[1:0]		BW_VAL[5:0]					
0x7B	TR3[7:0]	–	–	TX_SPLT_MASK_B	TX_SPLT_MASK_A	–	TX_SRC_ID[2:0]		

Address	Name	MSB							LSB
0x7C	TR4[7:0]	RX_SRC_SEL[7:0]							
CFGL SPI									
0x80	TR0[7:0]	TX_CRC_EN	RX_CRC_EN	–	–	PRIO_VAL[1:0]		PRIO_CFG[1:0]	
0x81	TR1[7:0]	BW_MULT[1:0]		BW_VAL[5:0]					
0x83	TR3[7:0]	–	–	TX_SPLT_MASK_B	TX_SPLT_MASK_A	–	TX_SRC_ID[2:0]		
0x84	TR4[7:0]	RX_SRC_SEL[7:0]							
0x85	ARQ0[7:0]	–	–	–	–	ARQ0_EN	–	–	–
0x86	ARQ1[7:0]	–	–	–	–	–	–	MAX_RT_ERR_OEN	RT_CNT_OEN
0x87	ARQ2[7:0]	MAX_RT_ERR	RT_CNT[6:0]						
CFGC CC									
0x88	TR0[7:0]	–	–	BURST_MULT[1:0]		PRIO_VAL[1:0]		PRIO_CFG[1:0]	
0x89	TR1[7:0]	BW_MULT[1:0]		BW_VAL[5:0]					
0x8B	TR3[7:0]	–	–	TX_SPLT_MASK_B	TX_SPLT_MASK_A	–	TX_SRC_ID[2:0]		
0x8C	TR4[7:0]	RX_SRC_SEL[7:0]							
0x8D	ARQ0[7:0]	–	–	–	–	ARQ0_EN	–	–	–
0x8E	ARQ1[7:0]	–	–	–	–	–	–	MAX_RT_ERR_OEN	RT_CNT_OEN
0x8F	ARQ2[7:0]	MAX_RT_ERR	RT_CNT[6:0]						
CFGL GPIO									
0x90	TR0[7:0]	TX_CRC_EN	RX_CRC_EN	–	–	PRIO_VAL[1:0]		PRIO_CFG[1:0]	
0x91	TR1[7:0]	BW_MULT[1:0]		BW_VAL[5:0]					
0x93	TR3[7:0]	–	–	TX_SPLT_MASK_B	TX_SPLT_MASK_A	–	TX_SRC_ID[2:0]		
0x94	TR4[7:0]	RX_SRC_SEL[7:0]							
0x95	ARQ0[7:0]	–	–	–	–	ARQ0_EN	–	–	–
0x96	ARQ1[7:0]	–	–	–	–	–	–	MAX_RT_ERR_OEN	RT_CNT_OEN
0x97	ARQ2[7:0]	MAX_RT_ERR	RT_CNT[6:0]						
CFGL IIC_X									
0xA0	TR0[7:0]	TX_CRC_EN	RX_CRC_EN	–	–	PRIO_VAL[1:0]		PRIO_CFG[1:0]	
0xA1	TR1[7:0]	BW_MULT[1:0]		BW_VAL[5:0]					

Address	Name	MSB							LSB
0xA3	TR3[7:0]	–	–	TX_SPLT_MASK_B	TX_SPLT_MASK_A	–	TX_SRC_ID[2:0]		
0xA4	TR4[7:0]	RX_SRC_SEL[7:0]							
0xA5	ARQ0[7:0]	–	–	–	–	ARQ0_EN	–	–	–
0xA6	ARQ1[7:0]	–	–	–	–	–	–	MAX_RT_ERR_OEN	RT_CNT_OEN
0xA7	ARQ2[7:0]	MAX_RT_ERR	RT_CNT[6:0]						
CFGL IIC_Y									
0xA8	TR0[7:0]	TX_CRC_EN	RX_CRC_EN	–	–	PRIO_VAL[1:0]		PRIO_CFG[1:0]	
0xA9	TR1[7:0]	BW_MULT[1:0]		BW_VAL[5:0]					
0xAB	TR3[7:0]	–	–	TX_SPLT_MASK_B	TX_SPLT_MASK_A	–	TX_SRC_ID[2:0]		
0xAC	TR4[7:0]	RX_SRC_SEL[7:0]							
0xAD	ARQ0[7:0]	–	–	–	–	ARQ0_EN	–	–	–
0xAE	ARQ1[7:0]	–	–	–	–	–	–	MAX_RT_ERR_OEN	RT_CNT_OEN
0xAF	ARQ2[7:0]	MAX_RT_ERR	RT_CNT[6:0]						
VID_TX X									
0x100	VIDEO_TX0[7:0]	LINE_CRC_SEL	LINE_CRC_EN	ENC_MODE[1:0]		AUTO_BPP	–	–	–
0x101	VIDEO_TX1[7:0]	–	–	BPP[5:0]					
0x102	VIDEO_TX2[7:0]	PCLKDET	DRIFT_ERR	OVERFLOW	FIFO_WARN	–	LIM_HEART	–	–
VID_TX Y									
0x108	VIDEO_TX0[7:0]	LINE_CRC_SEL	LINE_CRC_EN	ENC_MODE[1:0]		AUTO_BPP	–	–	–
0x109	VIDEO_TX1[7:0]	–	–	BPP[5:0]					
0x10A	VIDEO_TX2[7:0]	PCLKDET	DRIFT_ERR	OVERFLOW	FIFO_WARN	–	LIM_HEART	–	–
VID_TX Z									
0x110	VIDEO_TX0[7:0]	LINE_CRC_SEL	LINE_CRC_EN	ENC_MODE[1:0]		AUTO_BPP	–	–	–
0x111	VIDEO_TX1[7:0]	–	–	BPP[5:0]					
0x112	VIDEO_TX2[7:0]	PCLKDET	DRIFT_ERR	OVERFLOW	FIFO_WARN	–	LIM_HEART	–	–
VID_TX U									
0x118	VIDEO_TX0[7:0]	LINE_CRC_SEL	LINE_CRC_EN	ENC_MODE[1:0]		AUTO_BPP	–	–	–
0x119	VIDEO_TX1[7:0]	–	–	BPP[5:0]					

Address	**Name**	**MSB**							**LSB**
0x11A	VIDEO_TX2[7:0]	PCLKDET	DRIFT_ERR	OVERFLOW	FIFO_WARN	–	LIM_HEART	–	–
SPI									
0x170	SPI_0[7:0]	SPI_LOC_ID[1:0]		SPI_CC_TRG_ID[1:0]		SPI_IGNR_ID	SPI_CC_EN	MST_SLVN	SPI_EN
0x171	SPI_1[7:0]	SPI_LOC_N[5:0]						–	–
0x172	SPI_2[7:0]	REQ_HOLD_OFF[2:0]			FULL_SCK_SETUP	SPI_MOD3_F	SPI_MOD3	SPIM_SS2_ACT_H	SPIM_SS1_ACT_H
0x173	SPI_3[7:0]	SPIM_SS_DLY_CLKS[7:0]							
0x174	SPI_4[7:0]	SPIM_SCK_LO_CLKS[7:0]							
0x175	SPI_5[7:0]	SPIM_SCK_HI_CLKS[7:0]							
0x176	SPI_6[7:0]	–	–	BNE	SPIS_RWN	SS_IO_EN_2	SS_IO_EN_1	BNE_IO_EN	RWN_IO_EN
0x177	SPI_7[7:0]	SPI_RX_OVRFLW	SPI_TX_OVRFLW	–	SPIS_BYTE_CNT[4:0]				
0x178	SPI_8[7:0]	REQ_HOLD_OFF_TO[7:0]							
WM									
0x190	WM_0[7:0]	WM_LEN	WM_MODE[2:0]			WM_DET[1:0]		–	WM_EN
0x192	WM_2[7:0]	–	–	–	–	HsyncPol	VsyncPol	–	–
0x194	WM_4[7:0]	–	–	WM_YUV_IN[1:0]		WM_COLORADJ	–	WM_MASKMODE[1:0]	
0x195	WM_5[7:0]	–	–	–	–	–	WM_POLOUT	WM_DETOUT	WM_ERROR
0x196	WM_6[7:0]	WM_TIMER[7:0]							
0x1AE	WM_WREN_0[7:0]	WM_WREN_L[7:0]							
0x1AF	WM_WREN_1[7:0]	WM_WREN_H[7:0]							
VTX X									
0x1B0	CROSS_0[7:0]	–	CROSS0_I	CROSS0_F	CROSS0[4:0]				
0x1B1	CROSS_1[7:0]	–	CROSS1_I	CROSS1_F	CROSS1[4:0]				
0x1B2	CROSS_2[7:0]	–	CROSS2_I	CROSS2_F	CROSS2[4:0]				
0x1B3	CROSS_3[7:0]	–	CROSS3_I	CROSS3_F	CROSS3[4:0]				
0x1B4	CROSS_4[7:0]	–	CROSS4_I	CROSS4_F	CROSS4[4:0]				
0x1B5	CROSS_5[7:0]	–	CROSS5_I	CROSS5_F	CROSS5[4:0]				
0x1B6	CROSS_6[7:0]	–	CROSS6_I	CROSS6_F	CROSS6[4:0]				
0x1B7	CROSS_7[7:0]	–	CROSS7_	CROSS7_	CROSS7[4:0]				

Address	Name	MSB							LSB
			I	F					
0x1B8	CROSS_8[7:0]	–	CROSS8_I	CROSS8_F	CROSS8[4:0]				
0x1B9	CROSS_9[7:0]	–	CROSS9_I	CROSS9_F	CROSS9[4:0]				
0x1BA	CROSS_10[7:0]	–	CROSS10_I	CROSS10_F	CROSS10[4:0]				
0x1BB	CROSS_11[7:0]	–	CROSS11_I	CROSS11_F	CROSS11[4:0]				
0x1BC	CROSS_12[7:0]	–	CROSS12_I	CROSS12_F	CROSS12[4:0]				
0x1BD	CROSS_13[7:0]	–	CROSS13_I	CROSS13_F	CROSS13[4:0]				
0x1BE	CROSS_14[7:0]	–	CROSS14_I	CROSS14_F	CROSS14[4:0]				
0x1BF	CROSS_15[7:0]	–	CROSS15_I	CROSS15_F	CROSS15[4:0]				
0x1C0	CROSS_16[7:0]	–	CROSS16_I	CROSS16_F	CROSS16[4:0]				
0x1C1	CROSS_17[7:0]	–	CROSS17_I	CROSS17_F	CROSS17[4:0]				
0x1C2	CROSS_18[7:0]	–	CROSS18_I	CROSS18_F	CROSS18[4:0]				
0x1C3	CROSS_19[7:0]	–	CROSS19_I	CROSS19_F	CROSS19[4:0]				
0x1C4	CROSS_20[7:0]	–	CROSS20_I	CROSS20_F	CROSS20[4:0]				
0x1C5	CROSS_21[7:0]	–	CROSS21_I	CROSS21_F	CROSS21[4:0]				
0x1C6	CROSS_22[7:0]	–	CROSS22_I	CROSS22_F	CROSS22[4:0]				
0x1C7	CROSS_23[7:0]	–	CROSS23_I	CROSS23_F	CROSS23[4:0]				
0x1C8	VTX0[7:0]	GEN_VS	GEN_HS	GEN_DE	VS_INV	HS_INV	DE_INV	VTG_MODE[1:0]	
0x1C9	VTX1[7:0]	–	–	PCLKDET_VTX	–	–	–	–	VS_TRIG
0x1CA	VTX2[7:0]	VS_DLY_2[7:0]							
0x1CB	VTX3[7:0]	VS_DLY_1[7:0]							
0x1CC	VTX4[7:0]	VS_DLY_0[7:0]							
0x1CD	VTX5[7:0]	VS_HIGH_2[7:0]							
0x1CE	VTX6[7:0]	VS_HIGH_1[7:0]							
0x1CF	VTX7[7:0]	VS_HIGH_0[7:0]							
0x1D0	VTX8[7:0]	VS_LOW_2[7:0]							

Address	Name	MSB							LSB
0x1D1	VTX9[7:0]	VS_LOW_1[7:0]							
0x1D2	VTX10[7:0]	VS_LOW_0[7:0]							
0x1D3	VTX11[7:0]	V2H_2[7:0]							
0x1D4	VTX12[7:0]	V2H_1[7:0]							
0x1D5	VTX13[7:0]	V2H_0[7:0]							
0x1D6	VTX14[7:0]	HS_HIGH_1[7:0]							
0x1D7	VTX15[7:0]	HS_HIGH_0[7:0]							
0x1D8	VTX16[7:0]	HS_LOW_1[7:0]							
0x1D9	VTX17[7:0]	HS_LOW_0[7:0]							
0x1DA	VTX18[7:0]	HS_CNT_1[7:0]							
0x1DB	VTX19[7:0]	HS_CNT_0[7:0]							
0x1DC	VTX20[7:0]	V2D_2[7:0]							
0x1DD	VTX21[7:0]	V2D_1[7:0]							
0x1DE	VTX22[7:0]	V2D_0[7:0]							
0x1DF	VTX23[7:0]	DE_HIGH_1[7:0]							
0x1E0	VTX24[7:0]	DE_HIGH_0[7:0]							
0x1E1	VTX25[7:0]	DE_LOW_1[7:0]							
0x1E2	VTX26[7:0]	DE_LOW_0[7:0]							
0x1E3	VTX27[7:0]	DE_CNT_1[7:0]							
0x1E4	VTX28[7:0]	DE_CNT_0[7:0]							
0x1E5	VTX29[7:0]	VID_PRBS_EN	–	–	–	–	GRAD_MODE	PATGEN_MODE[1:0]	
0x1E6	VTX30[7:0]	GRAD_INC[7:0]							
0x1E7	VTX31[7:0]	CHKR_A_L[7:0]							
0x1E8	VTX32[7:0]	CHKR_A_M[7:0]							
0x1E9	VTX33[7:0]	CHKR_A_H[7:0]							
0x1EA	VTX34[7:0]	CHKR_B_L[7:0]							
0x1EB	VTX35[7:0]	CHKR_B_M[7:0]							
0x1EC	VTX36[7:0]	CHKR_B_H[7:0]							
0x1ED	VTX37[7:0]	CHKR_RPT_A[7:0]							
0x1EE	VTX38[7:0]	CHKR_RPT_B[7:0]							
0x1EF	VTX39[7:0]	CHKR_ALT[7:0]							
0x1F0	VTX40[7:0]	DIS_COLOR_CROSSBAR	CROSSHS_I	CROSSHS_F	CROSSHS[4:0]				

Address	Name	MSB							LSB
0x1F1	VTX41[7:0]	–	CROSSVS_I	CROSSVS_F	CROSSVS[4:0]				
0x1F2	VTX42[7:0]	–	CROSSDE_I	CROSSDE_F	CROSSDE[4:0]				
VTX Y									
0x1F3	CROSS_0[7:0]	–	CROSS0_I	CROSS0_F	CROSS0[4:0]				
0x1F4	CROSS_1[7:0]	–	CROSS1_I	CROSS1_F	CROSS1[4:0]				
0x1F5	CROSS_2[7:0]	–	CROSS2_I	CROSS2_F	CROSS2[4:0]				
0x1F6	CROSS_3[7:0]	–	CROSS3_I	CROSS3_F	CROSS3[4:0]				
0x1F7	CROSS_4[7:0]	–	CROSS4_I	CROSS4_F	CROSS4[4:0]				
0x1F8	CROSS_5[7:0]	–	CROSS5_I	CROSS5_F	CROSS5[4:0]				
0x1F9	CROSS_6[7:0]	–	CROSS6_I	CROSS6_F	CROSS6[4:0]				
0x1FA	CROSS_7[7:0]	–	CROSS7_I	CROSS7_F	CROSS7[4:0]				
0x1FB	CROSS_8[7:0]	–	CROSS8_I	CROSS8_F	CROSS8[4:0]				
0x1FC	CROSS_9[7:0]	–	CROSS9_I	CROSS9_F	CROSS9[4:0]				
0x1FD	CROSS_10[7:0]	–	CROSS10_I	CROSS10_F	CROSS10[4:0]				
0x1FE	CROSS_11[7:0]	–	CROSS11_I	CROSS11_F	CROSS11[4:0]				
0x1FF	CROSS_12[7:0]	–	CROSS12_I	CROSS12_F	CROSS12[4:0]				
0x200	CROSS_13[7:0]	–	CROSS13_I	CROSS13_F	CROSS13[4:0]				
0x201	CROSS_14[7:0]	–	CROSS14_I	CROSS14_F	CROSS14[4:0]				
0x202	CROSS_15[7:0]	–	CROSS15_I	CROSS15_F	CROSS15[4:0]				
0x203	CROSS_16[7:0]	–	CROSS16_I	CROSS16_F	CROSS16[4:0]				
0x204	CROSS_17[7:0]	–	CROSS17_I	CROSS17_F	CROSS17[4:0]				
0x205	CROSS_18[7:0]	–	CROSS18_I	CROSS18_F	CROSS18[4:0]				
0x206	CROSS_19[7:0]	–	CROSS19_I	CROSS19_F	CROSS19[4:0]				
0x207	CROSS_20[7:0]	–	CROSS20_I	CROSS20_F	CROSS20[4:0]				

Address	Name	MSB							LSB
0x208	CROSS_21[7:0]	–	CROSS21_I	CROSS21_F	CROSS21[4:0]				
0x209	CROSS_22[7:0]	–	CROSS22_I	CROSS22_F	CROSS22[4:0]				
0x20A	CROSS_23[7:0]	–	CROSS23_I	CROSS23_F	CROSS23[4:0]				
0x20B	VTX0[7:0]	GEN_VS	GEN_HS	GEN_DE	VS_INV	HS_INV	DE_INV	VTG_MODE[1:0]	
0x20C	VTX1[7:0]	–	–	PCLKDET_VTX	–	–	–	–	VS_TRIG
0x20D	VTX2[7:0]	VS_DLY_2[7:0]							
0x20E	VTX3[7:0]	VS_DLY_1[7:0]							
0x20F	VTX4[7:0]	VS_DLY_0[7:0]							
0x210	VTX5[7:0]	VS_HIGH_2[7:0]							
0x211	VTX6[7:0]	VS_HIGH_1[7:0]							
0x212	VTX7[7:0]	VS_HIGH_0[7:0]							
0x213	VTX8[7:0]	VS_LOW_2[7:0]							
0x214	VTX9[7:0]	VS_LOW_1[7:0]							
0x215	VTX10[7:0]	VS_LOW_0[7:0]							
0x216	VTX11[7:0]	V2H_2[7:0]							
0x217	VTX12[7:0]	V2H_1[7:0]							
0x218	VTX13[7:0]	V2H_0[7:0]							
0x219	VTX14[7:0]	HS_HIGH_1[7:0]							
0x21A	VTX15[7:0]	HS_HIGH_0[7:0]							
0x21B	VTX16[7:0]	HS_LOW_1[7:0]							
0x21C	VTX17[7:0]	HS_LOW_0[7:0]							
0x21D	VTX18[7:0]	HS_CNT_1[7:0]							
0x21E	VTX19[7:0]	HS_CNT_0[7:0]							
0x21F	VTX20[7:0]	V2D_2[7:0]							
0x220	VTX21[7:0]	V2D_1[7:0]							
0x221	VTX22[7:0]	V2D_0[7:0]							
0x222	VTX23[7:0]	DE_HIGH_1[7:0]							
0x223	VTX24[7:0]	DE_HIGH_0[7:0]							
0x224	VTX25[7:0]	DE_LOW_1[7:0]							
0x225	VTX26[7:0]	DE_LOW_0[7:0]							
0x226	VTX27[7:0]	DE_CNT_1[7:0]							
0x227	VTX28[7:0]	DE_CNT_0[7:0]							

Address	Name	MSB							LSB
0x228	VTX29[7:0]	VID_PRBS_EN	–	–	–	–	GRAD_MODE	PATGEN_MODE[1:0]	
0x229	VTX30[7:0]	GRAD_INC[7:0]							
0x22A	VTX31[7:0]	CHKR_A_L[7:0]							
0x22B	VTX32[7:0]	CHKR_A_M[7:0]							
0x22C	VTX33[7:0]	CHKR_A_H[7:0]							
0x22D	VTX34[7:0]	CHKR_B_L[7:0]							
0x22E	VTX35[7:0]	CHKR_B_M[7:0]							
0x22F	VTX36[7:0]	CHKR_B_H[7:0]							
0x230	VTX37[7:0]	CHKR_RPT_A[7:0]							
0x231	VTX38[7:0]	CHKR_RPT_B[7:0]							
0x232	VTX39[7:0]	CHKR_ALT[7:0]							
0x233	VTX40[7:0]	DIS_COLOR_CROSSBAR	CROSSHS_I	CROSSHS_F	CROSSHS[4:0]				
0x234	VTX41[7:0]	–	CROSSVS_I	CROSSVS_F	CROSSVS[4:0]				
0x235	VTX42[7:0]	–	CROSSDE_I	CROSSDE_F	CROSSDE[4:0]				
VTX Z									
0x236	CROSS_0[7:0]	–	CROSS0_I	CROSS0_F	CROSS0[4:0]				
0x237	CROSS_1[7:0]	–	CROSS1_I	CROSS1_F	CROSS1[4:0]				
0x238	CROSS_2[7:0]	–	CROSS2_I	CROSS2_F	CROSS2[4:0]				
0x239	CROSS_3[7:0]	–	CROSS3_I	CROSS3_F	CROSS3[4:0]				
0x23A	CROSS_4[7:0]	–	CROSS4_I	CROSS4_F	CROSS4[4:0]				
0x23B	CROSS_5[7:0]	–	CROSS5_I	CROSS5_F	CROSS5[4:0]				
0x23C	CROSS_6[7:0]	–	CROSS6_I	CROSS6_F	CROSS6[4:0]				
0x23D	CROSS_7[7:0]	–	CROSS7_I	CROSS7_F	CROSS7[4:0]				
0x23E	CROSS_8[7:0]	–	CROSS8_I	CROSS8_F	CROSS8[4:0]				
0x23F	CROSS_9[7:0]	–	CROSS9_I	CROSS9_F	CROSS9[4:0]				
0x240	CROSS_10[7:0]	–	CROSS10_I	CROSS10_F	CROSS10[4:0]				
0x241	CROSS_11[7:0]	–	CROSS11	CROSS11	CROSS11[4:0]				

Address	Name	MSB							LSB
			_I	_F					
0x242	CROSS_12[7:0]	–	CROSS12_I	CROSS12_F	CROSS12[4:0]				
0x243	CROSS_13[7:0]	–	CROSS13_I	CROSS13_F	CROSS13[4:0]				
0x244	CROSS_14[7:0]	–	CROSS14_I	CROSS14_F	CROSS14[4:0]				
0x245	CROSS_15[7:0]	–	CROSS15_I	CROSS15_F	CROSS15[4:0]				
0x246	CROSS_16[7:0]	–	CROSS16_I	CROSS16_F	CROSS16[4:0]				
0x247	CROSS_17[7:0]	–	CROSS17_I	CROSS17_F	CROSS17[4:0]				
0x248	CROSS_18[7:0]	–	CROSS18_I	CROSS18_F	CROSS18[4:0]				
0x249	CROSS_19[7:0]	–	CROSS19_I	CROSS19_F	CROSS19[4:0]				
0x24A	CROSS_20[7:0]	–	CROSS20_I	CROSS20_F	CROSS20[4:0]				
0x24B	CROSS_21[7:0]	–	CROSS21_I	CROSS21_F	CROSS21[4:0]				
0x24C	CROSS_22[7:0]	–	CROSS22_I	CROSS22_F	CROSS22[4:0]				
0x24D	CROSS_23[7:0]	–	CROSS23_I	CROSS23_F	CROSS23[4:0]				
0x24E	VTX0[7:0]	GEN_VS	GEN_HS	GEN_DE	VS_INV	HS_INV	DE_INV	VTG_MODE[1:0]	
0x24F	VTX1[7:0]	–	–	PCLKDET_VTX	–	–	–	–	VS_TRIG
0x250	VTX2[7:0]	VS_DLY_2[7:0]							
0x251	VTX3[7:0]	VS_DLY_1[7:0]							
0x252	VTX4[7:0]	VS_DLY_0[7:0]							
0x253	VTX5[7:0]	VS_HIGH_2[7:0]							
0x254	VTX6[7:0]	VS_HIGH_1[7:0]							
0x255	VTX7[7:0]	VS_HIGH_0[7:0]							
0x256	VTX8[7:0]	VS_LOW_2[7:0]							
0x257	VTX9[7:0]	VS_LOW_1[7:0]							
0x258	VTX10[7:0]	VS_LOW_0[7:0]							
0x259	VTX11[7:0]	V2H_2[7:0]							
0x25A	VTX12[7:0]	V2H_1[7:0]							
0x25B	VTX13[7:0]	V2H_0[7:0]							
0x25C	VTX14[7:0]	HS_HIGH_1[7:0]							

Address	**Name**	**MSB**							**LSB**
0x25D	VTX15[7:0]	HS_HIGH_0[7:0]							
0x25E	VTX16[7:0]	HS_LOW_1[7:0]							
0x25F	VTX17[7:0]	HS_LOW_0[7:0]							
0x260	VTX18[7:0]	HS_CNT_1[7:0]							
0x261	VTX19[7:0]	HS_CNT_0[7:0]							
0x262	VTX20[7:0]	V2D_2[7:0]							
0x263	VTX21[7:0]	V2D_1[7:0]							
0x264	VTX22[7:0]	V2D_0[7:0]							
0x265	VTX23[7:0]	DE_HIGH_1[7:0]							
0x266	VTX24[7:0]	DE_HIGH_0[7:0]							
0x267	VTX25[7:0]	DE_LOW_1[7:0]							
0x268	VTX26[7:0]	DE_LOW_0[7:0]							
0x269	VTX27[7:0]	DE_CNT_1[7:0]							
0x26A	VTX28[7:0]	DE_CNT_0[7:0]							
0x26B	VTX29[7:0]	VID_PRBS_EN	–	–	–	–	GRAD_MODE	PATGEN_MODE[1:0]	
0x26C	VTX30[7:0]	GRAD_INC[7:0]							
0x26D	VTX31[7:0]	CHKR_A_L[7:0]							
0x26E	VTX32[7:0]	CHKR_A_M[7:0]							
0x26F	VTX33[7:0]	CHKR_A_H[7:0]							
0x270	VTX34[7:0]	CHKR_B_L[7:0]							
0x271	VTX35[7:0]	CHKR_B_M[7:0]							
0x272	VTX36[7:0]	CHKR_B_H[7:0]							
0x273	VTX37[7:0]	CHKR_RPT_A[7:0]							
0x274	VTX38[7:0]	CHKR_RPT_B[7:0]							
0x275	VTX39[7:0]	CHKR_ALT[7:0]							
0x276	VTX40[7:0]	DIS_COLOR_CROSSBAR	CROSSHS_I	CROSSHS_F	CROSSHS[4:0]				
0x277	VTX41[7:0]	–	CROSSVS_I	CROSSVS_F	CROSSVS[4:0]				
0x278	VTX42[7:0]	–	CROSSDE_I	CROSSDE_F	CROSSDE[4:0]				
VTX U									
0x279	CROSS_0[7:0]	–	CROSS0_I	CROSS0_F	CROSS0[4:0]				
0x27A	CROSS_1[7:0]	–	CROSS1_	CROSS1_	CROSS1[4:0]				

Address	Name	MSB							LSB
			I	F					
0x27B	CROSS_2[7:0]	–	CROSS2_I	CROSS2_F	CROSS2[4:0]				
0x27C	CROSS_3[7:0]	–	CROSS3_I	CROSS3_F	CROSS3[4:0]				
0x27D	CROSS_4[7:0]	–	CROSS4_I	CROSS4_F	CROSS4[4:0]				
0x27E	CROSS_5[7:0]	–	CROSS5_I	CROSS5_F	CROSS5[4:0]				
0x27F	CROSS_6[7:0]	–	CROSS6_I	CROSS6_F	CROSS6[4:0]				
0x280	CROSS_7[7:0]	–	CROSS7_I	CROSS7_F	CROSS7[4:0]				
0x281	CROSS_8[7:0]	–	CROSS8_I	CROSS8_F	CROSS8[4:0]				
0x282	CROSS_9[7:0]	–	CROSS9_I	CROSS9_F	CROSS9[4:0]				
0x283	CROSS_10[7:0]	–	CROSS10_I	CROSS10_F	CROSS10[4:0]				
0x284	CROSS_11[7:0]	–	CROSS11_I	CROSS11_F	CROSS11[4:0]				
0x285	CROSS_12[7:0]	–	CROSS12_I	CROSS12_F	CROSS12[4:0]				
0x286	CROSS_13[7:0]	–	CROSS13_I	CROSS13_F	CROSS13[4:0]				
0x287	CROSS_14[7:0]	–	CROSS14_I	CROSS14_F	CROSS14[4:0]				
0x288	CROSS_15[7:0]	–	CROSS15_I	CROSS15_F	CROSS15[4:0]				
0x289	CROSS_16[7:0]	–	CROSS16_I	CROSS16_F	CROSS16[4:0]				
0x28A	CROSS_17[7:0]	–	CROSS17_I	CROSS17_F	CROSS17[4:0]				
0x28B	CROSS_18[7:0]	–	CROSS18_I	CROSS18_F	CROSS18[4:0]				
0x28C	CROSS_19[7:0]	–	CROSS19_I	CROSS19_F	CROSS19[4:0]				
0x28D	CROSS_20[7:0]	–	CROSS20_I	CROSS20_F	CROSS20[4:0]				
0x28E	CROSS_21[7:0]	–	CROSS21_I	CROSS21_F	CROSS21[4:0]				
0x28F	CROSS_22[7:0]	–	CROSS22_I	CROSS22_F	CROSS22[4:0]				
0x290	CROSS_23[7:0]	–	CROSS23_I	CROSS23_F	CROSS23[4:0]				
0x291	VTX0[7:0]	GEN_VS	GEN_HS	GEN_DE	VS_INV	HS_INV	DE_INV	VTG_MODE[1:0]	

Address	Name	MSB							LSB
0x292	VTX1[7:0]	–	–	PCLKDET_VTX	–	–	–	–	VS_TRIG
0x293	VTX2[7:0]	VS_DLY_2[7:0]							
0x294	VTX3[7:0]	VS_DLY_1[7:0]							
0x295	VTX4[7:0]	VS_DLY_0[7:0]							
0x296	VTX5[7:0]	VS_HIGH_2[7:0]							
0x297	VTX6[7:0]	VS_HIGH_1[7:0]							
0x298	VTX7[7:0]	VS_HIGH_0[7:0]							
0x299	VTX8[7:0]	VS_LOW_2[7:0]							
0x29A	VTX9[7:0]	VS_LOW_1[7:0]							
0x29B	VTX10[7:0]	VS_LOW_0[7:0]							
0x29C	VTX11[7:0]	V2H_2[7:0]							
0x29D	VTX12[7:0]	V2H_1[7:0]							
0x29E	VTX13[7:0]	V2H_0[7:0]							
0x29F	VTX14[7:0]	HS_HIGH_1[7:0]							
0x2A0	VTX15[7:0]	HS_HIGH_0[7:0]							
0x2A1	VTX16[7:0]	HS_LOW_1[7:0]							
0x2A2	VTX17[7:0]	HS_LOW_0[7:0]							
0x2A3	VTX18[7:0]	HS_CNT_1[7:0]							
0x2A4	VTX19[7:0]	HS_CNT_0[7:0]							
0x2A5	VTX20[7:0]	V2D_2[7:0]							
0x2A6	VTX21[7:0]	V2D_1[7:0]							
0x2A7	VTX22[7:0]	V2D_0[7:0]							
0x2A8	VTX23[7:0]	DE_HIGH_1[7:0]							
0x2A9	VTX24[7:0]	DE_HIGH_0[7:0]							
0x2AA	VTX25[7:0]	DE_LOW_1[7:0]							
0x2AB	VTX26[7:0]	DE_LOW_0[7:0]							
0x2AC	VTX27[7:0]	DE_CNT_1[7:0]							
0x2AD	VTX28[7:0]	DE_CNT_0[7:0]							
0x2AE	VTX29[7:0]	VID_PRBS_EN	–	–	–	–	GRAD_MODE	PATGEN_MODE[1:0]	
0x2AF	VTX30[7:0]	GRAD_INC[7:0]							
0x2B0	VTX31[7:0]	CHKR_A_L[7:0]							
0x2B1	VTX32[7:0]	CHKR_A_M[7:0]							
0x2B2	VTX33[7:0]	CHKR_A_H[7:0]							

Address	Name	MSB							LSB
0x2B3	VTX34[7:0]	CHKR_B_L[7:0]							
0x2B4	VTX35[7:0]	CHKR_B_M[7:0]							
0x2B5	VTX36[7:0]	CHKR_B_H[7:0]							
0x2B6	VTX37[7:0]	CHKR_RPT_A[7:0]							
0x2B7	VTX38[7:0]	CHKR_RPT_B[7:0]							
0x2B8	VTX39[7:0]	CHKR_ALT[7:0]							
0x2B9	VTX40[7:0]	DIS_COLOR_CROSSBAR	CROSSHS_I	CROSSHS_F	CROSSHS[4:0]				
0x2BA	VTX41[7:0]	–	CROSSVS_I	CROSSVS_F	CROSSVS[4:0]				
0x2BB	VTX42[7:0]	–	CROSSDE_I	CROSSDE_F	CROSSDE[4:0]				
GPIO0 0									
0x2BE	GPIO_A[7:0]	RES_CFG	TX_PRIO	TX_COMP_EN	GPIO_OUT	GPIO_IN	GPIO_RX_EN	GPIO_TX_EN	GPIO_OUT_DIS
0x2BF	GPIO_B[7:0]	PULL_UPDN_SEL[1:0]		OUT_TYPE	GPIO_TX_ID[4:0]				
0x2C0	GPIO_C[7:0]	OVR_RES_CFG	–	–	GPIO_RX_ID[4:0]				
GPIO1 1									
0x2C1	GPIO_A[7:0]	RES_CFG	TX_PRIO	TX_COMP_EN	GPIO_OUT	GPIO_IN	GPIO_RX_EN	GPIO_TX_EN	GPIO_OUT_DIS
0x2C2	GPIO_B[7:0]	PULL_UPDN_SEL[1:0]		OUT_TYPE	GPIO_TX_ID[4:0]				
0x2C3	GPIO_C[7:0]	OVR_RES_CFG	–	–	GPIO_RX_ID[4:0]				
GPIO2 2									
0x2C4	GPIO_A[7:0]	RES_CFG	TX_PRIO	TX_COMP_EN	GPIO_OUT	GPIO_IN	GPIO_RX_EN	GPIO_TX_EN	GPIO_OUT_DIS
0x2C5	GPIO_B[7:0]	PULL_UPDN_SEL[1:0]		OUT_TYPE	GPIO_TX_ID[4:0]				
0x2C6	GPIO_C[7:0]	OVR_RES_CFG	–	–	GPIO_RX_ID[4:0]				
GPIO3 3									
0x2C7	GPIO_A[7:0]	RES_CFG	TX_PRIO	TX_COMP_EN	GPIO_OUT	GPIO_IN	GPIO_RX_EN	GPIO_TX_EN	GPIO_OUT_DIS
0x2C8	GPIO_B[7:0]	PULL_UPDN_SEL[1:0]		OUT_TYPE	GPIO_TX_ID[4:0]				
0x2C9	GPIO_C[7:0]	OVR_RES_CFG	–	–	GPIO_RX_ID[4:0]				
GPIO4 4									
0x2CA	GPIO_A[7:0]	RES_CFG	TX_PRIO	TX_COMP	GPIO_OU	GPIO_IN	GPIO_RX	GPIO_TX_	GPIO_OU

Address	Name	MSB							LSB
				_EN	T		_EN	EN	T_DIS
0x2CB	GPIO_B[7:0]	PULL_UPDN_SEL[1:0]		OUT_TYPE	GPIO_TX_ID[4:0]				
0x2CC	GPIO_C[7:0]	OVR_RES_CFG	–	–	GPIO_RX_ID[4:0]				
GPIO5 5									
0x2CD	GPIO_A[7:0]	RES_CFG	TX_PRIO	TX_COMP_EN	GPIO_OUT	GPIO_IN	GPIO_RX_EN	GPIO_TX_EN	GPIO_OUT_DIS
0x2CE	GPIO_B[7:0]	PULL_UPDN_SEL[1:0]		OUT_TYPE	GPIO_TX_ID[4:0]				
0x2CF	GPIO_C[7:0]	OVR_RES_CFG	–	–	GPIO_RX_ID[4:0]				
GPIO6 6									
0x2D0	GPIO_A[7:0]	RES_CFG	TX_PRIO	TX_COMP_EN	GPIO_OUT	GPIO_IN	GPIO_RX_EN	GPIO_TX_EN	GPIO_OUT_DIS
0x2D1	GPIO_B[7:0]	PULL_UPDN_SEL[1:0]		OUT_TYPE	GPIO_TX_ID[4:0]				
0x2D2	GPIO_C[7:0]	OVR_RES_CFG	–	–	GPIO_RX_ID[4:0]				
GPIO7 7									
0x2D3	GPIO_A[7:0]	RES_CFG	TX_PRIO	TX_COMP_EN	GPIO_OUT	GPIO_IN	GPIO_RX_EN	GPIO_TX_EN	GPIO_OUT_DIS
0x2D4	GPIO_B[7:0]	PULL_UPDN_SEL[1:0]		OUT_TYPE	GPIO_TX_ID[4:0]				
0x2D5	GPIO_C[7:0]	OVR_RES_CFG	–	–	GPIO_RX_ID[4:0]				
GPIO8 8									
0x2D6	GPIO_A[7:0]	RES_CFG	TX_PRIO	TX_COMP_EN	GPIO_OUT	GPIO_IN	GPIO_RX_EN	GPIO_TX_EN	GPIO_OUT_DIS
0x2D7	GPIO_B[7:0]	PULL_UPDN_SEL[1:0]		OUT_TYPE	GPIO_TX_ID[4:0]				
0x2D8	GPIO_C[7:0]	OVR_RES_CFG	–	–	GPIO_RX_ID[4:0]				
GPIO9 9									
0x2D9	GPIO_A[7:0]	RES_CFG	TX_PRIO	TX_COMP_EN	GPIO_OUT	GPIO_IN	GPIO_RX_EN	GPIO_TX_EN	GPIO_OUT_DIS
0x2DA	GPIO_B[7:0]	PULL_UPDN_SEL[1:0]		OUT_TYPE	GPIO_TX_ID[4:0]				
0x2DB	GPIO_C[7:0]	OVR_RES_CFG	–	–	GPIO_RX_ID[4:0]				
GPIO10 10									
0x2DC	GPIO_A[7:0]	RES_CFG	TX_PRIO	TX_COMP_EN	GPIO_OUT	GPIO_IN	GPIO_RX_EN	GPIO_TX_EN	GPIO_OUT_DIS
0x2DD	GPIO_B[7:0]	PULL_UPDN_SEL[1:0]		OUT_TYP	GPIO_TX_ID[4:0]				

Address	Name	MSB							LSB
		]		E					
0x2DE	GPIO_C[7:0]	OVR_RES_CFG	–	–	GPIO_RX_ID[4:0]				
FRONTTOP									
0x308	FRONTTOP_0[7:0]	–	enable_line_info	START_PORTB	START_PORTA	CLK_SELU	CLK_SELZ	CLK_SELY	CLK_SELX
0x309	FRONTTOP_1[7:0]	VC_SELX_L[7:0]							
0x30A	FRONTTOP_2[7:0]	VC_SELX_H[7:0]							
0x30B	FRONTTOP_3[7:0]	VC_SELY_L[7:0]							
0x30C	FRONTTOP_4[7:0]	VC_SELY_H[7:0]							
0x30D	FRONTTOP_5[7:0]	VC_SELZ_L[7:0]							
0x30E	FRONTTOP_6[7:0]	VC_SELZ_H[7:0]							
0x30F	FRONTTOP_7[7:0]	VC_SELU_L[7:0]							
0x310	FRONTTOP_8[7:0]	VC_SELU_H[7:0]							
0x311	FRONTTOP_9[7:0]	START_PORTBU	START_PORTBZ	START_PORTBY	START_PORTBX	START_PORTAU	START_PORTAZ	START_PORTAY	START_PORTAX
0x312	FRONTTOP_10[7:0]	–	–	–	–	bpp8dblu	bpp8dblz	bpp8dbly	bpp8dblx
0x313	FRONTTOP_11[7:0]	bpp12dblu	bpp12dblz	bpp12dbly	bpp12dblx	bpp10dblu	bpp10dblz	bpp10dbly	bpp10dblx
0x314	FRONTTOP_12[7:0]	–	mem_dt1_selx[6:0]						
0x315	FRONTTOP_13[7:0]	–	mem_dt2_selx[6:0]						
0x316	FRONTTOP_14[7:0]	–	mem_dt1_sely[6:0]						
0x317	FRONTTOP_15[7:0]	–	mem_dt2_sely[6:0]						
0x318	FRONTTOP_16[7:0]	–	mem_dt1_selz[6:0]						
0x319	FRONTTOP_17[7:0]	–	mem_dt2_selz[6:0]						
0x31A	FRONTTOP_18[7:0]	–	mem_dt1_selu[6:0]						
0x31B	FRONTTOP_19[7:0]	–	mem_dt2_selu[6:0]						
0x31C	FRONTTOP_20[7:0]	soft_dtx_en	soft_vcx_en	soft_bppx_en	soft_bppx[4:0]				
0x31D	FRONTTOP_21[7:0]	soft_dty_en	soft_vcy_en	soft_bppy_en	soft_bppy[4:0]				
0x31E	FRONTTOP_22[7:0]	soft_dtz_en	soft_vcz_en	soft_bppz_en	soft_bppz[4:0]				
0x31F	FRONTTOP_23[7:0]	soft_dtu_en	soft_vcu_en	soft_bppu_en	soft_bppu[4:0]				
0x320	FRONTTOP_24[7:0]	soft_vcu[1:0]		soft_vcz[1:0]		soft_vcy[1:0]		soft_vcx[1:0]	
0x321	FRONTTOP_25[7:0]	–	–	soft_dtx[5:0]					
0x322	FRONTTOP_26[7:0]	–	–	soft_dty[5:0]					

Address	Name	MSB							LSB
0x323	FRONTTOP_27[7:0]	–	–	soft_dtz[5:0]					
0x324	FRONTTOP_28[7:0]	–	–	soft_dtu[5:0]					
DUALVIEW A									
0x32A	DV0[7:0]	RSVD	DV_SWP_AB	LINE_ALT	–	–	DV_CONV	DV_SPL	DV_EN
MIPI_RX									
0x330	MIPI_RX0[7:0]	–	–	ctrl1_vc_map_en	ctrl0_vc_map_en	mipi_rx_reset	phy_config[2:0]		
0x331	MIPI_RX1[7:0]	ctrl1_vcx_en	ctrl1_deskewen	ctrl1_num_lanes[1:0]		ctrl0_vcx_en	ctrl0_deskewen	ctrl0_num_lanes[1:0]	
0x332	MIPI_RX2[7:0]	phy1_lane_map[3:0]				phy0_lane_map[3:0]			
0x333	MIPI_RX3[7:0]	phy3_lane_map[3:0]				phy2_lane_map[3:0]			
0x334	MIPI_RX4[7:0]	–	phy1_pol_map[2:0]			–	phy0_pol_map[2:0]		
0x335	MIPI_RX5[7:0]	–	phy3_pol_map[2:0]			–	phy2_pol_map[2:0]		
0x338	MIPI_RX8[7:0]	t_term_en[1:0]		t_hs_settle[1:0]		t_clk_miss[1:0]		t_clk_settle[1:0]	
0x339	MIPI_RX9[7:0]	–	–	–	phy0_lp_err[4:0]				
0x33A	MIPI_RX10[7:0]	phy0_hs_err[7:0]							
0x33B	MIPI_RX11[7:0]	–	–	–	phy1_lp_err[4:0]				
0x33C	MIPI_RX12[7:0]	phy1_hs_err[7:0]							
0x33D	MIPI_RX13[7:0]	–	–	–	phy2_lp_err[4:0]				
0x33E	MIPI_RX14[7:0]	phy2_hs_err[7:0]							
0x33F	MIPI_RX15[7:0]	–	–	–	phy3_lp_err[4:0]				
0x340	MIPI_RX16[7:0]	phy3_hs_err[7:0]							
0x341	MIPI_RX17[7:0]	ctrl0_csi_err_l[7:0]							
0x342	MIPI_RX18[7:0]	–	–	–	–	–	ctrl0_csi_err_h[2:0]		
0x343	MIPI_RX19[7:0]	ctrl1_csi_err_l[7:0]							
0x344	MIPI_RX20[7:0]	–	–	–	–	–	ctrl1_csi_err_h[2:0]		
0x345	MIPI_RX21[7:0]	ctrl1_vc_map0[3:0]				ctrl0_vc_map0[3:0]			
0x346	MIPI_RX22[7:0]	ctrl1_vc_map1[3:0]				ctrl0_vc_map1[3:0]			
MIPI_LPB_PRBS									
0x370	MIPI_LPB0[7:0]	–	LPB_MIPI_RX_PRBSEN_1	LPB_MIPI_TX_PRBSEN_1	LPB_PATTERN_SEL[2:0]			LPB_MIPI_RX_PRBSEN_0	LPB_MIPI_TX_PRBSEN_0
0x371	MIPI_LPB1[7:0]	LPB_LINE_PF[7:0]							
0x372	MIPI_LPB2[7:0]	LPB_PIXEL_COUNT_L[7:0]							
0x373	MIPI_LPB3[7:0]	LPB_PIXEL_COUNT_H[7:0]							

Address	Name	MSB							LSB
0x374	MIPI_LPB4[7:0]	LPB_DATA_INVALID1[3:0]				LPB_DATA_INVALID0[3:0]			
0x375	MIPI_LPB5[7:0]	–	–	–	–	–	–	–	–
0x376	MIPI_LPB6[7:0]	–	–	–	–	–	–	–	–
FRONTTOP_EXT									
0x3C0	FRONTTOP_EXT0[7:0]	mem_dt3_selx[7:0]							
0x3C1	FRONTTOP_EXT1[7:0]	mem_dt4_selx[7:0]							
0x3C2	FRONTTOP_EXT2[7:0]	mem_dt5_selx[7:0]							
0x3C3	FRONTTOP_EXT3[7:0]	mem_dt6_selx[7:0]							
0x3C4	FRONTTOP_EXT4[7:0]	mem_dt3_sely[7:0]							
0x3C5	FRONTTOP_EXT5[7:0]	mem_dt4_sely[7:0]							
0x3C6	FRONTTOP_EXT6[7:0]	mem_dt5_sely[7:0]							
0x3C7	FRONTTOP_EXT7[7:0]	mem_dt6_sely[7:0]							
0x3C8	FRONTTOP_EXT8[7:0]	mem_dt3_selz[7:0]							
0x3C9	FRONTTOP_EXT9[7:0]	mem_dt4_selz[7:0]							
0x3CA	FRONTTOP_EXT10[7:0]	mem_dt5_selz[7:0]							
0x3CB	FRONTTOP_EXT11[7:0]	mem_dt6_selz[7:0]							
0x3CC	FRONTTOP_EXT12[7:0]	mem_dt3_selu[7:0]							
0x3CD	FRONTTOP_EXT13[7:0]	mem_dt4_selu[7:0]							
0x3CE	FRONTTOP_EXT14[7:0]	mem_dt5_selu[7:0]							
0x3CF	FRONTTOP_EXT15[7:0]	mem_dt6_selu[7:0]							
0x3D0	FRONTTOP_EXT16[7:0]	mem_dt6_sely_en	mem_dt5_sely_en	mem_dt4_sely_en	mem_dt3_sely_en	mem_dt6_selx_en	mem_dt5_selx_en	mem_dt4_selx_en	mem_dt3_selx_en
0x3D1	FRONTTOP_EXT17[7:0]	mem_dt6_selu_en	mem_dt5_selu_en	mem_dt4_selu_en	mem_dt3_selu_en	mem_dt6_selz_en	mem_dt5_selz_en	mem_dt4_selz_en	mem_dt3_selz_en
REF_VTG									
0x3E0	VTX0[7:0]	–	VS_TRIG	REF_VTG_MODE[1:0]		HS_INV	GEN_HS	VS_INV	GEN_VS
0x3E1	VTX1[7:0]	VS_HIGH_2[7:0]							
0x3E2	VTX2[7:0]	VS_HIGH_1[7:0]							
0x3E3	VTX3[7:0]	VS_HIGH_0[7:0]							
0x3E4	VTX4[7:0]	VS_LOW_2[7:0]							
0x3E5	VTX5[7:0]	VS_LOW_1[7:0]							
0x3E6	VTX6[7:0]	VS_LOW_0[7:0]							
0x3E7	VTX7[7:0]	V2H_2[7:0]							
0x3E8	VTX8[7:0]	V2H_1[7:0]							
0x3E9	VTX9[7:0]	V2H_0[7:0]							

Address	**Name**	**MSB**							**LSB**
0x3EA	VTX10[7:0]	HS_HIGH_1[7:0]							
0x3EB	VTX11[7:0]	HS_HIGH_0[7:0]							
0x3EC	VTX12[7:0]	HS_LOW_1[7:0]							
0x3ED	VTX13[7:0]	HS_LOW_0[7:0]							
0x3EE	VTX14[7:0]	HS_CNT_1[7:0]							
0x3EF	VTX15[7:0]	HS_CNT_0[7:0]							
0x3F0	REF_VTG0[7:0]	REFGEN_LOCKED	REFGEN_PREDEF_EN	REFGEN_PREDEF_FREQ[1:0]		–	–	REFGEN_RST	REFGEN_EN
0x3F1	REF_VTG1[7:0]	RCLKEN_Y	–	PCLK_GPIO[4:0]					PCLKEN
0x3F2	REF_VTG2[7:0]	–	–	HS_GPIO[4:0]					HSEN
0x3F3	REF_VTG3[7:0]	–	–	VS_GPIO[4:0]					VSEN
0x3F4	REF_VTG4[7:0]	REFGEN_FB_FRACT_L[7:0]							
0x3F5	REF_VTG5[7:0]	–	–	–	–	REFGEN_FB_FRACT_H[3:0]			
0x3F6	REF_VTG6[7:0]	VS_DLY_2[7:0]							
0x3F7	REF_VTG7[7:0]	VS_DLY_1[7:0]							
0x3F8	REF_VTG8[7:0]	VS_DLY_0[7:0]							
0x3F9	REF_VTG9[7:0]	REF_VTG_TRIG_EN	–	–	REF_VTG_TRIG_ID[4:0]				
GMSL1									
0x402	GMSL1_2[7:0]	–	–	SSEN	–	–	–	–	–
0x404	GMSL1_4[7:0]	SEREN	CLINKEN	PRBSEN	–	–	–	REVCCEN	FWDCCEN
0x405	GMSL1_5[7:0]	–	–	PRBS_LEN[1:0]		–	–	–	–
0x407	GMSL1_7[7:0]	DBL	HIBW	BWS	–	DRS	HVEN	–	PXL_CRC
0x40D	GMSL1_D[7:0]	I2C_LOC_ACK	–	–	–	–	–	–	–
0x40F	GMSL1_F[7:0]	CNTL_IN_EN[4:0]					GPO_RX_EN	GPO_OUT_SEL	SET_GPO
0x411	GMSL1_11[7:0]	ERRG_RATE2[1:0]		ERRG_TYPE2[1:0]		ERRG_CNT2[1:0]		ERRG_PER2	ERRG_EN
0x414	GMSL1_14[7:0]	–	–	–	I2C_TIMED_OUT2	CC_WBLOCK_LOST	REV_BIT_LOCK	CC_WBLOCK	REM_CCLOCK
0x415	GMSL1_15[7:0]	–	–	–	–	–	–	SEL_VESA	SEL_RGB888
0x416	GMSL1_16[7:0]	–	MAX_RT_ERR	–	–	–	–	–	–
0x41C	GMSL1_18[7:0]	CC_I2C_RETR_CNT[7:0]							
0x41D	GMSL1_19[7:0]	CC_CRC_ERRCNT[7:0]							

Address	Name	MSB							LSB
0x438	GMSL1_39[7:0]	–	CROSS24_I	CROSS24_F	CROSS24[4:0]				
0x439	GMSL1_3A[7:0]	–	CROSS25_I	CROSS25_F	CROSS25[4:0]				
0x43A	GMSL1_3B[7:0]	–	CROSS26_I	CROSS26_F	CROSS26[4:0]				
0x43B	GMSL1_3C[7:0]	–	CROSS27_I	CROSS27_F	CROSS27[4:0]				
0x43C	GMSL1_3D[7:0]	–	CROSS28_I	CROSS28_F	CROSS28[4:0]				
0x43D	GMSL1_3E[7:0]	–	CROSS29_I	CROSS29_F	CROSS29[4:0]				
0x43E	GMSL1_3F[7:0]	–	CROSS30_I	CROSS30_F	CROSS30[4:0]				
0x442	GMSL1_42[7:0]	–	–	–	MAX_RT_EN	I2C_RT_EN	GPI_COMP_EN	GPI_RT_EN	GPO_EN
0x44D	GMSL1_4D[7:0]	HIGHIMM	–	–	–	–	–	–	–
0x466	GMSL1_66[7:0]	–	–	PRBS_TYPE	REV_FAST	DIS_DE	–	–	–
0x467	GMSL1_67[7:0]	CNTL_TRIG[1:0]		AUTO_CLINK	DE_RGB888_DBL	–	DBL_ALIGN_TO[2:0]		
0x468	GMSL1_68[7:0]	–	–	–	–	–	–	CC_CRC_LENGTH[1:0]	
0x496	GMSL1_96[7:0]	–	–	–	–	BYPASS_HVD_ALIGN	–	–	–
0x499	GMSL1_99[7:0]	–	–	–	REV_FILT_EN	ALIGN_VS_MODE	ALIGN_INFO	–	–
0x49A	GMSL1_9A[7:0]	–	–	–	–	PKTCC_EN	–	–	–
0x4C8	GMSL1_C8[7:0]	–	ALIGNING	I2C_ACK_RECVED	I2C_ACK_ACKED	–	–	–	–
AFE									
0x500	ADC_CTRL_0[7:0]	buf_bypass	–	–	adc_chgpump_pu	adc_refbuf_pu	buf_pu	adc_pu	cpu_adc_start
0x501	ADC_CTRL_1[7:0]	adc_chsel[3:0]				adc_clk_en	adc_refsel	adc_scale	adc_refscl
0x502	ADC_CTRL_2[7:0]	–	–	–	–	adc_div[1:0]		adc_xref	lnmux_en
0x503	ADC_CTRL_3[7:0]	AfePwrUpDly[7:0]							
0x504	ADC_STATUS0[7:0]	–	–	–	–	–	–	–	adc_active
0x508	ADC_DATA0[7:0]	adc_data_l[7:0]							
0x509	ADC_DATA1[7:0]	–	–	–	–	–	–	adc_data_h[1:0]	
0x50C	ADC_INTRIE0[7:0]	–	–	–	adc_overflow_ie	adc_lo_limit_ie	adc_hi_limit_ie	adc_ref_ready_ie	adc_done_ie

Address	Name	MSB							LSB
0x50D	ADC_INTRIE1[7:0]	ch7_hi_limit_ie	ch6_hi_limit_ie	ch5_hi_limit_ie	ch4_hi_limit_ie	ch3_hi_limit_ie	ch2_hi_limit_ie	ch1_hi_limit_ie	ch0_hi_limit_ie
0x50E	ADC_INTRIE2[7:0]	ch7_lo_limit_ie	ch6_lo_limit_ie	ch5_lo_limit_ie	ch4_lo_limit_ie	ch3_lo_limit_ie	ch2_lo_limit_ie	ch1_lo_limit_ie	ch0_lo_limit_ie
0x50F	ADC_INTRIE3[7:0]	–	–	–	–	–	–	–	invalid_ch_sel_ie
0x510	ADC_INTR0[7:0]	–	–	–	adc_overflow_if	adc_lo_limit_if	adc_hi_limit_if	adc_ref_ready_if	adc_done_if
0x511	ADC_INTR1[7:0]	ch7_hi_limit_if	ch6_hi_limit_if	ch5_hi_limit_if	ch4_hi_limit_if	ch3_hi_limit_if	ch2_hi_limit_if	ch1_hi_limit_if	ch0_hi_limit_if
0x512	ADC_INTR2[7:0]	ch7_lo_limit_if	ch6_lo_limit_if	ch5_lo_limit_if	ch4_lo_limit_if	ch3_lo_limit_if	ch2_lo_limit_if	ch1_lo_limit_if	ch0_lo_limit_if
0x513	ADC_INTR3[7:0]	–	–	–	–	–	–	–	invalid_ch_sel_if
0x514	ADC_LIMIT0_0[7:0]	chLoLimit_l0[7:0]							
0x515	ADC_LIMIT0_1[7:0]	chHiLimit_l0[3:0]				–	–	chLoLimit_h0[1:0]	
0x516	ADC_LIMIT0_2[7:0]	–	–	chHiLimit_h0[5:0]					
0x517	ADC_LIMIT0_3[7:0]	–	–	div_sel0[1:0]		ch_sel0[3:0]			
0x518	ADC_LIMIT1_0[7:0]	chLoLimit_l1[7:0]							
0x519	ADC_LIMIT1_1[7:0]	chHiLimit_l1[3:0]				–	–	chLoLimit_h1[1:0]	
0x51A	ADC_LIMIT1_2[7:0]	–	–	chHiLimit_h1[5:0]					
0x51B	ADC_LIMIT1_3[7:0]	–	–	div_sel1[1:0]		ch_sel1[3:0]			
0x51C	ADC_LIMIT2_0[7:0]	chLoLimit_l2[7:0]							
0x51D	ADC_LIMIT2_1[7:0]	chHiLimit_l2[3:0]				–	–	chLoLimit_h2[1:0]	
0x51E	ADC_LIMIT2_2[7:0]	–	–	chHiLimit_h2[5:0]					
0x51F	ADC_LIMIT2_3[7:0]	–	–	div_sel2[1:0]		ch_sel2[3:0]			
0x520	ADC_LIMIT3_0[7:0]	chLoLimit_l3[7:0]							
0x521	ADC_LIMIT3_1[7:0]	chHiLimit_l3[3:0]				–	–	chLoLimit_h3[1:0]	
0x522	ADC_LIMIT3_2[7:0]	–	–	chHiLimit_h3[5:0]					
0x523	ADC_LIMIT3_3[7:0]	–	–	div_sel3[1:0]		ch_sel3[3:0]			
0x524	ADC_LIMIT4_0[7:0]	chLoLimit_l4[7:0]							
0x525	ADC_LIMIT4_1[7:0]	chHiLimit_l4[3:0]				–	–	chLoLimit_h4[1:0]	
0x526	ADC_LIMIT4_2[7:0]	–	–	chHiLimit_h4[5:0]					
0x527	ADC_LIMIT4_3[7:0]	–	–	div_sel4[1:0]		ch_sel4[3:0]			
0x528	ADC_LIMIT5_0[7:0]	chLoLimit_l5[7:0]							
0x529	ADC_LIMIT5_1[7:0]	chHiLimit_l5[3:0]				–	–	chLoLimit_h5[1:0]	
0x52A	ADC_LIMIT5_2[7:0]	–	–	chHiLimit_h5[5:0]					

Address	Name	MSB							LSB
0x52B	ADC_LIMIT5_3[7:0]	–	–	div_sel5[1:0]		ch_sel5[3:0]			
0x52C	ADC_LIMIT6_0[7:0]	chLoLimit_l6[7:0]							
0x52D	ADC_LIMIT6_1[7:0]	chHiLimit_l6[3:0]				–	–	chLoLimit_h6[1:0]	
0x52E	ADC_LIMIT6_2[7:0]	–	–	chHiLimit_h6[5:0]					
0x52F	ADC_LIMIT6_3[7:0]	–	–	div_sel6[1:0]		ch_sel6[3:0]			
0x530	ADC_LIMIT7_0[7:0]	chLoLimit_l7[7:0]							
0x531	ADC_LIMIT7_1[7:0]	chHiLimit_l7[3:0]				–	–	chLoLimit_h7[1:0]	
0x532	ADC_LIMIT7_2[7:0]	–	–	chHiLimit_h7[5:0]					
0x533	ADC_LIMIT7_3[7:0]	–	–	div_sel7[1:0]		ch_sel7[3:0]			
0x534	ADC_RR_CTRL0[7:0]	–	–	–	–	–	–	–	adc_rr_run
0x535	ADC_RR_CTRL1[7:0]	–	–	–	abus_adcsel	–	–	–	–
0x536	ADC_RR_CTRL2[7:0]	adc_rr_sleep_l[7:0]							
0x537	ADC_RR_CTRL3[7:0]	adc_rr_sleep_h[7:0]							
0x53E	ADC_CTRL_4[7:0]	–	–	–	–	–	adc_pin_en[2:0]		
SPI_CC_WR									
0x1300	SPI_CC_WR_[7:0]	–	–	–	–	–	–	–	–
SPI_CC_RD									
0x1380	SPI_CC_RD_[7:0]	–	–	–	–	–	–	–	–
RLMS A									
0x1404	RLMS4[7:0]	EOM_CHK_AMOUNT[3:0]				EOM_CHK_THR[1:0]		EOM_PER_MODE	EOM_EN
0x1405	RLMS5[7:0]	EOM_MAN_TRG_REQ	EOM_MIN_THR[6:0]						
0x1406	RLMS6[7:0]	EOM_PV_MODE	EOM_RST_THR[6:0]						
0x1407	RLMS7[7:0]	EOM_DONE	EOM[6:0]						
0x1464	RLMS64[7:0]	–	–	–	–	–	–	TxSSCMode[1:0]	
0x1470	RLMS70[7:0]	–	TxSSCFrqCtrl[6:0]						
0x1471	RLMS71[7:0]	–	TxSSCCenSprSt[5:0]						TxSSCEn
0x1472	RLMS72[7:0]	TxSSCPreSclL[7:0]							
0x1473	RLMS73[7:0]	–	–	–	–	–	TxSSCPreSclH[2:0]		
0x1474	RLMS74[7:0]	TxSSCPhL[7:0]							
0x1475	RLMS75[7:0]	–	TxSSCPhH[6:0]						
0x1476	RLMS76[7:0]	–	–	–	–	–	–	TxSSCPhQuad[1:0]	

Address	Name	MSB							LSB
0x1483	RLMS83[7:0]	–	–	–	–	–	–	–	–
0x14A7	RLMSA7[7:0]	–	–	–	–	–	–	–	–

Register Details

REG0 (0x0)

Bit	7	6	5	4	3	2	1	0
Field	DEV_ADDR[6:0]							CFG_BLOCK
Reset	1000000							0
Access Type	Write, Read							Write, Read

Bitfield	Bits	Description	Decode
DEV_ADDR	7:1	Device Address - Default value is set by the ADD[2:0] pins as follows: ADD[2:0] Device Address 000 0b1000000 001 0b1000010 010 0b1000100 (for splitter mode) 011 0b1100000 100 0b1100010 101 0b1100100 110 0b0100000 111 0b0100010	0b0000000: I2C write/read address is 0x00/0x01 0b0000001: I2C write/read address is 0x02/0x03 .: . .: . 0b1000000: I2C write/read address is 0x80/0x81 0b1000010: I2C write/read address is 0x84/0x85 0b1000100: I2C write/read address is 0x88/0x89 0b1100000: I2C write/read address is 0xC0/0xC1 0b1100010: I2C write/read address is 0xC4/0xC5 0b1100100: I2C write/read address is 0xC8/0xC9 0b0100000: I2C write/read address is 0x40/0x41 0b0100010: I2C write/read address is 0x44/0x45 .: . .: . 0b1111111: I2C write/read address is 0xFE/0xFF
CFG_BLOCK	0	Configuration Block. When set, all registers become non-writable (read-only). This bit can be used to freeze the chip configuration.	0b0: Not Blocked 0b1: Blocked

REG1 (0x1)

Bit	7	6	5	4	3	2	1	0
Field	IIC_2_EN	IIC_1_EN	DIS_LOCAL_CC	DIS_REM_CC	TX_RATE[1:0]		RX_RATE[1:0]	
Reset	0	0	0	0	10		00	
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read		Write, Read	

Bitfield	Bits	Description	Decode
IIC_2_EN	7	Enable pass-thru I2C channel 2 (SDA2/RX2, SCL2/TX2)	0b0: I2C pass through channel 2 disabled 0b1: I2C pass through channel 2 enabled
IIC_1_EN	6	Enable pass-thru I2C channel 1 (SDA1/RX1, SCL1/TX1)	0b0: I2C pass through channel 1 disabled

Bitfield	Bits	Description	Decode
			0b1: I2C pass through channel 1 enabled
DIS_LOCAL_CC	5	Disable control channel connection to RX/SDA and TX/SCL pins	0b0: RX/SDA and TX/XCL connected to control channel 0b1: RX/SDA and TX/SCL disconnected from control channel
DIS_REM_CC	4	Disable remote control channel link over GMSL2 connection	0b0: Remote control channel enabled 0b1: Remote control channel disabled
TX_RATE	3:2	Transmitter (forward channel) bit rate (when changed, becomes active after next link reset) Default value is set by the CXTP pin at power up: 6Gbps when CXTP=1 (Coax cable) and 3Gbps when CXTP=0 (Twisted pair cable)	0b00: Reserved 0b01: 3Gbps 0b10: 6Gbps 0b11: Reserved
RX_RATE	1:0	Receiver (reverse channel) bit rate (when changed, becomes active after next link reset)	0b00: 187.5Mbps 0b01: 375Mbps 0b10: 750Mbps 0b11: 1.5Gbps

REG2 (0x2)

Bit	7	6	5	4	3	2	1	0
Field	VID_TX_EN_U	VID_TX_EN_Z	VID_TX_EN_Y	VID_TX_EN_X	AUD_TX_EN_Y	AUD_TX_EN_X	–	–
Reset	0	1	0	1	0	0	–	–
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	–	–

Bitfield	Bits	Description	Decode
VID_TX_EN_U	7	Video transmit enable for port U	0b0: Video transmit channel U disabled 0b1: Video transmit channel U enabled
VID_TX_EN_Z	6	Video transmit enable for port Z	0b0: Video transmit channel Z disabled 0b1: Video transmit channel Z enabled
VID_TX_EN_Y	5	Video transmit enable for port Y	0b0: Video transmit channel Y disabled 0b1: Video transmit channel Y enabled
VID_TX_EN_X	4	Video transmit enable for port X	0b0: Video transmit channel X disabled 0b1: Video transmit channel X enabled
AUD_TX_EN_Y	3	Audio transmit enable for port Y	0b0: Audio transmit channel Y disabled 0b1: Audio transmit channel Y enabled
AUD_TX_EN_X	2	Audio transmit enable for port X	0b0: Audio transmit channel X disabled 0b1: Audio transmit channel X enabled

REG3 (0x3)

Bit	7	6	5	4	3	2	1	0
Field	–	–	UART_2_EN	UART_1_EN	AUD_RX_SRC	RCLK_ALT	RCLKSEL[1:0]	
Reset	–	–	0	0	0	0	00	
Access Type	–	–	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	

Bitfield	Bits	Description	Decode
UART_2_EN	5	Enable pass-thru UART channel #2 (SDA2/RX2, SCL2/TX2)	0b0: Pass through UART channel 2 disabled 0b1: Pass through UART channel 2 enabled
UART_1_EN	4	Enable pass-thru UART channel #1 (SDA1/RX1, SCL1/TX1)	0b0: Pass through UART channel 1 disabled 0b1: Pass through UART channel 1 enabled
AUD_RX_SRC	3	Select reverse channel port to receive audio from	0b0: Reverse audio received on GMSL2A 0b1: Reverse audio received on GMSL2B
RCLK_ALT	2	Eanble RCLK output from alternative MFP pin	0b0: RCLK output disabled 0b1: RCLK output enabled
RCLKSEL	1:0	RCLKOUT clock selection	0b00: XTAL/1 = 25MHz 0b01: XTAL/2 = 12.5MHz 0b10: XTAL/4 = 6.25MHz 0b11: Reference PLL output

REG5 (0x5)

Bit	7	6	5	4	3	2	1	0
Field	LOCK_EN	ERRB_EN	–	–	PU_LF3	PU_LF2	PU_LF1	PU_LF0
Reset	1	1	–	–	0	0	0	0
Access Type	Write, Read	Write, Read	–	–	Write, Read	Write, Read	Write, Read	Write, Read

Bitfield	Bits	Description	Decode
LOCK_EN	7	Enable LOCK output (disabled by default in 9295D)	0b0: LOCK output disabled 0b1: LOCK output enabled
ERRB_EN	6	Enable ERRB output (disabled by default in 9295D)	0b0: ERRB output disabled 0b1: ERRB output enabled
PU_LF3	3	Power up Line Faullt monitor 3	0b0: Line fault monitor 3 disabled 0b1: Line fault monitor 3 enabled
PU_LF2	2	Power up Line Faullt monitor 2	0b0: Line fault monitor 2 disabled 0b1: Line fault monitor 2 enabled
PU_LF1	1	Power up Line Faullt monitor 1	0b0: Line fault monitor 1 disabled 0b1: Line fault monitor 1 enabled

Bitfield	Bits	Description	Decode
PU_LF0	0	Power up Line Faullt monitor 0	0b0: Line fault monitor 0 disabled 0b1: Line fault monitor 0 enabled

REG6 (0x6)

Bit	7	6	5	4	3	2	1	0
Field	–	–	RCLKEN	–	–	–	–	–
Reset	–	–	0	–	–	–	–	–
Access Type	–	–	Write, Read	–	–	–	–	–

Bitfield	Bits	Description	Decode
RCLKEN	5	Enable RCLK output. Bit is set by the RCLKEN pin at power up.	0b0: RCLK output disabled 0b1: RCLK output enabled

REG7 (0x7)

Bit	7	6	5	4	3	2	1	0
Field	PAR_DIN_EN[3:0]				PAR_INV_PCLK	PAR_VS_EN	PAR_HS_EN	PAR_VID_EN
Reset	0xF				0	1	1	0
Access Type	Write, Read				Write, Read	Write, Read	Write, Read	Write, Read

Bitfield	Bits	Description	Decode
PAR_DIN_EN	7:4	Enable extra parallel data inputs (valid only if PAR_VID_EN=1)	0bXXX0: 32-pin package - disable D12; 48-pin package - disable D2 0bXXX1: 32-pin package - enable D12; 48-pin package - enable D2 0bXX0X: 32-pin package - disable D13; 48-pin package - disable D3 0bXX0X: 32-pin package - enable D13; 48-pin package - enable D3 0bX0XX: 32-pin packge - Reserved; 48-pin package - disable D20 0bX1XX: 32-pin packge - Reserved; 48-pin package - enable D20 0b0XXX: 32-pin packge - Reserved; 48-pin package - disable D21 0b1XXX: 32-pin packge - Reserved; 48-pin package - enable D21
PAR_INV_PCLK	3	Parallel PCLK sampling edge select	0b0: Rising edge 0b1: Falling edge
PAR_VS_EN	2	Enable parallel vertical sync (VS) input (valid only if PAR_VID_EN=1)	0b0: Parallel VS disabled 0b1: Parallel VS enabled

Bitfield	Bits	Description	Decode
PAR_HS_EN	1	Enable parallel horizontal sync (HS) input (valid only if PAR_VID_EN=1)	0b0: Parallel HS disabled 0b1: Parallel HS enabled
PAR_VID_EN	0	Enable parallel video input	0b0: Parallel video disabled 0b1: Parallel video enabled

REG13 (0xD)

Bit	7	6	5	4	3	2	1	0
Field	DEV_ID[7:0]							
Reset	10011011							
Access Type	Read Only							

Bitfield	Bits	Description	Decode
DEV_ID	7:0	Device Identifier	0x91: MAX9295A (T32) 0x93: MAX9295B (T32) 0x95: MAX9295D (T48) 0x99: MAX9283 (T40) 0x9A: MAX9285 (T40) 0x9B: MAX96755 (T56) 0x9C: MAX96757 (T56) 0x9D: MAX96773 (A76) 0x9E: MAX96775 (A76)

REG14 (0xE)

Bit	7	6	5	4	3	2	1	0
Field	–	–	–	–	DEV_REV[3:0]			
Reset	–	–	–	–	0000			
Access Type	–	–	–	–	Read Only			

Bitfield	Bits	Description	Decode
DEV_REV	3:0	Device revision	0xX: Device revision number

REG26 (0x26)

Bit	7	6	5	4	3	2	1	0
Field	–	LF_1[2:0]			–	LF_0[2:0]		
Reset	–	010			–	010		
Access Type	–	Read Only			–	Read Only		

Bitfield	Bits	Description	Decode
LF_1	6:4	Line Fault status of wire connected to LMN1 pin	0b000: Short to battery

Bitfield	Bits	Description	Decode
			0b001: Short to GND 0b010: Normal Operation 0b011: Line Open 0b1XX: Line-to-line short
LF_0	2:0	Line Fault status of wire connected to LMN0 pin	0b000: Short to battery 0b001: Short to GND 0b010: Normal Operation 0b011: Line Open 0b1XX: Line-to-line short

REG27 (0x27)

Bit	7	6	5	4	3	2	1	0
Field	–	LF_3[2:0]			–	LF_2[2:0]		
Reset	–	010			–	010		
Access Type	–	Read Only			–	Read Only		

Bitfield	Bits	Description	Decode
LF_3	6:4	Line Fault status of wire connected to LMN3 pin	0b000: Short to battery 0b001: Short to GND 0b010: Normal Operation 0b011: Line Open 0b1XX: Line-to-line short
LF_2	2:0	Line Fault status of wire connected to LMN2 pin	0b000: Short to battery 0b001: Short to GND 0b010: Normal Operation 0b011: Line Open 0b1XX: Line-to-line short

IO_CHK0 (0x38)

Bit	7	6	5	4	3	2	1	0
Field	PIN_DRV_EN_0[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
PIN_DRV_EN_0	7:0	Enables individual pin output buffers and drives the value selected by PIN_DRV_SEL to that pin so that the driven value can be read from the PIN_IN_0 register. This allows testing of IO buffer input and output paths. When pin is driven by using this register, it should not be driven by any other strong driver externally.	0bXXXXXXX0: Disable RG_TXCTL output buffer 0bXXXXXXX1: Enable RG_TXCTL output buffer 0bXXXXXX0X: Disable RG_RXCTL output buffer 0bXXXXXX1X: Enable RG_RXCTL output buffer 0bXXXXX0XX: Disable RG_MDC output buffer 0bXXXXX1XX: Enable RG_MDC output buffer 0bXXXX0XXX: Disable RG_MDIO output buffer

Bitfield	Bits	Description	Decode
			0bXXXX1XXX: Enable RG_MDIO output buffer 0bXXX0XXXX: Disable RG-REFCLK output buffer 0bXXX1XXXX: Enable RG-REFCLK output buffer 0bXX0XXXXX: Disable RG_RD0 output buffer 0bXX1XXXXX: Enable RG_RD0 output buffer 0bX0XXXXXX: Disable RG_RD1 output buffer 0bX1XXXXXX: Enable RG_RD1 output buffer 0b0XXXXXXX: Disable RG-RXC output buffer 0b1XXXXXXX: Enable RG_RXC output buffer

IO_CHK1 (0x39)

Bit	7	6	5	4	3	2	1	0
Field	PIN_DRV_SEL	PIN_DRV_EN_1[6:0]						
Reset	0	0000000						
Access Type	Write, Read	Write, Read						

Bitfield	Bits	Description	Decode
PIN_DRV_SEL	7	Select polarity for driving dedicated IOs	0b0: Drive low level 0b1: Drive high level
PIN_DRV_EN_1	6:0	Enables individual pin output buffers and drives the value selected by PIN_DRV_SEL to that pin so that the driven value can be read from the PIN_IN_1 register. This allows testing of IO buffer input and output paths. When pin is driven by using this register, it should not be driven by any other strong driver externally.	0bXXXXXXX0: Disable RG_RD2 output buffer 0bXXXXXXX1: Enable RG_RD2 output buffer 0bXXXXXX0X: Disable RG_RD3 output buffer 0bXXXXXX1X: Enable RG_RD3 output buffer 0bXXXXX0XX: Disable RG_TD3 output buffer 0bXXXXX1XX: Enable RG_TD3 output buffer 0bXXX0XXX: Disable RG_TD2 output buffer 0bXXX1XXX: Enable RG_TD2 output buffer 0bXX0XXXX: Disable RG-TXC output buffer 0bXX1XXXX: Enable RG-TXC output buffer 0bX0XXXXX: Disable RG_TD1 output buffer 0bX1XXXXX: Enable RG_TD1 output buffer 0b0XXXXXX: Disable RG_TD0 output buffer 0b1XXXXXX: Enable RG_TD0 output buffer

IO_CHK3 (0x3B)

Bit	7	6	5	4	3	2	1	0
Field	PIN_IN_0[7:0]							
Reset	00000000							
Access Type	Read Only							

Bitfield	Bits	Description	Decode
PIN_IN_0	7:0	Pin level readback. Each bit contains the level at the pin, either driven externally, or by the	Bit 0: RG_TXCTL

Bitfield	Bits	Description	Decode
		PIN_DRV_SEL value if the equivalent bit is enabled in the PIN_DRV_EN_0 register.	Bit 1: RG_RXCTL Bit 2: RG_MDC Bit 3: RG_MDIO Bit 4: RG-REFCLK Bit 5: RG_RD0 Bit 6: RG_RD1 Bit 7: RG_RXC

IO_CHK4 (0x3C)

Bit	7	6	5	4	3	2	1	0
Field	–	PIN_IN_1[6:0]						
Reset	–	0000000						
Access Type	–	Read Only						

Bitfield	Bits	Description	Decode
PIN_IN_1	6:0	Read dedicated IO pin level [14:8] = RG_TD0, RG_TD1, RG_TXC, RG_TD2, RG_TD3, RG_RD3, RG_RD2	Bit 0: RG_RD2 Bit 1: RG_RD3 Bit 2: RG_TD3 Bit 3: RG_TD2 Bit 4: RG_TXC Bit 5: RG_TD1 Bit 6: RG_TD0

PWR4 (0xC)

Bit	7	6	5	4	3	2	1	0
Field	–	DIS_LOCAL_WAKE	WAKE_EN_B	WAKE_EN_A	–	–	–	–
Reset	–	0	0	1	–	–	–	–
Access Type	–	Write, Read	Write, Read	Write, Read	–	–	–	–

Bitfield	Bits	Description	Decode
DIS_LOCAL_WAKE	6	Disable wake up by local uC from SDA_RX pin	0b0: Local wake up disabled 0b1: Local wake up enabled
WAKE_EN_B	5	Enable wake up by remote chip connected to Link B	0b0: Link B remote wake up disabled 0b1: Link B remote wake up enabled
WAKE_EN_A	4	Enable wake up by remote chip connected to Link A	0b0: Link A remote wake up disabled 0b1: Link A remote wake up enabled

CTRL0 (0x10)

Bit	7	6	5	4	3	2	1	0
Field	RESET_ALL	RESET_LINK	RESET_ONESHOT	AUTO_LINK	SLEEP	REG_ENABLE	LINK_CFG[1:0]	
Reset	0	0	0	1	0	0	01	
Access Type	Write, Read	Write, Read	Write Clears All, Read	Write, Read	Write, Read	Write, Read	Write, Read	

Bitfield	Bits	Description	Decode
RESET_ALL	7	Writing 1 to this bit resets the device including all blocks, and registers are reset to defaults. This is equivalent to toggling the PWDNB pin. The bit is cleared when written.	0b0: No action 0b1: Activate chip reset
RESET_LINK	6	Reset whole data path (keep register settings). Write 1 to activate reset, write 0 to release reset.	0b0: Release link reset 0b1: Activate link reset
RESET_ONESHOT	5	Reset whole data path (keep register settings) one shot. Write 1 to activate reset, bit self clears and automatically releases reset.	0b0: No action 0b1: Reset data path
AUTO_LINK	4	Automatically select link configuration (single A, single B or dua). Splitter mode is not automatic. **For Splitter mode, set LINK_CFG=11.**	0b0: Disable auto link configuration 0b1: Enable auto link configuration
SLEEP	3	Activate sleep mode	0b0: Sleep mode disabled 0b1: Sleep mode enabled
REG_ENABLE	2	Regulator enable	0b0: Regulator disabled 0b1: Regulator enabled
LINK_CFG	1:0	AUTO_LINK and this bitfield select the link configuration per the decode table. When ADD2, ADD1, ADD0 pins are “010” this register’s default value is “11” enabling splitter mode.	0b00: If AUTO_LINK=0, Dual Link selected. IF AUTO_LINK=1, Link mode is automatically selected 0b01: If AUTO_LINK=0, Link A is selected. IF AUTO_LINK=1, Link mode is automatically selected 0b10: If AUTO_LINK=0, Link B is selected. IF AUTO_LINK=1, Link mode is automatically selected 0b11: Splitter mode

CTRL1 (0x11)

Bit	7	6	5	4	3	2	1	0
Field	–	–	–	BACK_COMP_SPLTR	–	CXTP_B	–	CXTP_A
Reset	–	–	–	0	–	0	–	0
Access Type	–	–	–	Write, Read	–	Write, Read	–	Write, Read

Bitfield	Bits	Description	Decode
BACK_COMP_SPLTR	4	Enable splitter mode compatibility with HS84 Rev1	0b0: Backward compatibility disabled 0b1: Backward compatibility enabled
CXTP_B	2	Co-ax / Twisted-pair cable select for Link B Bit is set according to the latched CFG pin value at power up	0b0: Shielded twisted pair drive 0b1: Coax drive
CXTP_A	0	Co-ax / Twisted-pair cable select for Link A Bit is set according to the latched CFG pin value at power up	0b0: Shielded twisted pair drive 0b1: Coax drive

CTRL2 (0x12)

Bit	7	6	5	4	3	2	1	0
Field	IBLEED_OFF	–	–	–	–	–	–	–
Reset	0	–	–	–	–	–	–	–
Access Type	Write, Read	–	–	–	–	–	–	–

Bitfield	Bits	Description	Decode
IBLEED_OFF	7	Turn off regulator bleeder current	0b0: Regulator bleeder current on 0b1: Regulator bleeder current off

CTRL3 (0x13)

Bit	7	6	5	4	3	2	1	0
Field	–	–	LINK_MODE[1:0]		LOCKED	ERROR	CMU_LOCKED	–
Reset	–	–	01		0	0	0	–
Access Type	–	–	Read Only		Read Only	Read Only	Read Only	–

Bitfield	Bits	Description	Decode
LINK_MODE	5:4	Active link mode	0b00: Dual link 0b01: Link A 0b10: Link B 0b11: Splitter mode
LOCKED	3	GMSL-2 Link Locked (bi-directional)	0b0: GMSL2 link not locked 0b1: GMSL2 link locked
ERROR	2	Reflects error status (inverse of ERRB pin value)	0b0: ERRB not asserted (ERRB pin = 1) 0b1: ERRB asserted (ERRB pin = 0)
CMU_LOCKED	1	CMU (Clock Multiplier Unit) locked	0b0: CMU not locked 0b1: CMU locked

INTR0 (0x18)

Bit	7	6	5	4	3	2	1	0
Field	–	–	–	–	AUTO_ERR_RST_EN	DEC_ERR_THR[2:0]		
Reset	–	–	–	–	0	000		
Access Type	–	–	–	–	Write, Read	Write, Read		

Bitfield	Bits	Description	Decode
AUTO_ERR_RST_EN	3	Automatically reset DEC_ERR_A, DEC_ERR_B and IDLE_ERR registers after ERRB pin is asserted for 1us	0b0: Auto reset disabled 0b1: Auto reset enabled
DEC_ERR_THR	2:0	Decoding and idle error reporting threshold. • DEC_ERR_FLAG_A is asserted when DEC_ERR_A >= DEC_ERR_THR. • DEC_ERR_FLAG_B is asserted when DEC_ERR_B >= DEC_ERR_THR. • IDLE_ERR_FLAG is asserted when IDLE_ERR >= DEC_ERR_THR.	0b000: 1 0b001: 2 0b010: 4 0b011: 8 0b100: 16 0b101: 32 0b110: 64 0b111: 128

INTR1 (0x19)

Bit	7	6	5	4	3	2	1	0
Field	PKT_CNT_EXP[3:0]				AUTO_CNT_RST_EN	–	–	–
Reset	0000				0	–	–	–
Access Type	Write, Read				Write, Read	–	–	–

Bitfield	Bits	Description	Decode
PKT_CNT_EXP	7:4	Packet count multiplier exponent See the description of PKT_CNT register.	0bXXX: PKT_CNT exponent
AUTO_CNT_RST_EN	3	Automatically reset PKT_CNT register after ERRB pin is asserted for 1us	0b0: Auto reset disabled 0b1: Auto reset enabled

INTR2 (0x1A)

Bit	7	6	5	4	3	2	1	0
Field	PHY_INT_OEN_B	PHY_INT_OEN_A	REM_ERR_OEN	–	LFLT_INT_OEN	IDLE_ERR_OEN	DEC_ERR_OEN_B	DEC_ERR_OEN_A
Reset	0	0	0	–	1	0	1	1
Access Type	Write, Read	Write, Read	Write, Read	–	Write, Read	Write, Read	Write, Read	Write, Read

Bitfield	Bits	Description	Decode
PHY_INT_OEN_B	7	Enable reporting of PHY interrupt	0b0: Reporting disabled

Bitfield	Bits	Description	Decode
		(PHY_INT_B) at ERRB pin	0b1: Reporting enabled
PHY_INT_OEN_A	6	Enable reporting of PHY interrupt (PHY_INT_A) at ERRB pin	0b0: Reporting disabled 0b1: Reporting enabled
REM_ERR_OEN	5	Enable reporting of remote error status (REM_ERR) at ERRB pin	0b0: Reporting disabled 0b1: Reporting enabled
LFLT_INT_OEN	3	Enable reporting of Line Fault interrupt (LFLT_INT) at ERRB pin	0b0: Reporting disabled 0b1: Reporting enabled
IDLE_ERR_OEN	2	Enable reporting of idle word errors (IDLE_ERR_FLAG) at ERRB pin	0b0: Reporting disabled 0b1: Reporting enabled
DEC_ERR_OEN_B	1	Enable reporting of decoding errors (DEC_ERR_FLAG_B) at ERRB pin	0b0: Reporting disabled 0b1: Reporting enabled
DEC_ERR_OEN_A	0	Enable reporting of decoding errors (DEC_ERR_FLAG_A) at ERRB pin	0b0: Reporting disabled 0b1: Reporting enabled

INTR3 (0x1B)

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Field	PHY_INT_B	PHY_INT_A	REM_ERR_FLAG	–	LFLT_INT	IDLE_ERR_FLAG	DEC_ERR_FLAG_B	DEC_ERR_FLAG_A
Reset	0	0	0	–	0	0	0	0
Access Type	Read Only	Read Only	Read Only	–	Read Only	Read Only	Read Only	Read Only

Bitfield	Bits	Description	Decode
PHY_INT_B	7	PHY Interrupt of Link B. See RLMS_B page, registers 0x4C, 0x4D, 0x4E, 0x4F.	0b0: No Link B PHY interrupt 0b1: Link B PHY interrupt
PHY_INT_A	6	PHY Interrupt of Link A. See RLMS_A page, registers 0x4C, 0x4D, 0x4E, 0x4F.	0b0: No Link A PHY interrupt 0b1: Link A PHY interrupt
REM_ERR_FLAG	5	Received remote side error status (inverse of remote side ERRB pin level)	0b0: No remote side error 0b1: Remote side error
LFLT_INT	3	Line Fault interrupt Asserted when either one of Line Fault monitors indicate a fault status. See LF_0 and LF_1 registers for more info.	0b0: No line fault 0b1: Line fault
IDLE_ERR_FLAG	2	Idle word error flag, asserted when IDLE_ERR >= DEC_ERR_THR	0b0: Flag not asserted 0b1: Flag asserted
DEC_ERR_FLAG_B	1	Decoding error flag for Link B, asserted when DEC_ERR_B >= DEC_ERR_THR	0b0: Flag not asserted 0b1: Flag asserted
DEC_ERR_FLAG_A	0	Decoding error flag for Link A, asserted when	0b0: Flag not asserted

Bitfield	Bits	Description	Decode
		DEC_ERR_A >= DEC_ERR_THR	0b1: Flag asserted

INTR4 (0x1C)

Bit	7	6	5	4	3	2	1	0
Field	EOM_ERR_OEN_B	EOM_ERR_OEN_A	–	HDCP_INT_OEN	MAX_RT_OEN	RT_CNT_OEN	PKT_CNT_OEN	WM_ERR_OEN
Reset	0	0	–	0	1	0	0	1
Access Type	Write, Read	Write, Read	–	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read

Bitfield	Bits	Description	Decode
EOM_ERR_OEN_B	7	Enable reporting of Eye Opening Monitor error(EOM_ERR_FLAG_B) for Link B at ERRB pin	0b0: Reporting disabled 0b1: Reporting enabled
EOM_ERR_OEN_A	6	Enable reporting of Eye Opening Monitor error(EOM_ERR_FLAG_A) for Link A at ERRB pin	0b0: Reporting disabled 0b1: Reporting enabled
HDCP_INT_OEN	4	Enable reporting of GMSL HDCP interrupt (HDCP_INT_FLAG) at ERRB pin	0b0: Reporting disabled 0b1: Reporting enabled
MAX_RT_OEN	3	Enable reporting of combined ARQ maximum re-transmission limit error flag (MAX_RT_FLAG) at ERRB pin	0b0: Reporting disabled 0b1: Reporting enabled
RT_CNT_OEN	2	Enable reporting of combined ARQ re-transmission event flag (RT_CNT_FLAG) at ERRB pin	0b0: Reporting disabled 0b1: Reporting enabled
PKT_CNT_OEN	1	Enable reporting of packet count flag (PKT_CNT_FLAG) at ERRB pin	0b0: Reporting disabled 0b1: Reporting enabled
WM_ERR_OEN	0	Enable reporting of Watermark errors (WM_ERR_FLAG) at ERRB pin	0b0: Reporting disabled 0b1: Reporting enabled

INTR5 (0x1D)

Bit	7	6	5	4	3	2	1	0
Field	EOM_ERR_FLAG_B	EOM_ERR_FLAG_A	–	HDCP_INT_FLAG	MAX_RT_FLAG	RT_CNT_FLAG	PKT_CNT_FLAG	WM_ERR_FLAG
Reset	0	0	–	0	0	0	0	0
Access Type	Read Only	Read Only	–	Read Only	Read Only	Read Only	Read Only	Read Clears All

Bitfield	Bits	Description	Decode
EOM_ERR_FLAG_B	7	Eye Opening is below configured threshold for Link B	0b0: No EOM error on Link B 0b1: EOM error on Link B

Bitfield	Bits	Description	Decode
EOM_ERR_FLAG_A	6	Eye Opening is below configured threshold for Link A	0b0: No EOM error on Link A 0b1: EOM error on Link A
HDCP_INT_FLAG	4	Asserted when there is a GMSL HDCP interrupt. See HDCP interrupts in HDCP register page for more detail.	0b0: No HDCP interrupt 0b1: HDCP interrupt
MAX_RT_FLAG	3	Combined ARQ maximum re-transmission limit error flag, asserted when any of the selected channels ARQ re-transmission limit is reached. Selection is done by each channel's MAX_RT_ERR_OEN register bit.	0b0: Flag not asserted 0b1: Flag asserted
RT_CNT_FLAG	2	Combined ARQ re-transmission event flag, asserted when any of the selected channels have done at least one ARQ re-transmission. Selection is done by each channel's RT_CNT_OEN register bit.	0b0: Flag not asserted 0b1: Flag asserted
PKT_CNT_FLAG	1	Packet count flag, asserted when PKT_CNT >= PKT_CNT_THR	0b0: Flag not asserted 0b1: Flag asserted
WM_ERR_FLAG	0	Watermark error flag, asserted when a Watermark error is detected	0b0: Flag not asserted 0b1: Flag asserted

INTR6 (0x1E)

Bit	7	6	5	4	3	2	1	0
Field	VDDCMP_INT_OEN	PORZ_INT_OEN	VDDBAD_INT_OEN	–	–	ADC_INT_OEN	APRBS_ERR_OEN	MIPI_ERR_OEN
Reset	1	1	1	–	–	0	1	1
Access Type	Write, Read	Write, Read	Write, Read	–	–	Write, Read	Write, Read	Write, Read

Bitfield	Bits	Description	Decode
VDDCMP_INT_OEN	7	Enable reporting of combined VDD comparator output (VDDCMP_INT_FLAG) at ERRB pin	0x0: Disable reporting 0x1: Enable reporting
PORZ_INT_OEN	6	Enable reporting of porz interrupt (PORZ_INT_FLAG) at ERRB pin	0x0: Disable reporting 0x1: Enable reporting
VDDBAD_INT_OEN	5	Enable reporting of vddbad interrupt (VDDBAD_INT_FLAG) at ERRB pin	0x0: Disable reporting 0x1: Enable reporting
ADC_INT_OEN	2	Enable reporting of ADC interrupts (ADC_INT_FLAG) at ERRB pin	0b0: Reporting disabled 0b1: Reporting enabled
APRBS_ERR_OEN	1	Enable reporting of Audio PRBS errors (APRBS_ERR_FLAG) at ERRB pin	0b0: Reporting disabled 0b1: Reporting enabled

Bitfield	Bits	Description	Decode
MIPI_ERR_OEN	0	Enable reporting of MIPI RX errors (MIPI_ERR_FLAG) at ERRB pin	0b0: Reporting disabled 0b1: Reporting enabled

INTR7 (0x1F)

Bit	7	6	5	4	3	2	1	0
Field	VDDCMP_INT_FLAG	PORZ_INT_FLAG	VDDBAD_INT_FLAG	–	–	ADC_INT_FLAG	APRBS_ERR_FLAG	MIPI_ERR_FLAG
Reset	0	0	0	–	–	0	0	0
Access Type	Read Clears All	Read Clears All	Read Clears All	–	–	Read Only	Read Only	Read Only

Bitfield	Bits	Description	Decode
VDDCMP_INT_FLAG	7	Combined VDD comparator output See CMP_STATUS_MASK register	0x0: Disable reporting 0x1: Enable reporting
PORZ_INT_FLAG	6	PORZ interrupt flag	0x0: Disable reporting 0x1: Enable reporting
VDDBAD_INT_FLAG	5	Combined VDD bad indicator	0x0: Disable reporting 0x1: Enable reporting
ADC_INT_FLAG	2	ADC Interrupt	0b0: No ADC interrupt 0b1: ADC interrupt
APRBS_ERR_FLAG	1	Audio PRBS error flag, asserted when APRBS_ERR > 0	0b0: Flag not asserted 0b1: Flag asserted
MIPI_ERR_FLAG	0	MIPI RX error flag, asserted when any of these is asserted: phy0_hs_err [0],[1],[4],[5] phy1_hs_err [0],[1],[4],[5] phy2_hs_err [0],[1],[4],[5] phy3_hs_err [0],[1],[4],[5] ctrl0_csi_err_l [0],[1],[7] ctrl0_csi_err_h [0] ctrl1_csi_err_l [0],[1],[7] ctrl1_csi_err_h [0] ctrl0_dsi_err_l [0],[1],[2],[3],[4],[5],[6] ctrl1_dsi_err_l [0],[1],[2],[3],[4],[5],[6]	0b0: Flag not asserted 0b1: Flag asserted

INTR8 (0x20)

Bit	7	6	5	4	3	2	1	0
Field	ERR_TX_EN	–	–	ERR_TX_ID[4:0]				
Reset	1	–	–	11111				
Access Type	Write, Read	–	–	Write, Read				

Bitfield	Bits	Description	Decode
ERR_TX_EN	7	Transmit local error status (inverse of ERRB pin level) to remote side through GPIO channel	0b0: Transmit error status disabled 0b1: Transmit error status enabled
ERR_TX_ID	4:0	GPIO ID used for transmitting ERR_TX	0bXXXXX: Value of GPIO ID for transmitting ERR_TX

INTR9 (0x21)

Bit	7	6	5	4	3	2	1	0
Field	ERR_RX_EN	–	–	ERR_RX_ID[4:0]				
Reset	1	–	–	11111				
Access Type	Write, Read	–	–	Write, Read				

Bitfield	Bits	Description	Decode
ERR_RX_EN	7	Receive remote error status (inverse of ERRB pin level) through GPIO channel	0b0: Receive error status disabled 0b1: Receive error status enabled
ERR_RX_ID	4:0	GPIO ID used for receiving ERR_RX	0bXXXXX: Value of GPIO ID for receiving ERR_TX

CNT0 (0x22)

Bit	7	6	5	4	3	2	1	0
Field	DEC_ERR_A[7:0]							
Reset	00000000							
Access Type	Read Clears All							

Bitfield	Bits	Description	Decode
DEC_ERR_A	7:0	Number of decoding (disparity) errors detected at Link A Reset after reading or with the rising edge of LOCK.	0xXX: Number of Link A decoding errors detected

CNT1 (0x23)

Bit	7	6	5	4	3	2	1	0
Field	DEC_ERR_B[7:0]							
Reset	00000000							
Access Type	Read Clears All							

Bitfield	Bits	Description	Decode
DEC_ERR_B	7:0	Number of decoding (disparity) errors detected at Link B	0xXX: Number of Link B decoding errors detected

Bitfield	Bits	Description	Decode
		Reset after reading or with the rising edge of LOCK.	

CNT2 (0x24)

Bit	7	6	5	4	3	2	1	0
Field	IDLE_ERR[7:0]							
Reset	00000000							
Access Type	Read Clears All							

Bitfield	Bits	Description	Decode
IDLE_ERR	7:0	Number of idle word errors detected. Reset after reading or with the rising edge of LOCK.	0xXX: Number of idle word errors detected

CNT3 (0x25)

Bit	7	6	5	4	3	2	1	0
Field	PKT_CNT[7:0]							
Reset	00000000							
Access Type	Read Clears All							

Bitfield	Bits	Description	Decode
PKT_CNT	7:0	Number of received packets of a selected type. Packet type is selected with PKT_CNT_SEL register. Reported packet count is a scaled value, such that actual packet count is >= PKT_CNT x (2^PKT_CNT_EXP) and < (PKT_CNT+1) x (2^PKT_CNT_EXP). When max value is reported, packet count is greater or equal to the reported value.	0xXX: Scaled number of received packets

TX1 (0x29)

Bit	7	6	5	4	3	2	1	0
Field	–	–	ERRG_EN_B	ERRG_EN_A	–	–	–	–
Reset	–	–	0	0	–	–	–	–
Access Type	–	–	Write, Read	Write, Read	–	–	–	–

Bitfield	Bits	Description	Decode
ERRG_EN_B	5	Error generator enable for Link B	0b0: Link B error generator disabled 0b1: Link B error generator enabled

Bitfield	Bits	Description	Decode
ERRG_EN_A	4	Error generator enable for Link A	0b0: Link A error generator disabled 0b1: Link A error generator enabled

TX2 (0x2A)

Bit	7	6	5	4	3	2	1	0
Field	ERRG_CNT[1:0]		ERRG_RATE[1:0]		ERRG_BURST[2:0]			ERRG_PER
Reset	00		10		000			0
Access Type	Write, Read		Write, Read		Write, Read			Write, Read

Bitfield	Bits	Description	Decode
ERRG_CNT	7:6	Number of errors to be generated	0b00: Continuous 0b01: 16 0b10: 128 0b11: 1024
ERRG_RATE	5:4	Error generator average bit error rate	0b00: 1 in 5120 bits 0b01: 1in 81920 bits 0b10: 1in 1310720 bits 0b11: 1 in 20971520 bits
ERRG_BURST	3:1	Error generator burst error length	0b000: 1 0b001: 2 0b010: 3 0b011: 4 0b100: 8 0b101: 12 0b110: 16 0b111: 20
ERRG_PER	0	Error generator error distribution selection	0b0: Pseudo random 0b1: Periodic

TX3 (0x2B)

Bit	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	TIMEOUT[2:0]		
Reset	–	–	–	–	–	100		
Access Type	–	–	–	–	–	Write, Read		

Bitfield	Bits	Description	Decode
TIMEOUT	2:0	ARQ Timeout duration multiplier. Multiplies a Timeout Base constant to set the ARQ Timeout. The Timeout Base is set by the reverse channel link rate (RX_RATE) as follows: RX_RATE Timeout Base	0b000: 0.5 x Timeout Base 0b001: 1.0 x Timeout Base 0b010: 1.5 x Timeout Base 0b011: 2.0 x Timeout Base 0b100: 2.5 x Timeout Base

Bitfield	Bits	Description	Decode
		187.5 Mbps 8µs 350 Mbps 4µs 750 Mbps 3µs 1.5 Gbps 2µs	0b101: 3.0 x Timeout Base 0b110: 3.5 x Timeout Base 0b111: 4.0 x Timeout Base

RX0 (0x2C)

Bit	7	6	5	4	3	2	1	0
Field	PKT_CNT_LBW[1:0]		–	–	PKT_CNT_SEL[3:0]			
Reset	00		–	–	0000			
Access Type	Write, Read		–	–	Write, Read			

Bitfield	Bits	Description	Decode
PKT_CNT_LBW	7:6	Select the sub-type of Low Bandwidth packets to count at PKT_CNT register	0b00: Count LBW data packets only 0b01: Count LBW acknowledge packets only 0b10: Count LBW data and acknowledge packets 0b11: Reserved
PKT_CNT_SEL	3:0	Select the type of received packets to count at PKT_CNT register	0x0: None 0x1: VIDEO 0x2: AUDIO 0x3: INFO Frame 0x4: SPI 0x5: I2C 0x6: UART 0x7: GPIO 0x8: AHDCP 0x9: RGMII 0xA: Reserved 0xB: Reserved 0xC: Reserved 0xD: Reserved 0xE: All 0xF: Unknown and packets with error

GPIOA (0x30)

Bit	7	6	5	4	3	2	1	0
Field	GPIO_RX_FAST_BIDIR_EN	GPIO_TX_CASC	GPIO_FWD_CDLY[5:0]					
Reset	0	1	000001					
Access Type	Write, Read	Write, Read	Write, Read					

Bitfield	Bits	Description	Decode
GPIO_RX_FAST_BIDIR_EN	7	GPIO fast direction switch for bi-directional IO	0b0: Fast direction switch disabled 0b1: Fast direction switch enabled

Bitfield	Bits	Description	Decode
GPIO_TX_CASC	6	Allow multiple pin transitions to be transmitted in the same packet	0b0: Multiple pin transitions not allowed 0b1: Multiple pin transitions allowed
GPIO_FWD_CDLY	5:0	Compensation delay multiplier for the forward direction. This must be same value as GPIO_FWD_CDLY of the chip on the other side of the link. Total delay is the (value+1) multiplied by 1.7us. Default delay is 3.4us.	0bXXXXXX: Forward compensation delay multiplier value

GPIOB (0x31)

Bit	7	6	5	4	3	2	1	0
Field	GPIO_TX_WNDW[1:0]		GPIO_REV_CDLY[5:0]					
Reset	10		001000					
Access Type	Write, Read		Write, Read					

Bitfield	Bits	Description	Decode
GPIO_TX_WNDW	7:6	Wait time after a GPIO transition to create a packet. This allows grouping transitions of different GPIO inputs in a single packet and so increases GPIO bandwdith usage efficiency.	0b00: Disabled 0b01: 200ns 0b10: 500ns 0b11: 1000ns
GPIO_REV_CDLY	5:0	Compensation delay multiplier for the reverse direction. This must be same value as GPIO_REV_CDLY of the chip on the other side of the link. Total delay is the (value+1) multiplied by 1.7us. Default delay is 15.3us.	0bXXXXXX: Reverse compensation delay multiplier value

I2C_0 (0x40)

Bit	7	6	5	4	3	2	1	0
Field	–	–	SLV_SH[1:0]		–	SLV_TO[2:0]		
Reset	–	–	10		–	110		
Access Type	–	–	Write, Read		–	Write, Read		

Bitfield	Bits	Description	Decode
SLV_SH	5:4	I2C-to-I2C master bit rate setting, configures the I2C bit rate used by the internal I2C master (in the device on remote side from the external I2C master) Set this according to the I2C speed mode.	0b00: Set for I2C Fast-mode Plus speed 0b01: Set for I2C Fast-mode speed 0b10: Set for I2C standard-mode speed 0b11: Reserved
SLV_TO	2:0	I2C-to-I2C slave timeout setting Internal GMSL2 I2C slave times out after the configured duration if it does not receive any response while waiting for a packet from remote device.	0b000: 16µs 0b001: 1ms 0b010: 2ms 0b011: 4ms

Bitfield	Bits	Description	Decode
			0b100: 8ms 0b101: 16ms 0b110: 32ms 0b111: Disabled

I2C_1 (0x41)

Bit	7	6	5	4	3	2	1	0
Field	–	MST_BT[2:0]			–	MST_TO[2:0]		
Reset	–	101			–	110		
Access Type	–	Write, Read			–	Write, Read		

Bitfield	Bits	Description	Decode
MST_BT	6:4	I2C-to-I2C master bit rate setting, configures the I2C bit rate used by the internal I2C master (in the device on remote side from the external I2C master) Set this according to the I2C speed mode.	0b000: 9.92Kbps - Set for I2C Standard mode speed 0b001: 33.2Kbps - Set for I2C Standard mode speed 0b010: 99.2Kbps - Set for I2C standard or Fast-mode speed 0b011: 123Kbps - Set for I2C Fast-mode speed 0b100: 203Kbps - Set for I2C Fast-mode speed 0b101: 397Kbps - Set for I2C Fast or Fast-mode Plus speed 0b110: 625Kbps - Set for I2C Fast-mode Plus speed 0b111: 980Kbps - Set for I2C Fast-mode Plus speed
MST_TO	2:0	I2C-to-I2C master timeout setting. Internal GMSL2 I2C master times out after the configured duration if it does not receive any response while waiting for a packet from remote device.	0b000: 16μs 0b001: 1ms 0b010: 2ms 0b011: 4ms 0b100: 8ms 0b101: 16ms 0b110: 32ms 0b111: Disabled

I2C_2 (0x42)

Bit	7	6	5	4	3	2	1	0
Field	SRC_A[6:0]							–
Reset	0000000							–
Access Type	Write, Read							–

Bitfield	Bits	Description	Decode
SRC_A	7:1	I2C address translator source A When I2C device address matches	0bXXXXXXX: Value of I2C SRC_A

Bitfield	Bits	Description	Decode
		I2C_SRC_A, internal I2C master (on remote side) replaces the device address by I2C_DST_A	

I2C_3 (0x43)

Bit	7	6	5	4	3	2	1	0
Field	DST_A[6:0]							–
Reset	0000000							–
Access Type	Write, Read							–

Bitfield	Bits	Description	Decode
DST_A	7:1	I2C address translator destination A See the description of I2C_SRC_A.	0bXXXXXXX: Value of I2C DST_A

I2C_4 (0x44)

Bit	7	6	5	4	3	2	1	0
Field	SRC_B[6:0]							–
Reset	0000000							–
Access Type	Write, Read							–

Bitfield	Bits	Description	Decode
SRC_B	7:1	I2C address translator source B When I2C device address matches I2C_SRC_B, internal I2C master (on remote side) replaces the device address by I2C_DST_B	0bXXXXXXX: Value of I2C SRC_B

I2C_5 (0x45)

Bit	7	6	5	4	3	2	1	0
Field	DST_B[6:0]							–
Reset	0000000							–
Access Type	Write, Read							–

Bitfield	Bits	Description	Decode
DST_B	7:1	I2C address translator destination B See the description of I2C_SRC_B.	0bXXXXXXX: Value of I2C DST_B

I2C_6 (0x46)

Bit	7	6	5	4	3	2	1	0
Field	–	–	–	–	I2C_AUTO_CFG	I2C_SRC_CNT[2:0]		
Reset	–	–	–	–	1	000		
Access Type	–	–	–	–	Write, Read	Write, Read		

Bitfield	Bits	Description	Decode
I2C_AUTO_CFG	3	When 1, I2C-to-I2C number of links is determined based on splitter mode automatically. In splitter mode, response from 2 I2C channels are expected, otherwise response from 1 I2C channel is expected.	0b0: Number of I2C-to-I2C links set by I2C_SRC_CNT[2:0] bits 0b1: Splitter mode automatically determines the number of I2C-to-I2C links
I2C_SRC_CNT	2:0	I2C-to-I2C number of links (valid when I2C_AUTO_SRC=0) Set this to N-1 when expecting I2C response from N remote I2C transmitters. (usually same as the number of remote devices connected to this device)	0b000: 1 deserializer connected 0b001: 2 deserializers connected 0b010: Reserved 0b011: Reserved 0b100: Reserved 0b101: Reserved 0b110: Reserved 0b111: Reserved

UART_0 (0x48)

Bit	7	6	5	4	3	2	1	0
Field	–	–	REM_MS_EN	LOC_MS_EN	BYPASS_DIS_PAR	BYPASS_TO[1:0]		BYPASS_EN
Reset	–	–	0	0	0	01		0
Access Type	–	–	Write, Read	Write, Read	Write, Read	Write, Read		Write, Read

Bitfield	Bits	Description	Decode
REM_MS_EN	5	Enable UART bypass mode control by remote GPIO pin. When set, remote chip's GPIO is used as MS pin (UART Mode Select). When MS is High, chip is in bypass mode, otherwise chip is in base mode.	0b0: UART bypass mode not controlled by remote MS pin 0b1: UART bypass mode controlled by remote MS pin
LOC_MS_EN	4	Enable UART bypass mode control by local GPIO pin. Set to use GPIO2 pin as MS pin (UART Mode Select). When MS is High, chip is in bypass mode, otherwise chip is in base mode.	0b0: UART bypass mode not controlled by local MS pin 0b1: UART bypass mode controlled by local MS pin
BYPASS_DIS_PAR	3	Do not receive and send parity bit in bypass mode	0b0: Receive and send parity bit in bypass mode 0b1: Do not receive and send parity bit in bypass mode
BYPASS_TO	2:1	UART soft bypass time-out duration. When set to 11, BYPASS_EN is never	0b00: 2ms

Bitfield	Bits	Description	Decode
		cleared, so the device stays in bypass mode until next power down.	0b01: 8ms 0b10: 32ms 0b11: Disabled
BYPASS_EN	0	Enable UART soft bypass mode. Bypass mode remains active as long as there is UART activity. When there is no UART activity for selected duration (configured by BYPASS_TO register), device exits bypass mode and the bit is automatically cleared.	0b0: UART soft bypass mode disabled 0b1: UART soft bypass mode enabled

I2C_PT_0 (0x4C)

Bit	7	6	5	4	3	2	1	0
Field	–	–	SLV_SH_PT[1:0]		–	SLV_TO_PT[2:0]		
Reset	–	–	10		–	110		
Access Type	–	–	Write, Read		–	Write, Read		

Bitfield	Bits	Description	Decode
SLV_SH_PT	5:4	Pass-thru I2C-to-I2C slave setup and hold time setting (setup, hold). Configures the interval between SDA and SCL transitions when driven by the internal I2C slave. Set this according to the I2C speed mode: Fast-mode Plus = 00 Fast-mode = 01 Standard-mode = 10	0b00: Set for I2C Fast-mode Plus speed 0b01: Set for I2C Fast-mode speed 0b10: Set for I2C standard-mode speed 0b11: Reserved
SLV_TO_PT	2:0	Pass-thru I2C-to-I2C slave timeout setting Internal GMSL2 I2C slave times out after the configured duration if it does not receive any response while waiting for a packet from remote device.	0b000: 16µs 0b001: 1ms 0b010: 2ms 0b011: 4ms 0b100: 8ms 0b101: 16ms 0b110: 32ms 0b111: Disabled

I2C_PT_1 (0x4D)

Bit	7	6	5	4	3	2	1	0
Field	–	MST_BT_PT[2:0]			–	MST_TO_PT[2:0]		
Reset	–	101			–	110		
Access Type	–	Write, Read			–	Write, Read		

Bitfield	Bits	Description	Decode
MST_BT_PT	6:4	Pass-thru I2C-to-I2C master bit rate setting, configures the I2C bit rate used by the internal I2C master (in the device on remote side from	0b000: 9.92Kbps - Set for I2C Standard mode speed

Bitfield	Bits	Description	Decode
		the external I2C master) Set this according to the I2C speed mode: Fast-mode Plus = 101 to 111 Fast-mode = 010 to 101 Standard-mode = 000 to 010	0b001: 33.2Kbps - Set for I2C Standard mode speed 0b010: 99.2Kbps - Set for I2C standard or Fast-mode speed 0b011: 123Kbps - Set for I2C Fast-mode speed 0b100: 203Kbps - Set for I2C Fast-mode speed 0b101: 397Kbps - Set for I2C Fast or Fast-mode Plus speed 0b110: 625Kbps - Set for I2C Fast-mode Plus speed 0b111: 980Kbps - Set for I2C Fast-mode Plus speed
MST_TO_PT	2:0	Pass-thru I2C-to-I2C master timeout setting Internal GMSL2 I2C master times out after the configured duration if it does not receive any response while waiting for a packet from the remote device.	0b000: 16μs 0b001: 1ms 0b010: 2ms 0b011: 4ms 0b100: 8ms 0b101: 16ms 0b110: 32ms 0b111: Disabled

UART_PT_0 (0x4F)

Bit	7	6	5	4	3	2	1	0
Field	BITLEN_MAN_CFG_2	DIS_PAR_2	–	–	BITLEN_MAN_CFG_1	DIS_PAR_1	–	–
Reset	0	0	–	–	0	0	–	–
Access Type	Write, Read	Write, Read	–	–	Write, Read	Write, Read	–	–

Bitfield	Bits	Description	Decode
BITLEN_MAN_CFG_2	7	Use the custom UART bit rate (selected by the BITLEN_PT_2_L and BITLEN_PT_2_H registers) in pass-thru UART channel #1	0b0: Use standard bit rate 0b1: Use custom bit rate
DIS_PAR_2	6	Disable parity bit in pass-thru UART channel #2	0b0: Parity bit enabled 0b1: Parity bit disabled
BITLEN_MAN_CFG_1	3	Use the custom UART bit rate (selected by the BITLEN_PT_1_L and BITLEN_PT_1_H registers) in pass-thru UART channel #1	0b0: Use standard bit rate 0b1: Use custom bit rate
DIS_PAR_1	2	Disable parity bit in pass-thru UART channel #1	0b0: Parity bit enabled 0b1: Parity bit disabled

TX0 (0x50)

Bit	7	6	5	4	3	2	1	0
Field	TX_CRC_EN	–	–	–	PRIO_VAL[1:0]		PRIO_CFG[1:0]	
Reset	0	–	–	–	00		00	
Access Type	Write, Read	–	–	–	Write, Read		Write, Read	

Bitfield	Bits	Description	Decode
TX_CRC_EN	7	Transmit CRC enable	0b0: Transmit CRC disabled 0b1: Transmit CRC enabled
PRIO_VAL	3:2	Sets the priority for this channel's packet requests	0b00: Low priority 0b01: Normal priority 0b10: High priority 0b11: Urgent priority
PRIO_CFG	1:0	Adjust the priority used for this channel	0b00: Current channel priority is used 0b01: Increase current channel priority 0b10: Decrease current channel priority 0b11: Reset to channel PRIO_VAL[1:0] setting

TX1 (0x51)

Bit	7	6	5	4	3	2	1	0
Field	BW_MULT[1:0]		BW_VAL[5:0]					
Reset	10		110000					
Access Type	Write, Read		Write, Read					

Bitfield	Bits	Description	Decode
BW_MULT	7:6	Channel bandwidth allocation multiplication factor	0b00: Multiply BW_VAL by 1 0b01: Multiply BW_VAL by 4 0b10: Multiply BW_VAL by 16 0b11: Multiply BW_VAL by 16
BW_VAL	5:0	Channel bandwidth allocation base. Fair bandwidth use ratio = BW_VAL*BW_MULT/10 as a percentage of total link bandwidth	0bXXXXXX: Channel base BW value

TX3 (0x53)

Bit	7	6	5	4	3	2	1	0
Field	–	–	TX_SPLT_MASK_B	TX_SPLT_MASK_A	–	–	TX_STR_SEL[1:0]	
Reset	–	–	0	1	–	–	00	
Access Type	–	–	Write, Read	Write, Read	–	–	Write, Read	

Bitfield	Bits	Description	Decode
TX_SPLT_MASK_B	5	In splitter mode: When 0, packets from this port are NOT transmitted from port B, otherwise packets are transmitted from port B.	0b0: Packets are not transmitted over GMSLB in splitter mode 0b1: Packets are transmitted over GMSLB in splitter mode
TX_SPLT_MASK_A	4	In splitter mode: When 0, packets from this port are NOT transmitted from port A, otherwise packets are transmitted from port A.	0b0: Packets are not transmitted over GMSLA in splitter mode 0b1: Packets are transmitted over GMSLA in splitter mode
TX_STR_SEL	1:0	Stream ID used in packets transmitted from this channel	0bXX: Stream ID for packets from this channel

TX0 (0x54)

Bit	7	6	5	4	3	2	1	0
Field	TX_CRC_EN	–	–	–	PRIO_VAL[1:0]		PRIO_CFG[1:0]	
Reset	0	–	–	–	00		00	
Access Type	Write, Read	–	–	–	Write, Read		Write, Read	

Bitfield	Bits	Description	Decode
TX_CRC_EN	7	Transmit CRC enable	0b0: Transmit CRC disabled 0b1: Transmit CRC enabled
PRIO_VAL	3:2	Sets the priority for this channel's packet requests	0b00: Low priority 0b01: Normal priority 0b10: High priority 0b11: Urgent priority
PRIO_CFG	1:0	Adjust the priority used for this channel	0b00: Current channel priority is used 0b01: Increase current channel priority 0b10: Decrease current channel priority 0b11: Reset to channel PRIO_VAL[1:0] setting

TX1 (0x55)

Bit	7	6	5	4	3	2	1	0
Field	BW_MULT[1:0]		BW_VAL[5:0]					
Reset	10		110000					
Access Type	Write, Read		Write, Read					

Bitfield	Bits	Description	Decode
BW_MULT	7:6	Channel bandwidth allocation multiplication factor	0b00: Multiply BW_VAL by 1 0b01: Multiply BW_VAL by 4 0b10: Multiply BW_VAL by 16 0b11: Multiply BW_VAL by 16

Bitfield	Bits	Description	Decode
BW_VAL	5:0	Channel bandwidth allocation base. Fair bandwidth use ratio = BW_VAL*BW_MULT/10 as a percentage of total link bandwidth	0bXXXXXX: Channel base BW value

TX3 (0x57)

Bit	7	6	5	4	3	2	1	0
Field	–	–	TX_SPLT_MASK_B	TX_SPLT_MASK_A	–	–	TX_STR_SEL[1:0]	
Reset	–	–	0	1	–	–	01	
Access Type	–	–	Write, Read	Write, Read	–	–	Write, Read	

Bitfield	Bits	Description	Decode
TX_SPLT_MASK_B	5	In splitter mode: When 0, packets from this port are NOT transmitted from port B, otherwise packets are transmitted from port B.	0b0: Packets are not transmitted over GMSLB in splitter mode 0b1: Packets are transmitted over GMSLB in splitter mode
TX_SPLT_MASK_A	4	In splitter mode: When 0, packets from this port are NOT transmitted from port A, otherwise packets are transmitted from port A.	0b0: Packets are not transmitted over GMSLA in splitter mode 0b1: Packets are transmitted over GMSLA in splitter mode
TX_STR_SEL	1:0	Stream ID used in packets transmitted from this channel	0bXX: Stream ID for packets from this channel

TX0 (0x58)

Bit	7	6	5	4	3	2	1	0
Field	TX_CRC_EN	–	–	–	PRIO_VAL[1:0]		PRIO_CFG[1:0]	
Reset	0	–	–	–	00		00	
Access Type	Write, Read	–	–	–	Write, Read		Write, Read	

Bitfield	Bits	Description	Decode
TX_CRC_EN	7	Transmit CRC enable	0b0: Transmit CRC disabled 0b1: Transmit CRC enabled
PRIO_VAL	3:2	Sets the priority for this channel's packet requests	0b00: Low priority 0b01: Normal priority 0b10: High priority 0b11: Urgent priority
PRIO_CFG	1:0	Adjust the priority used for this channel	0b00: Current channel priority is used 0b01: Increase current channel priority 0b10: Decrease current channel priority 0b11: Reset to channel PRIO_VAL[1:0] setting

TX1 (0x59)

Bit	7	6	5	4	3	2	1	0
Field	BW_MULT[1:0]		BW_VAL[5:0]					
Reset	10		110000					
Access Type	Write, Read		Write, Read					

Bitfield	Bits	Description	Decode
BW_MULT	7:6	Channel bandwidth allocation multiplication factor	0b00: Multiply BW_VAL by 1 0b01: Multiply BW_VAL by 4 0b10: Multiply BW_VAL by 16 0b11: Multiply BW_VAL by 16
BW_VAL	5:0	Channel bandwidth allocation base. Fair bandwidth use ratio = BW_VAL*BW_MULT/10 as a percentage of total link bandwidth	0bXXXXXX: Channel base BW value

TX3 (0x5B)

Bit	7	6	5	4	3	2	1	0
Field	–	–	TX_SPLT_MASK_B	TX_SPLT_MASK_A	–	–	TX_STR_SEL[1:0]	
Reset	–	–	0	1	–	–	10	
Access Type	–	–	Write, Read	Write, Read	–	–	Write, Read	

Bitfield	Bits	Description	Decode
TX_SPLT_MASK_B	5	In splitter mode: When 0, packets from this port are NOT transmitted from port B, otherwise packets are transmitted from port B.	0b0: Packets are not transmitted over GMSLB in splitter mode 0b1: Packets are transmitted over GMSLB in splitter mode
TX_SPLT_MASK_A	4	In splitter mode: When 0, packets from this port are NOT transmitted from port A, otherwise packets are transmitted from port A.	0b0: Packets are not transmitted over GMSLA in splitter mode 0b1: Packets are transmitted over GMSLA in splitter mode
TX_STR_SEL	1:0	Stream ID used in packets transmitted from this channel	0bXX: Stream ID for packets from this channel

TX0 (0x5C)

Bit	7	6	5	4	3	2	1	0
Field	TX_CRC_EN	–	–	–	PRIO_VAL[1:0]		PRIO_CFG[1:0]	
Reset	0	–	–	–	00		00	
Access Type	Write, Read	–	–	–	Write, Read		Write, Read	

Bitfield	Bits	Description	Decode
TX_CRC_EN	7	Transmit CRC enable	0b0: Transmit CRC disabled 0b1: Transmit CRC enabled
PRIO_VAL	3:2	Sets the priority for this channel's packet requests	0b00: Low priority 0b01: Normal priority 0b10: High priority 0b11: Urgent priority
PRIO_CFG	1:0	Adjust the priority used for this channel	0b00: Current channel priority is used 0b01: Increase current channel priority 0b10: Decrease current channel priority 0b11: Reset to channel PRIO_VAL[1:0] setting

TX1 (0x5D)

Bit	7	6	5	4	3	2	1	0
Field	BW_MULT[1:0]		BW_VAL[5:0]					
Reset	10		110000					
Access Type	Write, Read		Write, Read					

Bitfield	Bits	Description	Decode
BW_MULT	7:6	Channel bandwidth allocation multiplication factor	0b00: Multiply BW_VAL by 1 0b01: Multiply BW_VAL by 4 0b10: Multiply BW_VAL by 16 0b11: Multiply BW_VAL by 16
BW_VAL	5:0	Channel bandwidth allocation base. Fair bandwidth use ratio = BW_VAL*BW_MULT/10 as a percentage of total link bandwidth	0bXXXXXX: Channel base BW value

TX3 (0x5F)

Bit	7	6	5	4	3	2	1	0
Field	–	–	TX_SPLT_MASK_B	TX_SPLT_MASK_A	–	–	TX_STR_SEL[1:0]	
Reset	–	–	0	1	–	–	11	
Access Type	–	–	Write, Read	Write, Read	–	–	Write, Read	

Bitfield	Bits	Description	Decode
TX_SPLT_MASK_B	5	In splitter mode: When 0, packets from this port are NOT transmitted from port B, otherwise packets are transmitted from port B.	0b0: Packets are not transmitted over GMSLB in splitter mode 0b1: Packets are transmitted over GMSLB in splitter mode
TX_SPLT_MASK_A	4	In splitter mode: When 0, packets from this port are NOT transmitted from port A, otherwise packets are transmitted from port A.	0b0: Packets are not transmitted over GMSLA in splitter mode 0b1: Packets are transmitted over GMSLA in splitter

Bitfield	Bits	Description	Decode
			mode
TX_STR_SEL	1:0	Stream ID used in packets transmitted from this channel	0bXX: Stream ID for packets from this channel

TR0 (0x78)

Bit	7	6	5	4	3	2	1	0
Field	TX_CRC_EN	RX_CRC_EN	–	–	PRIO_VAL[1:0]		PRIO_CFG[1:0]	
Reset	1	1	–	–	00		00	
Access Type	Write, Read	Write, Read	–	–	Write, Read		Write, Read	

Bitfield	Bits	Description	Decode
TX_CRC_EN	7	When set, calculate and append CRC to each packet transmitted from this port	0b0: Transmit CRC disabled 0b1: Transmit CRC enabled
RX_CRC_EN	6	When set, indicates that packets received at this port have appended CRC and CRC checking should be performed at each packet.	0b0: Receive CRC disabled 0b1: Receive CRC enabled
PRIO_VAL	3:2	Sets the priority for this channel's packet requests	0b00: Low priority 0b01: Normal priority 0b10: High priority 0b11: Urgent priority
PRIO_CFG	1:0	Adjust the priority used for this channel	0b00: Current channel priority is used 0b01: Increase current channel priority 0b10: Decrease current channel priority 0b11: Reset to channel PRIO_VAL[1:0] setting

TR1 (0x79)

Bit	7	6	5	4	3	2	1	0
Field	BW_MULT[1:0]		BW_VAL[5:0]					
Reset	10		110000					
Access Type	Write, Read		Write, Read					

Bitfield	Bits	Description	Decode
BW_MULT	7:6	Channel bandwidth allocation multiplication factor	0b00: Multiply BW_VAL by 1 0b01: Multiply BW_VAL by 4 0b10: Multiply BW_VAL by 16 0b11: Multiply BW_VAL by 16
BW_VAL	5:0	Channel bandwidth allocation base. Fair bandwidth use ratio = BW_VAL*BW_MULT/10 as a percentage of total link bandwidth.	0bXXXXXX: Channel base BW value

TR3 (0x7B)

Bit	7	6	5	4	3	2	1	0
Field	–	–	TX_SPLT_MASK_B	TX_SPLT_MASK_A	–	TX_SRC_ID[2:0]		
Reset	–	–	1	1	–	000		
Access Type	–	–	Write, Read	Write, Read	–	Write, Read		

Bitfield	Bits	Description	Decode
TX_SPLT_MASK_B	5	In splitter mode: When 0, packets from this port are NOT transmitted from port B, otherwise packets are transmitted from port B.	0b0: Packets are not transmitted over GMSLB in splitter mode 0b1: Packets are transmitted over GMSLB in splitter mode
TX_SPLT_MASK_A	4	In splitter mode: When 0, packets from this port are NOT transmitted from port A, otherwise packets are transmitted from port A.	0b0: Packets are not transmitted over GMSLA in splitter mode 0b1: Packets are transmitted over GMSLA in splitter mode
TX_SRC_ID	2:0	Source identifier used in packets transmitted from this channel. Default value of the 2 MSBs are set by ADD2, ADD1. Default value of the LSB is 0.	0bXXX: Source ID for packets from this channel

TR4 (0x7C)

Bit	7	6	5	4	3	2	1	0
Field	RX_SRC_SEL[7:0]							
Reset	0xFF							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
RX_SRC_SEL	7:0	Receive packets from selected sources Each bit indicates whether packets with that source id should be received or not. For example, when SRC_SEL=00001001, then packets with source id equal to 0 and 3 will be received.	0x00: No packets received 0x01: Packets from source ID 1 received 0x02: Packets from source ID 2 received 0x03: Packets from source ID 1 and 2 receviced . . 0xFF: Packets from all source IDs received

TR0 (0x80, 0x90, 0xA0, 0xA8)

Bit	7	6	5	4	3	2	1	0
Field	TX_CRC_EN	RX_CRC_EN	–	–	PRIO_VAL[1:0]		PRIO_CFG[1:0]	
Reset	1	1	–	–	00		00	
Access Type	Write, Read	Write, Read	–	–	Write, Read		Write, Read	

Bitfield	Bits	Description	Decode
TX_CRC_EN	7	When set, calculate and append CRC to each packet transmitted from this port	0b0: Transmit CRC disabled 0b1: Transmit CRC enabled
RX_CRC_EN	6	When set, indicates that packets received at this port have appended CRC and CRC checking should be performed at each packet.	0b0: Receive CRC disabled 0b1: Receive CRC enabled
PRIO_VAL	3:2	Sets the priority for this channel's packet requests	0b00: Low priority 0b01: Normal priority 0b10: High priority 0b11: Urgent priority
PRIO_CFG	1:0	Adjust the priority used for this channel	0b00: Current channel priority is used 0b01: Increase current channel priority 0b10: Decrease current channel priority 0b11: Reset to channel PRIO_VAL[1:0] setting

TR1 (0x81, 0x91, 0xA1, 0xA9)

Bit	7	6	5	4	3	2	1	0
Field	BW_MULT[1:0]		BW_VAL[5:0]					
Reset	10		110000					
Access Type	Write, Read		Write, Read					

Bitfield	Bits	Description	Decode
BW_MULT	7:6	Channel bandwidth allocation multiplication factor	0b00: Multiply BW_VAL by 1 0b01: Multiply BW_VAL by 4 0b10: Multiply BW_VAL by 16 0b11: Multiply BW_VAL by 16
BW_VAL	5:0	Channel bandwidth allocation base. Fair bandwidth use ratio = BW_VAL*BW_MULT/10 as a percentage of total link bandwidth.	0bXXXXXX: Channel base BW value

TR3 (0x83, 0x93, 0xA3, 0xAB)

Bit	7	6	5	4	3	2	1	0
Field	–	–	TX_SPLT_MASK_B	TX_SPLT_MASK_A	–	TX_SRC_ID[2:0]		
Reset	–	–	1	1	–	000		
Access Type	–	–	Write, Read	Write, Read	–	Write, Read		

Bitfield	Bits	Description	Decode
TX_SPLT_MASK_B	5	In splitter mode: When 0, packets from this port are NOT transmitted from port B, otherwise packets are transmitted from port B.	0b0: Packets are not transmitted over GMSLB in splitter mode 0b1: Packets are transmitted over GMSLB in splitter mode

Bitfield	Bits	Description	Decode
TX_SPLT_MASK_A	4	In splitter mode: When 0, packets from this port are NOT transmitted from port A, otherwise packets are transmitted from port A.	0b0: Packets are not transmitted over GMSLA in splitter mode 0b1: Packets are transmitted over GMSLA in splitter mode
TX_SRC_ID	2:0	Source identifier used in packets transmitted from this channel. Default value of the 2 MSBs are set by ADD2, ADD1. Default value of the LSB is 0.	0bXXX: Source ID for packets from this channel

TR4 (0x84, 0x94, 0xA4, 0xAC)

Bit	7	6	5	4	3	2	1	0
Field	RX_SRC_SEL[7:0]							
Reset	0xFF							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
RX_SRC_SEL	7:0	Receive packets from selected sources Each bit indicates whether packets with that source id should be received or not. For example, when SRC_SEL=00001001, then packets with source id equal to 0 and 3 will be received.	0x00: No packets received 0x01: Packets from source ID 1 received 0x02: Packets from source ID 2 received 0x03: Packets from source ID 1 and 2 receviced . . 0xFF: Packets from all source IDs received

ARQ0 (0x85, 0x95, 0xA5, 0xAD)

Bit	7	6	5	4	3	2	1	0
Field	–	–	–	–	ARQ0_EN	–	–	–
Reset	–	–	–	–	1	–	–	–
Access Type	–	–	–	–	Write, Read	–	–	–

Bitfield	Bits	Description	Decode
ARQ0_EN	3	Enable ARQ	0b0: ARQ disabled 0b1: ARQ enabled

ARQ1 (0x86, 0x96, 0xA6, 0xAE)

Bit	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	–	MAX_RT_ERR_OEN	RT_CNT_OEN
Reset	–	–	–	–	–	–	1	0
Access Type	–	–	–	–	–	–	Write, Read	Write, Read

Bitfield	Bits	Description	Decode
MAX_RT_ERR_OEN	1	Enable reporting of ARQ max re-transmission limit errors (MAX_RT_ERR) for this channel at ERRB pin.	0b0: ARQ max re-transmit limit errors reporting at ERRB pin disabled 0b1: ARQ max re-transmit limit errors reporting at ERRB pin enabled
RT_CNT_OEN	0	Enable reporting of ARQ re-transmission event for this channel at ERRB pin. When enabled, ERRB is asserted when RT_CNT of this channel is greater than 0.	0b0: ARQ re-transmission count reporting at ERRB pin disabled 0b1: ARQ re-transmission count reporting at ERRB pin enabled

ARQ2 (0x87, 0x97, 0xA7, 0xAF)

Bit	7	6	5	4	3	2	1	0
Field	MAX_RT_ERR	RT_CNT[6:0]						
Reset	0	0000000						
Access Type	Read Clears All	Read Clears All						

Bitfield	Bits	Description	Decode
MAX_RT_ERR	7	Reached maximum re-tx limit (MAX_RT) for one packet in this channel	0b0: Max re-transmission limit not reached 0b1: Max re-transmission limit reached
RT_CNT	6:0	Total re-transmission count in this channel	0xXX: Count of re-transmissions for this channel

TR0 (0x88)

Bit	7	6	5	4	3	2	1	0
Field	–	–	BURST_MULT[1:0]		PRIO_VAL[1:0]		PRIO_CFG[1:0]	
Reset	–	–	11		00		00	
Access Type	–	–	Write, Read		Write, Read		Write, Read	

Bitfield	Bits	Description	Decode
BURST_MULT	5:4		
PRIO_VAL	3:2	Sets the priority for this channel's packet requests	0b00: Low priority 0b01: Normal priority 0b10: High priority 0b11: Urgent priority
PRIO_CFG	1:0	Adjust the priority used for this channel	0b00: Current channel priority is used 0b01: Increase current channel priority 0b10: Decrease current channel priority 0b11: Reset to channel PRIO_VAL[1:0] setting

TR1 (0x89)

Bit	7	6	5	4	3	2	1	0
Field	BW_MULT[1:0]		BW_VAL[5:0]					
Reset	10		110000					
Access Type	Write, Read		Write, Read					

Bitfield	Bits	Description	Decode
BW_MULT	7:6	Channel bandwidth allocation multiplication factor	0b00: Multiply BW_VAL by 1 0b01: Multiply BW_VAL by 4 0b10: Multiply BW_VAL by 16 0b11: Multiply BW_VAL by 16
BW_VAL	5:0	Channel bandwidth allocation base. Fair bandwidth use ratio = BW_VAL*BW_MULT/10 as a percentage of total link bandwidth.	0bXXXXXX: Channel base BW value

TR3 (0x8B)

Bit	7	6	5	4	3	2	1	0
Field	–	–	TX_SPLT_MASK_B	TX_SPLT_MASK_A	–	TX_SRC_ID[2:0]		
Reset	–	–	1	1	–	000		
Access Type	–	–	Write, Read	Write, Read	–	Write, Read		

Bitfield	Bits	Description	Decode
TX_SPLT_MASK_B	5	In splitter mode: When 0, packets from this port are NOT transmitted from port B, otherwise packets are transmitted from port B.	0b0: Packets are not transmitted over GMSLB in splitter mode 0b1: Packets are transmitted over GMSLB in splitter mode
TX_SPLT_MASK_A	4	In splitter mode: When 0, packets from this port are NOT transmitted from port A, otherwise packets are transmitted from port A.	0b0: Packets are not transmitted over GMSLA in splitter mode 0b1: Packets are transmitted over GMSLA in splitter mode
TX_SRC_ID	2:0	Source identifier used in packets transmitted from this channel. Default value of the 2 MSBs are set by ADD2, ADD1. Default value of the LSB is 0.	0bXXX: Source ID for packets from this channel

TR4 (0x8C)

Bit	7	6	5	4	3	2	1	0
Field	RX_SRC_SEL[7:0]							
Reset	0xFF							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
RX_SRC_SEL	7:0	Receive packets from selected sources Each bit indicates whether packets with that source id should be received or not. For example, when SRC_SEL=00001001, then packets with source id equal to 0 and 3 will be received.	0x00: No packets received 0x01: Packets from source ID 1 received 0x02: Packets from source ID 2 received 0x03: Packets from source ID 1 and 2 receviced . . 0xFF: Packets from all source IDs received

ARQ0 (0x8D)

Bit	7	6	5	4	3	2	1	0
Field	–	–	–	–	ARQ0_EN	–	–	–
Reset	–	–	–	–	1	–	–	–
Access Type	–	–	–	–	Write, Read	–	–	–

Bitfield	Bits	Description	Decode
ARQ0_EN	3	Enable ARQ	0b0: ARQ disabled 0b1: ARQ enabled

ARQ1 (0x8E)

Bit	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	–	MAX_RT_ERR_OEN	RT_CNT_OEN
Reset	–	–	–	–	–	–	1	0
Access Type	–	–	–	–	–	–	Write, Read	Write, Read

Bitfield	Bits	Description	Decode
MAX_RT_ERR_OEN	1	Enable reporting of ARQ max re-transmission limit errors (MAX_RT_ERR) for this channel at ERRB pin.	0b0: ARQ max re-transmit limit errors reporting at ERRB pin disabled 0b1: ARQ max re-transmit limit errors reporting at ERRB pin enabled
RT_CNT_OEN	0	Enable reporting of ARQ re-transmission event for this channel at ERRB pin. When enabled, ERRB is asserted when RT_CNT of this channel is greater than 0.	0b0: ARQ re-transmission count reporting at ERRB pin disabled 0b1: ARQ re-transmission count reporting at ERRB pin enabled

ARQ2 (0x8F)

Bit	7	6	5	4	3	2	1	0
Field	MAX_RT_ERR	RT_CNT[6:0]						
Reset	0	0000000						
Access Type	Read Clears All	Read Clears All						

Bitfield	Bits	Description	Decode
MAX_RT_ERR	7	Reached maximum re-tx limit (MAX_RT) for one packet in this channel	0b0: Max re-transmission limit not reached 0b1: Max re-transmission limit reached
RT_CNT	6:0	Total re-transmission count in this channel	0xXX: Count of re-transmissions for this channel

VIDEO_TX0 (0x100, 0x108, 0x110, 0x118)

Bit	7	6	5	4	3	2	1	0
Field	LINE_CRC_SEL	LINE_CRC_EN	ENC_MODE[1:0]		AUTO_BPP	–	–	–
Reset	0	1	10		1	–	–	–
Access Type	Write, Read	Write, Read	Write, Read		Write, Read	–	–	–

Bitfield	Bits	Description	Decode
LINE_CRC_SEL	7	Line CRC checksum generation with DE or HS	0b0: Use DE for Line CRC 0b1: Use HS for Line CRC
LINE_CRC_EN	6	Line CRC Enable, generate a CRC code for the video line and send it to the receiver side for comparison	0b0: Line CRC disabled 0b1: Line CRC enabled
ENC_MODE	5:4	HS,VS,DE Encoding Mode	0b00: HS, VS, DE encoding off 0b01: HS, VS, DE encoding on, color bits always sent 0b10: HS, VS, DE encoding on, color bits sent only when DE is high 0b11: HS, VS, DE encoding on, color bits sent only when HS is high
AUTO_BPP	3	Select BPP source	0b0: Use BPP from BPP register 0b1: Use BPP from MIPI receiver

VIDEO_TX1 (0x101, 0x109, 0x111, 0x119)

Bit	7	6	5	4	3	2	1	0
Field	–	–	BPP[5:0]					
Reset	–	–	011000					
Access Type	–	–	Write, Read					

Bitfield	Bits	Description	Decode
BPP	5:0	Color bits per pixel (RGB888 = 24)	0bXXXXXX: Number of bits per pixel

VIDEO_TX2 (0x102, 0x10A, 0x112, 0x11A)

Bit	7	6	5	4	3	2	1	0
Field	PCLKDET	DRIFT_ERR	OVERFLOW	FIFO_WARN	–	LIM_HEART	–	–
Reset	0	0	0	0	–	0	–	–
Access Type	Read Only	Read Clears All	Read Clears All	Read Clears All	–	Write, Read	–	–

Bitfield	Bits	Description	Decode
PCLKDET	7	PCLK detected	0b0: Video transmit PCLK not detected 0b1: Video transmit PCLK detected
DRIFT_ERR	6	VID_TX PCLK drift error detected. After the video pipeline starts, PCLK cannot drift more than a certain amount without restarting the sub system.	0b0: Video transmit PCLK drift error not detected 0b1: Video transmit PCLK drift error detected
OVERFLOW	5	VID_TX FIFO has overflowed, video input throughput may be too high, BW allocation on GMSL2 link for video may not be enough.	0b0: Video transmit FIFO has not overflowed 0b1: Video transmit FIFO has overflowed
FIFO_WARN	4	VID_TX FIFO is more than half full, video data coming from HDMI receiver is read by the scheduler	0b0: Video transmit FIFO is less than or equal to half full 0b1: Video transmit FIFO is more than half full
LIM_HEART	2	Disable heartbeat during blanking Use together with SEQ_MISS_EN and DIS_PKT_DET registers in Deserializer.	0b0: Heartbeat enabled during blanking 0b1: Hearbeat disabled during blanking

SPI_0 (0x170)

Bit	7	6	5	4	3	2	1	0
Field	SPI_LOC_ID[1:0]		SPI_CC_TRG_ID[1:0]		SPI_IGNR_ID	SPI_CC_EN	MST_SLVN	SPI_EN
Reset	00		00		1	0	0	0
Access Type	Write, Read		Write, Read		Write, Read	Write, Read	Write, Read	Write, Read

Bitfield	Bits	Description	Decode
SPI_LOC_ID	7:6	Program to local ID if filtering packets based on header ID.	0b00: ID = 0 0b01: ID = 1 0b10: ID = 2 0b11: ID = 3
SPI_CC_TRG_ID	5:4	ID for GMSL2 Header in SPI control channel Bridge Mode	0b00: ID = 0 0b01: ID = 1 0b10: ID = 2 0b11: ID = 3

Bitfield	Bits	Description	Decode
SPI_IGNR_ID	3	Selects if SPI should use or ignore header ID to decide on packet acceptance	0b0: Accept only packets with proper ID 0b1: Ignore ID and accept all packets
SPI_CC_EN	2	Enable control channel SPI bridge function	0b0: SPI bridge disabled 0b1: SPI bridge enabled
MST_SLVN	1	Selects if SPI is master or slave	0b0: SPI slave 0b1: SPI master
SPI_EN	0	Enable SPI channel	0b0: SPI channel disabled 0b1: SPI channel enabled

SPI_1 (0x171)

Bit	7	6	5	4	3	2	1	0
Field	SPI_LOC_N[5:0]						–	–
Reset	000111						–	–
Access Type	Write, Read						–	–

Bitfield	Bits	Description	Decode
SPI_LOC_N	7:2	Sets the packet size ((2N+1)bytes) for GMSL2 SPI packets. If this is programmed to a value more than 7, ARQ of the SPI channel must be disabled.	0b000000: Packet size is 1 byte 0b000001: Packet size is 3 bytes . . 0b111111: Packet size is 127 bytes

SPI_2 (0x172)

Bit	7	6	5	4	3	2	1	0
Field	REQ_HOLD_OFF[2:0]			FULL_SCK_SETUP	SPI_MOD3_F	SPI_MOD3	SPIM_SS2_ACT_H	SPIM_SS1_ACT_H
Reset	000			0	0	0	1	1
Access Type	Write, Read			Write, Read	Write, Read	Write, Read	Write, Read	Write, Read

Bitfield	Bits	Description	Decode
REQ_HOLD_OFF	7:5	Hold off GMSL2 Request until this number of extra bytes are received on SPI port.	0b00: No extra bytes 0b01: 1 extra byte 0b10: 2 extra bytes 0b11: 3 extra bytes
FULL_SCK_SETUP	4	Sample MISO after half or full SCK period.	0b0: MISO sampled after half SCK period 0b1: MISO sampled after full SCK period

Bitfield	Bits	Description	Decode
SPI_MOD3_F	3	Allows the supression of an extra SCK prior to SS deassertion when SPI mode 3 is selected.	0b0: Extra SCK present prior to SS deassertion when in SPI mode 3 0b1: Extra SCK suppressed prior to SS deassertion when in SPI mode 3
SPI_MOD3	2	Selects SPI mode 0 or 3	0b0: SPI mode 0 0b1: SPI mode 3
SPIM_SS2_ACT_H	1	Sets the polarity for SS2 when the SPI is a master.	0b0: Slave select 2 is active low 0b1: Slave select 2 is active high
SPIM_SS1_ACT_H	0	Sets the polarity for SS1 when the SPI is a master.	0b0: Slave select 1 is active low 0b1: Slave select 1 is active high

SPI_3 (0x173)

Bit	7	6	5	4	3	2	1	0
Field	SPIM_SS_DLY_CLKS[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
SPIM_SS_DLY_CLKS	7:0	Number of 300MHz clocks to delay between: 1) Assertion of SS and Start of SCK pulses 2) End of SCK pulses and De-assertion of SS 3) De-assertion of SS and Re-assertion of SS (if necessary)	0xXX: Number of clock delays

SPI_4 (0x174)

Bit	7	6	5	4	3	2	1	0
Field	SPIM_SCK_LO_CLKS[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
SPIM_SCK_LO_CLKS	7:0	Number of 300MHz clocks for SCK Low time	0xXX: Number of clocks for SCK low time

SPI_5 (0x175)

Bit	7	6	5	4	3	2	1	0
Field	SPIM_SCK_HI_CLKS[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
SPIM_SCK_HI_CLKS	7:0	Number of 300MHz clocks for SCK High time	0xXX: Number of clocks for SCK high time

SPI_6 (0x176)

Bit	7	6	5	4	3	2	1	0
Field	–	–	BNE	SPIS_RWN	SS_IO_EN_2	SS_IO_EN_1	BNE_IO_EN	RWN_IO_EN
Reset	–	–	0	0	0	0	0	0
Access Type	–	–	Read Only	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read

Bitfield	Bits	Description	Decode
BNE	5	Alternate GPU Status register to use for BNE status if GPIO is not avaiable	0b0: Buffer empty 0b1: Buffer not empty
SPIS_RWN	4	Alternate GPU Control register to use for Rd/WrN control if GPIO is not available.	0b0: Write 0b1: Read
SS_IO_EN_2	3	Enable GPIO for use as Slave Select 2 output.	0b0: GPIO not used for SPI SS2 function 0b1: GPIO used for SPI SS2 function
SS_IO_EN_1	2	Enable GPIO for use as Slave Select 1 output.	0b0: GPIO not used for SPI SS1 function 0b1: GPIO used for SPI SS1 function
BNE_IO_EN	1	Enable GPIO for use as BNE output for SPI data available status.	0b0: GPIO not used for SPI BNE function 0b1: GPIO used for SPI BNE function
RWN_IO_EN	0	Enable GPIO for use as RO input for control of SPI data movement.	0b0: GPIO not used for SPI RO function 0b1: GPIO used for SPI RO function

SPI_7 (0x177)

Bit	7	6	5	4	3	2	1	0
Field	SPI_RX_OVRFLW	SPI_TX_OVRFLW	–	SPIS_BYTE_CNT[4:0]				
Reset	0	0	–	00000				
Access Type	Read Clears All	Read Clears All	–	Read Only				

Bitfield	Bits	Description	Decode
SPI_RX_OVRFLW	7	SPI Rx Buffer overflow flag	0b0: No SPI Rx buffer overflow 0b1: SPI Rx buffer overflow
SPI_TX_OVRFLW	6	SPI Tx Buffer overflow flag	0b0: No SPI Tx buffer overflow 0b1: SPI Tx buffer overflow
SPIS_BYTE_CNT	4:0	Number of SPI data bytes available for reading from Rx Buffer.	0bXXXXX: Number of bytes available

SPI_8 (0x178)

Bit	7	6	5	4	3	2	1	0
Field	REQ_HOLD_OFF_TO[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
REQ_HOLD_OFF_TO	7:0	Timeout Delay (in 100nS increments) for GMSL2 Req Hold off (0 is disable).	0xXX: Number of 100nS delay increments for GMSL2 request hold off

WM_0 (0x190)

Bit	7	6	5	4	3	2	1	0
Field	WM_LEN	WM_MODE[2:0]			WM_DET[1:0]		–	WM_EN
Reset	0	000			00		–	0
Access Type	Write, Read	Write, Read			Write, Read		–	Write, Read

Bitfield	Bits	Description	Decode
WM_LEN	7	Watermark length	0b0: 32-bit 0b1: 63-bit
WM_MODE	6:4	Watermark mode 000: Insert WM0/WM1 every frame Temporal Dither enabled (default) 001: Insert WM0 every frame Temporal Dither enabled < Error Injection Test Mode > 010: Insert WM0/WM1 every frame Temp Dither enabled polarity disabled < Error Injection Test Mode > 011: Insert WM0/WM1 every frame, do not reset WM, Temporal Dither enabled	0b000: Insert WM-/WM+ every frame, temporal dither enabled 0b001: Reserved 0b010: Insert WM0/WM1 every frame, temporal dither enabled, polarity disabled (Error generation mode) 0b011: Insert WM0/WM1 every frame, do not reset WM, temporal dither enabled 0b100: Insert WM0/WM1 every frame, temporal dither disabled

Bitfield	Bits	Description	Decode
		100: Insert WM0/WM1 every frame Temporal Dither disabled 101: Insert WM0 every frame Temporal Dither disabled 110: reserved 111: reserved	0b101: Reserved 0b110: Reserved 0b111: Reserved
WM_DET	3:2	Watermark Detection/ Generation	0b00: Insert watermark in video stream 0b01: Detect watermark and remove from outgoing video stream 0b10: Reserved 0b11: Reserved
WM_EN	0	Watermark Enable	0b0: Watermarking disabled 0b1: Watermarking enabled

WM_2 (0x192)

Bit	7	6	5	4	3	2	1	0
Field	–	–	–	–	HsyncPol	VsyncPol	–	–
Reset	–	–	–	–	0	0	–	–
Access Type	–	–	–	–	Write, Read	Write, Read	–	–

Bitfield	Bits	Description	Decode
HsyncPol	3	HS Polarity (Only effective for Watermark Block)	0b0: Non-inverting 0x1: Invert
VsyncPol	2	VS Polarity (Only effective for Watermark Block)	0b0: Non-inverting 0x1: Invert

WM_4 (0x194)

Bit	7	6	5	4	3	2	1	0
Field	–	–	WM_YUV_IN[1:0]		WM_COLOR ADJ	–	WM_MASKMODE[1:0]	
Reset	–	–	01		0	–	00	
Access Type	–	–	Write, Read		Write, Read	–	Write, Read	

Bitfield	Bits	Description	Decode
WM_YUV_IN	5:4	Input color format	0b0: RGB - 8-bit color data 0b1: YUV444 8-bits each
WM_COLORADJ	3	Color Adjust	0b0: Masked off LSBs of U. Use if temporal dithering is enabled. 0b1: Set U[Ko:0] = 1/2 WM gain. Use this mode if temporal dithering is disabled

Bitfield	Bits	Description	Decode
WM_MASKMODE	1:0	Sets watermark mask for the device	0b00: Mask if WM is detected 0b01: Mask if WM is detected, blank if error is detected 0b10: Reserved b011: Reserved

WM_5 (0x195)

Bit	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	WM_POLOUT	WM_DETOUT	WM_ERROR
Reset	–	–	–	–	–	0	0	0
Access Type	–	–	–	–	–	Read Only	Read Only	Read Only

Bitfield	Bits	Description	Decode
WM_POLOUT	2	Live frame based polarity	0b0: Watermark polarity 0 0b1: Watermark polarity 1
WM_DETOUT	1	Live frame based detection output	0b0: Watermark not detected 0b1: Watermark detected
WM_ERROR	0	Live active high watermark error	0b0: No watermark error 0b1: Watermark error active, bit automatically clears when error clears.

WM_6 (0x196)

Bit	7	6	5	4	3	2	1	0
Field	WM_TIMER[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
WM_TIMER	7:0	Time in 2 msec steps the frozen frame condition must be observed before an Error is generated. 0 = No filter	0xXX: Number of milliseconds

WM_WREN_0 (0x1AE)

Bit	7	6	5	4	3	2	1	0
Field	WM_WREN_L[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
WM_WREN_L	7:0	Write 0xBA to WM_WREN_L and 0xDC to WM_WREN_H registers to enable writing to Watermark registers. Otherwise Watermark registers are read-only.	0xBA: Enables writing to WM registers Others: WM registers remain read only

WM_WREN_1 (0x1AF)

Bit	7	6	5	4	3	2	1	0
Field	WM_WREN_H[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
WM_WREN_H	7:0	Write 0xBA to WM_WREN_L and 0xDC to WM_WREN_H registers to enable writing to Watermark registers. Otherwise Watermark registers are read-only.	0xDC: Enables writing to WM registers Others: WM registers remain read only

CROSS_0 (0x1B0, 0x1F3, 0x236, 0x279)

Bit	7	6	5	4	3	2	1	0
Field	–	CROSS0_I	CROSS0_F	CROSS0[4:0]				
Reset	–	0	0	00000				
Access Type	–	Write, Read	Write, Read	Write, Read				

Bitfield	Bits	Description	Decode
CROSS0_I	6	Invert CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS0_F	5	Force CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS0	4:0	Maps incoming bit position set by this field to the outgoing bit position 0	0bXXXXX: Incoming bit position

CROSS_1 (0x1B1, 0x1F4, 0x237, 0x27A)

Bit	7	6	5	4	3	2	1	0
Field	–	CROSS1_I	CROSS1_F	CROSS1[4:0]				
Reset	–	0	0	00001				
Access Type	–	Write, Read	Write, Read	Write, Read				

Bitfield	Bits	Description	Decode
CROSS1_I	6	Invert CrossX	0b0: Do not invert bit 0b1: Invert bit

Bitfield	Bits	Description	Decode
CROSS1_F	5	Force CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS1	4:0	Maps incoming bit position set by this field to the outgoing bit position 1	0bXXXXX: Incoming bit position

CROSS_2 (0x1B2, 0x1F5, 0x238, 0x27B)

Bit	7	6	5	4	3	2	1	0
Field	–	CROSS2_I	CROSS2_F	CROSS2[4:0]				
Reset	–	0	0	00010				
Access Type	–	Write, Read	Write, Read	Write, Read				

Bitfield	Bits	Description	Decode
CROSS2_I	6	Invert CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS2_F	5	Force CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS2	4:0	Maps incoming bit position set by this field to the outgoing bit position 2	0bXXXXX: Incoming bit position

CROSS_3 (0x1B3, 0x1F6, 0x239, 0x27C)

Bit	7	6	5	4	3	2	1	0
Field	–	CROSS3_I	CROSS3_F	CROSS3[4:0]				
Reset	–	0	0	00011				
Access Type	–	Write, Read	Write, Read	Write, Read				

Bitfield	Bits	Description	Decode
CROSS3_I	6	Invert CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS3_F	5	Force CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS3	4:0	Maps incoming bit position set by this field to the outgoing bit position 3	0bXXXXX: Incoming bit position

CROSS_4 (0x1B4, 0x1F7, 0x23A, 0x27D)

Bit	7	6	5	4	3	2	1	0
Field	–	CROSS4_I	CROSS4_F	CROSS4[4:0]				
Reset	–	0	0	00100				
Access Type	–	Write, Read	Write, Read	Write, Read				

Bitfield	Bits	Description	Decode
CROSS4_I	6	Invert CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS4_F	5	Force CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS4	4:0	Maps incoming bit position set by this field to the outgoing bit position 4	0bXXXXX: Incoming bit position

CROSS_5 (0x1B5, 0x1F8, 0x23B, 0x27E)

Bit	7	6	5	4	3	2	1	0
Field	–	CROSS5_I	CROSS5_F	CROSS5[4:0]				
Reset	–	0	0	00101				
Access Type	–	Write, Read	Write, Read	Write, Read				

Bitfield	Bits	Description	Decode
CROSS5_I	6	Invert CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS5_F	5	Force CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS5	4:0	Maps incoming bit position set by this field to the outgoing bit position 5	0bXXXXX: Incoming bit position

CROSS_6 (0x1B6, 0x1F9, 0x23C, 0x27F)

Bit	7	6	5	4	3	2	1	0
Field	–	CROSS6_I	CROSS6_F	CROSS6[4:0]				
Reset	–	0	0	00110				
Access Type	–	Write, Read	Write, Read	Write, Read				

Bitfield	Bits	Description	Decode
CROSS6_I	6	Invert CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS6_F	5	Force CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS6	4:0	Maps incoming bit position set by this field to the outgoing bit position 6	0bXXXXX: Incoming bit position

CROSS_7 (0x1B7, 0x1FA, 0x23D, 0x280)

Bit	7	6	5	4	3	2	1	0
Field	–	CROSS7_I	CROSS7_F	CROSS7[4:0]				
Reset	–	0	0	00111				
Access Type	–	Write, Read	Write, Read	Write, Read				

Bitfield	Bits	Description	Decode
CROSS7_I	6	Invert CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS7_F	5	Force CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS7	4:0	Maps incoming bit position set by this field to the outgoing bit position 7	0bXXXXX: Incoming bit position

CROSS_8 (0x1B8, 0x1FB, 0x23E, 0x281)

Bit	7	6	5	4	3	2	1	0
Field	–	CROSS8_I	CROSS8_F	CROSS8[4:0]				
Reset	–	0	0	01000				
Access Type	–	Write, Read	Write, Read	Write, Read				

Bitfield	Bits	Description	Decode
CROSS8_I	6	Invert CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS8_F	5	Force CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS8	4:0	Maps incoming bit position set by this field to the outgoing bit position 8	0bXXXXX: Incoming bit position

CROSS_9 (0x1B9, 0x1FC, 0x23F, 0x282)

Bit	7	6	5	4	3	2	1	0
Field	–	CROSS9_I	CROSS9_F	CROSS9[4:0]				
Reset	–	0	0	01001				
Access Type	–	Write, Read	Write, Read	Write, Read				

Bitfield	Bits	Description	Decode
CROSS9_I	6	Invert CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS9_F	5	Force CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero

Bitfield	Bits	Description	Decode
CROSS9	4:0	Maps incoming bit position set by this field to the outgoing bit position 9	0bXXXXX: Incoming bit position

CROSS_10 (0x1BA, 0x1FD, 0x240, 0x283)

Bit	7	6	5	4	3	2	1	0
Field	–	CROSS10_I	CROSS10_F	CROSS10[4:0]				
Reset	–	0	0	0xA				
Access Type	–	Write, Read	Write, Read	Write, Read				

Bitfield	Bits	Description	Decode
CROSS10_I	6	Invert CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS10_F	5	Force CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS10	4:0	Maps incoming bit position set by this field to the outgoing bit position 10	0bXXXXX: Incoming bit position

CROSS_11 (0x1BB, 0x1FE, 0x241, 0x284)

Bit	7	6	5	4	3	2	1	0
Field	–	CROSS11_I	CROSS11_F	CROSS11[4:0]				
Reset	–	0	0	0xB				
Access Type	–	Write, Read	Write, Read	Write, Read				

Bitfield	Bits	Description	Decode
CROSS11_I	6	Invert CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS11_F	5	Force CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS11	4:0	Maps incoming bit position set by this field to the outgoing bit position 11	0bXXXXX: Incoming bit position

CROSS_12 (0x1BC, 0x1FF, 0x242, 0x285)

Bit	7	6	5	4	3	2	1	0
Field	–	CROSS12_I	CROSS12_F	CROSS12[4:0]				
Reset	–	0	0	0xC				
Access Type	–	Write, Read	Write, Read	Write, Read				

Bitfield	Bits	Description	Decode
CROSS12_I	6	Invert CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS12_F	5	Force CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS12	4:0	Maps incoming bit position set by this field to the outgoing bit position 12	0bXXXXX: Incoming bit position

CROSS_13 (0x1BD, 0x200, 0x243, 0x286)

Bit	7	6	5	4	3	2	1	0
Field	–	CROSS13_I	CROSS13_F	CROSS13[4:0]				
Reset	–	0	0	0xD				
Access Type	–	Write, Read	Write, Read	Write, Read				

Bitfield	Bits	Description	Decode
CROSS13_I	6	Invert CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS13_F	5	Force CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS13	4:0	Maps incoming bit position set by this field to the outgoing bit position 13	0bXXXXX: Incoming bit position

CROSS_14 (0x1BE, 0x201, 0x244, 0x287)

Bit	7	6	5	4	3	2	1	0
Field	–	CROSS14_I	CROSS14_F	CROSS14[4:0]				
Reset	–	0	0	0xE				
Access Type	–	Write, Read	Write, Read	Write, Read				

Bitfield	Bits	Description	Decode
CROSS14_I	6	Invert CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS14_F	5	Force CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS14	4:0	Maps incoming bit position set by this field to the outgoing bit position 14	0bXXXXX: Incoming bit position

CROSS_15 (0x1BF, 0x202, 0x245, 0x288)

Bit	7	6	5	4	3	2	1	0
Field	–	CROSS15_I	CROSS15_F	CROSS15[4:0]				
Reset	–	0	0	0xF				
Access Type	–	Write, Read	Write, Read	Write, Read				

Bitfield	Bits	Description	Decode
CROSS15_I	6	Invert CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS15_F	5	Force CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS15	4:0	Maps incoming bit position set by this field to the outgoing bit position 15	0bXXXXX: Incoming bit position

CROSS_16 (0x1C0, 0x203, 0x246, 0x289)

Bit	7	6	5	4	3	2	1	0
Field	–	CROSS16_I	CROSS16_F	CROSS16[4:0]				
Reset	–	0	0	10000				
Access Type	–	Write, Read	Write, Read	Write, Read				

Bitfield	Bits	Description	Decode
CROSS16_I	6	Invert CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS16_F	5	Force CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS16	4:0	Maps incoming bit position set by this field to the outgoing bit position 16	0bXXXXX: Incoming bit position

CROSS_17 (0x1C1, 0x204, 0x247, 0x28A)

Bit	7	6	5	4	3	2	1	0
Field	–	CROSS17_I	CROSS17_F	CROSS17[4:0]				
Reset	–	0	0	10001				
Access Type	–	Write, Read	Write, Read	Write, Read				

Bitfield	Bits	Description	Decode
CROSS17_I	6	Invert CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS17_F	5	Force CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero

Bitfield	Bits	Description	Decode
CROSS17	4:0	Maps incoming bit position set by this field to the outgoing bit position 17	0bXXXXX: Incoming bit position

CROSS_18 (0x1C2, 0x205, 0x248, 0x28B)

Bit	7	6	5	4	3	2	1	0
Field	–	CROSS18_I	CROSS18_F	CROSS18[4:0]				
Reset	–	0	0	10010				
Access Type	–	Write, Read	Write, Read	Write, Read				

Bitfield	Bits	Description	Decode
CROSS18_I	6	Invert CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS18_F	5	Force CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS18	4:0	Maps incoming bit position set by this field to the outgoing bit position 18	0bXXXXX: Incoming bit position

CROSS_19 (0x1C3, 0x206, 0x249, 0x28C)

Bit	7	6	5	4	3	2	1	0
Field	–	CROSS19_I	CROSS19_F	CROSS19[4:0]				
Reset	–	0	0	10011				
Access Type	–	Write, Read	Write, Read	Write, Read				

Bitfield	Bits	Description	Decode
CROSS19_I	6	Invert CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS19_F	5	Force CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS19	4:0	Maps incoming bit position set by this field to the outgoing bit position 19	0bXXXXX: Incoming bit position

CROSS_20 (0x1C4, 0x207, 0x24A, 0x28D)

Bit	7	6	5	4	3	2	1	0
Field	–	CROSS20_I	CROSS20_F	CROSS20[4:0]				
Reset	–	0	0	10100				
Access Type	–	Write, Read	Write, Read	Write, Read				

Bitfield	Bits	Description	Decode
CROSS20_I	6	Invert CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS20_F	5	Force CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS20	4:0	Maps incoming bit position set by this field to the outgoing bit position 20	0bXXXXX: Incoming bit position

CROSS_21 (0x1C5, 0x208, 0x24B, 0x28E)

Bit	7	6	5	4	3	2	1	0
Field	–	CROSS21_I	CROSS21_F	CROSS21[4:0]				
Reset	–	0	0	10101				
Access Type	–	Write, Read	Write, Read	Write, Read				

Bitfield	Bits	Description	Decode
CROSS21_I	6	Invert CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS21_F	5	Force CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS21	4:0	Maps incoming bit position set by this field to the outgoing bit position 21	0bXXXXX: Incoming bit position

CROSS_22 (0x1C6, 0x209, 0x24C, 0x28F)

Bit	7	6	5	4	3	2	1	0
Field	–	CROSS22_I	CROSS22_F	CROSS22[4:0]				
Reset	–	0	0	10110				
Access Type	–	Write, Read	Write, Read	Write, Read				

Bitfield	Bits	Description	Decode
CROSS22_I	6	Invert CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS22_F	5	Force CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS22	4:0	Maps incoming bit position set by this field to the outgoing bit position 22	0bXXXXX: Incoming bit position

CROSS_23 (0x1C7, 0x20A, 0x24D, 0x290)

Bit	7	6	5	4	3	2	1	0
Field	–	CROSS23_I	CROSS23_F	CROSS23[4:0]				
Reset	–	0	0	10111				
Access Type	–	Write, Read	Write, Read	Write, Read				

Bitfield	Bits	Description	Decode
CROSS23_I	6	Invert CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS23_F	5	Force CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS23	4:0	Maps incoming bit position set by this field to the outgoing bit position 23	0bXXXXX: Incoming bit position

VTX0 (0x1C8, 0x20B, 0x24E, 0x291)

Bit	7	6	5	4	3	2	1	0
Field	GEN_VS	GEN_HS	GEN_DE	VS_INV	HS_INV	DE_INV	VTG_MODE[1:0]	
Reset	0	0	0	0	0	0	11	
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	

Bitfield	Bits	Description	Decode
GEN_VS	7	Enable to generate VS output according to the timing definition	0b0: Do not generate VS 0b1: Generate VS
GEN_HS	6	Enable to generate HS output according to the timing definition	0b0: Do not generate HS 0b1: Generate HS
GEN_DE	5	Enable to generate DE output according to the timing definition	0b0: Do not generate DE 0b1: Generate DE
VS_INV	4	Invert VSYNC output of Video Timing Generator	0b0: Do not invert VS 0b1: Invert VS
HS_INV	3	Invert HSYNC output of Video Timing Generator	0b0: Do not invert HS 0b1: Invert HS
DE_INV	2	Invert DE output of Video Timing Generator	0b0: Do not invert DE 0b1: Invert DE
VTG_MODE	1:0	Video interface timing generation mode. 00 or 11= VS tracking mode. VS input's period (VS_HIGH+VS_LOW) is tracked. After VS tracking is locked, any VS input edge (glitches) not in the expected pclk cycle is ignored. VS tracking is locked with 3	0b00: VS tracking mode 0b01: VS on trigger mode 0b10: Auto repeat mode 0b11: Free running mode

Bitfield	Bits	Description	Decode
		consecutive matches and unlocked by 3 consecutive mismatches. When unlocked or power up, the next VS input edge is assumed to be the right VS edge. 01 = VS one trigger mode (default) One VS input edge will trigger the generation of ONE frame of VSO/HSO/DEO. If next VS input edge comes earlier or later than expected by VS period, the newly generated frame will be correct. The current VSO/HSO/DEO will be cut or extended at the time point of rising edge of the newly generated VSO/HSO/DEO. 10 = Auto repeat mode VS input edge will trigger the generation of continuous frames of VSO/HSO/DEO even if no more VS input edges. If next VS input edge comes earlier or later than expected by VS period, the newly generated frame will be correct. The current VSO/HSO/DEO will be cut or extended at the time point of rising edge of the newly generated VSO/HSO/DEO.	

VTX1 (0x1C9, 0x20C, 0x24F, 0x292)

Bit	7	6	5	4	3	2	1	0
Field	–	–	PCLKDET_VTX	–	–	–	–	VS_TRIG
Reset	–	–	0	–	–	–	–	1
Access Type	–	–	Read Only	–	–	–	–	Write, Read

Bitfield	Bits	Description	Decode
PCLKDET_VTX	5	PCLK detected	0b0: PCLK not detected 0b1: PCLK detected
VS_TRIG	0	Select VS trigger edge	0b0: Falling edge 0b1: Rising edge

VTX2 (0x1CA, 0x20D, 0x250, 0x293)

Bit	7	6	5	4	3	2	1	0
Field	VS_DLY_2[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
VS_DLY_2	7:0	VS Delay in terms of PCLK cycles. The output VS is delayed by VS_DELAY cycles from the input VS. (Bits [23:16])	0xXX: Most significant byte of VS delay

VTX3 (0x1CB, 0x20E, 0x251, 0x294)

Bit	7	6	5	4	3	2	1	0
Field	VS_DLY_1[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
VS_DLY_1	7:0	VS Delay in terms of PCLK cycles. The output VS is delayed by VS_DELAY cycles from the input VS. (Bits [15:8])	0xXX: Middle significant byte of VS delay

VTX4 (0x1CC, 0x20F, 0x252, 0x295)

Bit	7	6	5	4	3	2	1	0
Field	VS_DLY_0[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
VS_DLY_0	7:0	VS Delay in terms of PCLK cycles. The output VS is delayed by VS_DELAY cycles from the input VS. (Bits [7:0])	0xXX: Least significant byte of VS delay

VTX5 (0x1CD, 0x210, 0x253, 0x296)

Bit	7	6	5	4	3	2	1	0
Field	VS_HIGH_2[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
VS_HIGH_2	7:0	VS High Period in terms of PCLK cycles (Bits [23:16])	0xXX: Most significant byte of VS high period

VTX6 (0x1CE, 0x211, 0x254, 0x297)

Bit	7	6	5	4	3	2	1	0
Field	VS_HIGH_1[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
VS_HIGH_1	7:0	VS High Period in terms of PCLK cycles (Bits [15:8])	0xXX: Middle significant byte of VS high period

VTX7 (0x1CF, 0x212, 0x255, 0x298)

Bit	7	6	5	4	3	2	1	0
Field	VS_HIGH_0[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
VS_HIGH_0	7:0	VS High Period in terms of PCLK cycles (Bits [7:0])	0xXX: Least significant byte of VS high period

VTX8 (0x1D0, 0x213, 0x256, 0x299)

Bit	7	6	5	4	3	2	1	0
Field	VS_LOW_2[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
VS_LOW_2	7:0	VS Low Period in terms of PCLK cycles (Bits [23:16])	0xXX: Most significant byte of VS low period

VTX9 (0x1D1, 0x214, 0x257, 0x29A)

Bit	7	6	5	4	3	2	1	0
Field	VS_LOW_1[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
VS_LOW_1	7:0	VS Low Period in terms of PCLK cycles (Bits [15:8])	0xXX: Middle significant byte of VS low period

VTX10 (0x1D2, 0x215, 0x258, 0x29B)

Bit	7	6	5	4	3	2	1	0
Field	VS_LOW_0[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
VS_LOW_0	7:0	VS Low Period in terms of PCLK cycles (Bits [7:0])	0xXX: Least significant byte of VS low period

VTX11 (0x1D3, 0x216, 0x259, 0x29C)

Bit	7	6	5	4	3	2	1	0
Field	V2H_2[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
V2H_2	7:0	VS edge to the rising edge of the first HS in terms of PCLK cycles (Bits [23:16])	0xXX: Most significant byte of VS edge to 1st HS rising edge

VTX12 (0x1D4, 0x217, 0x25A, 0x29D)

Bit	7	6	5	4	3	2	1	0
Field	V2H_1[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
V2H_1	7:0	VS edge to the rising edge of the first HS in terms of PCLK cycles (Bits [15:8])	0xXX: Middle significant byte of VS edge to 1st HS rising edge

VTX13 (0x1D5, 0x218, 0x25B, 0x29E)

Bit	7	6	5	4	3	2	1	0
Field	V2H_0[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
V2H_0	7:0	VS edge to the rising edge of the first HS in terms of PCLK cycles (Bits [7:0])	0xXX: Least significant byte of VS edge to 1st HS rising edge

VTX14 (0x1D6, 0x219, 0x25C, 0x29F)

Bit	7	6	5	4	3	2	1	0
Field	HS_HIGH_1[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
HS_HIGH_1	7:0	HS High Period in terms of PCLK cycles (Bits [15:8])	0xXX: Most significant byte of HS high period

VTX15 (0x1D7, 0x21A, 0x25D, 0x2A0)

Bit	7	6	5	4	3	2	1	0
Field	HS_HIGH_0[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
HS_HIGH_0	7:0	HS High Period in terms of PCLK cycles (Bits [7:0])	0xXX: Least significant byte of HS high period

VTX16 (0x1D8, 0x21B, 0x25E, 0x2A1)

Bit	7	6	5	4	3	2	1	0
Field	HS_LOW_1[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
HS_LOW_1	7:0	HS Low Period in terms of PCLK cycles (Bits [15:8])	0xXX: Most significant byte of HS low period

VTX17 (0x1D9, 0x21C, 0x25F, 0x2A2)

Bit	7	6	5	4	3	2	1	0
Field	HS_LOW_0[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
HS_LOW_0	7:0	HS Low Period in terms of PCLK cycles (Bits [7:0])	0xXX: Least significant byte of HS low period

VTX18 (0x1DA, 0x21D, 0x260, 0x2A3)

Bit	7	6	5	4	3	2	1	0
Field	HS_CNT_1[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
HS_CNT_1	7:0	HS pulses per frame (Bits [15:8])	0xXX: Most significant byte of HS pulses per frame

VTX19 (0x1DB, 0x21E, 0x261, 0x2A4)

Bit	7	6	5	4	3	2	1	0
Field	HS_CNT_0[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
HS_CNT_0	7:0	HS pulses per frame (Bits [7:0])	0xXX: Least significant byte of HS pulses per frame

VTX20 (0x1DC, 0x21F, 0x262, 0x2A5)

Bit	7	6	5	4	3	2	1	0
Field	V2D_2[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
V2D_2	7:0	VS edge to the rising edge of the first DE in terms of PCLK cycles (Bits [23:16])	0xXX: Most significant byte of VS edge to 1st DE

VTX21 (0x1DD, 0x220, 0x263, 0x2A6)

Bit	7	6	5	4	3	2	1	0
Field	V2D_1[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
V2D_1	7:0	VS edge to the rising edge of the first DE in terms of PCLK cycles (Bits [15:8])	0xXX: Middle significant byte of VS edge to 1st DE

VTX22 (0x1DE, 0x221, 0x264, 0x2A7)

Bit	7	6	5	4	3	2	1	0
Field	V2D_0[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
V2D_0	7:0	VS edge to the rising edge of the first DE in terms of PCLK cycles (Bits [7:0])	0xXX: Least significant byte of VS edge to 1st DE

VTX23 (0x1DF, 0x222, 0x265, 0x2A8)

Bit	7	6	5	4	3	2	1	0
Field	DE_HIGH_1[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
DE_HIGH_1	7:0	DE High Period in terms of PCLK cycles (Bits [15:8])	0xXX: Most significant byte of DE high period

VTX24 (0x1E0, 0x223, 0x266, 0x2A9)

Bit	7	6	5	4	3	2	1	0
Field	DE_HIGH_0[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
DE_HIGH_0	7:0	DE High Period in terms of PCLK cycles (Bits [7:0])	0xXX: Least significant byte of DE high period

VTX25 (0x1E1, 0x224, 0x267, 0x2AA)

Bit	7	6	5	4	3	2	1	0
Field	DE_LOW_1[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
DE_LOW_1	7:0	DE Low Period in terms of PCLK cycles (Bits [15:8])	0xXX: Most significant byte of DE low period

VTX26 (0x1E2, 0x225, 0x268, 0x2AB)

Bit	7	6	5	4	3	2	1	0
Field	DE_LOW_0[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
DE_LOW_0	7:0	DE Low Period in terms of PCLK cycles (Bits [7:0])	0xXX: Least significant byte of DE low period

VTX27 (0x1E3, 0x226, 0x269, 0x2AC)

Bit	7	6	5	4	3	2	1	0
Field	DE_CNT_1[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
DE_CNT_1	7:0	Active lines per frame (Bits [15:8])	0xXX: Most significant byte of DE pulses per frame

VTX28 (0x1E4, 0x227, 0x26A, 0x2AD)

Bit	7	6	5	4	3	2	1	0
Field	DE_CNT_0[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
DE_CNT_0	7:0	Active lines per frame (Bits [7:0])	0xXX: Least significant byte of DE pulses per frame

VTX29 (0x1E5, 0x228, 0x26B, 0x2AE)

Bit	7	6	5	4	3	2	1	0
Field	VID_PRBS_EN	–	–	–	–	GRAD_MODE	PATGEN_MODE[1:0]	
Reset	0	–	–	–	–	0	00	
Access Type	Write, Read	–	–	–	–	Write, Read	Write, Read	

Bitfield	Bits	Description	Decode
VID_PRBS_EN	7	Enable Video PRBS generator	0b0: Video PRBS generator disabled 0b1: Video PRBS generator enabled
GRAD_MODE	2	Gradient pattern generator mode	0b0: Gradient mode disabled

Bitfield	Bits	Description	Decode
			0b1: Gradient mode enabled
PATGEN_MODE	1:0	Pattern Generator Mode	0b00: Reserved 0b01: Generate checkerboard pattern 0b10: Generate gradient pattern 0b11: Reserved

VTX30 (0x1E6, 0x229, 0x26C, 0x2AF)

Bit	7	6	5	4	3	2	1	0
Field	GRAD_INC[7:0]							
Reset	00000100							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
GRAD_INC	7:0	Gradient mode increment amount (increment amount is the register value divided by 4)	0xXX: Gradient increment base

VTX31 (0x1E7, 0x22A, 0x26D, 0x2B0)

Bit	7	6	5	4	3	2	1	0
Field	CHKR_A_L[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
CHKR_A_L	7:0	Checkerboard mode color A Low byte	0xXX: Least significant byte of checkerboard mode color A

VTX32 (0x1E8, 0x22B, 0x26E, 0x2B1)

Bit	7	6	5	4	3	2	1	0
Field	CHKR_A_M[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
CHKR_A_M	7:0	Checkerboard mode color A Mid byte	0xXX: Middle significant byte of checkerboard mode color A

VTX33 (0x1E9, 0x22C, 0x26F, 0x2B2)

Bit	7	6	5	4	3	2	1	0
Field	CHKR_A_H[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
CHKR_A_H	7:0	Checkerboard mode color A High byte	0xXX: Most significant byte of checkerboard mode color A

VTX34 (0x1EA, 0x22D, 0x270, 0x2B3)

Bit	7	6	5	4	3	2	1	0
Field	CHKR_B_L[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
CHKR_B_L	7:0	Checkerboard mode color B Low byte	0xXX: Least significant byte of checkerboard mode color B

VTX35 (0x1EB, 0x22E, 0x271, 0x2B4)

Bit	7	6	5	4	3	2	1	0
Field	CHKR_B_M[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
CHKR_B_M	7:0	Checkerboard mode color B Mid byte	0xXX: Middle significant byte of checkerboard mode color B

VTX36 (0x1EC, 0x22F, 0x272, 0x2B5)

Bit	7	6	5	4	3	2	1	0
Field	CHKR_B_H[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
CHKR_B_H	7:0	Checkerboard mode color B High byte	0xXX: Most significant byte of checkerboard mode color B

VTX37 (0x1ED, 0x230, 0x273, 0x2B6)

Bit	7	6	5	4	3	2	1	0
Field	CHKR_RPT_A[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
CHKR_RPT_A	7:0	Checkerboard mode color A repeat count	0xXX: Repeat count of checkerboard mode color A

VTX38 (0x1EE, 0x231, 0x274, 0x2B7)

Bit	7	6	5	4	3	2	1	0
Field	CHKR_RPT_B[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
CHKR_RPT_B	7:0	Checkerboard mode color B repeat count	0xXX: Repeat count of checkerboard mode color B

VTX39 (0x1EF, 0x232, 0x275, 0x2B8)

Bit	7	6	5	4	3	2	1	0
Field	CHKR_ALT[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
CHKR_ALT	7:0	Checkerboard mode alternate line count	0xXX: Checkerboard mode alternate line count

VTX40 (0x1F0, 0x233, 0x276, 0x2B9)

Bit	7	6	5	4	3	2	1	0
Field	DIS_COLOR_CROSSBAR	CROSSHS_I	CROSSHS_F	CROSSHS[4:0]				
Reset	0	0	0	11000				
Access Type	Write, Read	Write, Read	Write, Read	Write, Read				

Bitfield	Bits	Description	Decode
DIS_COLOR_CROSSBAR	7	Disable this crossbar and use the gmsl1 crossbar	0b0: Enable this crossbar 0b1: Disable this crossbar, use GMSL1crossbar

Bitfield	Bits	Description	Decode
CROSSHS_I	6	Invert Cross_HS	0b0: Do not invert bit 0b1: Invert bit
CROSSHS_F	5	Force CROSS_HS to 0 before inversion	
CROSSHS	4:0	Map selected internal signal to HS	0bXXXXX: Incoming bit position

VTX41 (0x1F1, 0x234, 0x277, 0x2BA)

Bit	7	6	5	4	3	2	1	0
Field	–	CROSSVS_I	CROSSVS_F	CROSSVS[4:0]				
Reset	–	0	0	11001				
Access Type	–	Write, Read	Write, Read	Write, Read				

Bitfield	Bits	Description	Decode
CROSSVS_I	6	Invert Cross_VS	0b0: Do not invert bit 0b1: Invert bit
CROSSVS_F	5	Force CROSS_VS to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSSVS	4:0	Map selected internal signal to VS	0bXXXXX: Incoming bit position

VTX42 (0x1F2, 0x235, 0x278, 0x2BB)

Bit	7	6	5	4	3	2	1	0
Field	–	CROSSDE_I	CROSSDE_F	CROSSDE[4:0]				
Reset	–	0	0	11010				
Access Type	–	Write, Read	Write, Read	Write, Read				

Bitfield	Bits	Description	Decode
CROSSDE_I	6	Invert Cross_DE	0b0: Do not invert bit 0b1: Invert bit
CROSSDE_F	5	Force Cross_DE to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSSDE	4:0	Map selected internal signal to DE	0bXXXXX: Incoming bit position

GPIO_A (0x2BE)

Bit	7	6	5	4	3	2	1	0
Field	RES_CFG	TX_PRIO	TX_COMP_EN	GPIO_OUT	GPIO_IN	GPIO_RX_EN	GPIO_TX_EN	GPIO_OUT_DIS
Reset	0	0	0	1	1	1	0	0
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	Read Only	Write, Read	Write, Read	Write, Read

Bitfield	Bits	Description	Decode
RES_CFG	7	Resistor pull-up/pull-down strength	0b0: 40kΩ 0b1: 1MΩ
TX_PRIO	6	Priority for GPIO scheduling	0b0: Low priority 0b1: High priority
TX_COMP_EN	5	Jitter minimization compensation enable	0b0: Jitter compensation disabled 0b1: Jitter compensation enabled
GPIO_OUT	4	GPIO pin output drive value when gpio_rx_en=0	0b0: This GPIO pin output is driven to 0 0b1: This GPIO pin output is driven to 1
GPIO_IN	3	GPIO pin input level	0b0: This GPIO pin value is 0 0b1: This GPIO pin value is 1
GPIO_RX_EN	2	GPIO Out source control	0b0: This GPIO source disabled for GMSL2 reception 0b1: This GPIO source enabled for GMSL2 reception
GPIO_TX_EN	1	GPIO TX source control	0b0: This GPIO source disabled for GMSL2 transmission 0b1: This GPIO source enabled for GMSL2 transmission
GPIO_OUT_DIS	0	Disable GPIO output driver	0b0: Output driver enabled 0b1: Output driver disabled

GPIO_B (0x2BF)

Bit	7	6	5	4	3	2	1	0
Field	PULL_UPDN_SEL[1:0]		OUT_TYPE	GPIO_TX_ID[4:0]				
Reset	01		0	00000				
Access Type	Write, Read		Write, Read	Write, Read				

Bitfield	Bits	Description	Decode
PULL_UPDN_SEL	7:6	Buffer pull up/down configuration	0b00: None 0b01: Pull-up 0b10: Pull-down 0b11: Reserved

Bitfield	Bits	Description	Decode
OUT_TYPE	5	Driver type selection	0b0: Open-drain 0b1: Push-pull
GPIO_TX_ID	4:0	GPIO ID for pin while transmitting	0bXXXXX: This GPIO transmit ID

GPIO_C (0x2C0)

Bit	7	6	5	4	3	2	1	0
Field	OVR_RES_CFG	–	–	GPIO_RX_ID[4:0]				
Reset	0	–	–	00000				
Access Type	Write, Read	–	–	Write, Read				

Bitfield	Bits	Description	Decode
OVR_RES_CFG	7	Override non-GPIO port function IO setting. When set, res_cfg and pull_updn_sel are effective when pin is configured as non-GPIO. When cleared, non-GPIO pin function determines IO type.	0b0: Non-GPIO function determines IO type when alternate function is selected 0b1: RES_CFG and PULL_UPDN_SEL determine IO type for non-GPIO configuration
GPIO_RX_ID	4:0	GPIO ID for pin while receiving	0bXXXXX: This GPIO receive ID

GPIO_A (0x2C1)

Bit	7	6	5	4	3	2	1	0
Field	RES_CFG	TX_PRIO	TX_COMP_EN	GPIO_OUT	GPIO_IN	GPIO_RX_EN	GPIO_TX_EN	GPIO_OUT_DIS
Reset	1	0	0	0	0	0	0	1
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	Read Only	Write, Read	Write, Read	Write, Read

Bitfield	Bits	Description	Decode
RES_CFG	7	Resistor pull-up/pull-down strength	0b0: 40kΩ 0b1: 1MΩ
TX_PRIO	6	Priority for GPIO scheduling	0b0: Low priority 0b1: High priority
TX_COMP_EN	5	Jitter minimization compensation enable	0b0: Jitter compensation disabled 0b1: Jitter compensation enabled
GPIO_OUT	4	GPIO pin output drive value when gpio_rx_en=0	0b0: This GPIO pin output is driven to 0 0b1: This GPIO pin output is driven to 1
GPIO_IN	3	GPIO pin input level	0b0: This GPIO pin value is 0 0b1: This GPIO pin value is 1
GPIO_RX_EN	2	GPIO Out source control	0b0: This GPIO source disabled for GMSL2

Bitfield	Bits	Description	Decode
			reception 0b1: This GPIO source enabled for GMSL2 reception
GPIO_TX_EN	1	GPIO TX source control	0b0: This GPIO source disabled for GMSL2 transmission 0b1: This GPIO source enabled for GMSL2 transmission
GPIO_OUT_DIS	0	Disable GPIO output driver	0b0: Output driver enabled 0b1: Output driver disabled

GPIO_B (0x2C2)

Bit	7	6	5	4	3	2	1	0
Field	PULL_UPDN_SEL[1:0]		OUT_TYPE	GPIO_TX_ID[4:0]				
Reset	00		1	00001				
Access Type	Write, Read		Write, Read	Write, Read				

Bitfield	Bits	Description	Decode
PULL_UPDN_SEL	7:6	Buffer pull up/down configuration	0b00: None 0b01: Pull-up 0b10: Pull-down 0b11: Reserved
OUT_TYPE	5	Driver type selection	0b0: Open-drain 0b1: Push-pull
GPIO_TX_ID	4:0	GPIO ID for pin while transmitting	0bXXXXX: This GPIO transmit ID

GPIO_C (0x2C3)

Bit	7	6	5	4	3	2	1	0
Field	OVR_RES_CFG	–	–	GPIO_RX_ID[4:0]				
Reset	0	–	–	00001				
Access Type	Write, Read	–	–	Write, Read				

Bitfield	Bits	Description	Decode
OVR_RES_CFG	7	Override non-GPIO port function IO setting. When set, res_cfg and pull_updn_sel are effective when pin is configured as non-GPIO. When cleared, non-GPIO pin function determines IO type.	0b0: Non-GPIO function determines IO type when alternate function is selected 0b1: RES_CFG and PULL_UPDN_SEL determine IO type for non-GPIO configuration
GPIO_RX_ID	4:0	GPIO ID for pin while receiving	0bXXXXX: This GPIO receive ID

GPIO_A (0x2C4)

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Field	RES_CFG	TX_PRIO	TX_COMP_EN	GPIO_OUT	GPIO_IN	GPIO_RX_EN	GPIO_TX_EN	GPIO_OUT_DIS
Reset	1	0	0	1	1	0	0	1
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	Read Only	Write, Read	Write, Read	Write, Read

Bitfield	Bits	Description	Decode
RES_CFG	7	Resistor pull-up/pull-down strength	0b0: 40kΩ 0b1: 1MΩ
TX_PRIO	6	Priority for GPIO scheduling	0b0: Low priority 0b1: High priority
TX_COMP_EN	5	Jitter minimization compensation enable	0b0: Jitter compensation disabled 0b1: Jitter compensation enabled
GPIO_OUT	4	GPIO pin output drive value when gpio_rx_en=0	0b0: This GPIO pin output is driven to 0 0b1: This GPIO pin output is driven to 1
GPIO_IN	3	GPIO pin input level	0b0: This GPIO pin value is 0 0b1: This GPIO pin value is 1
GPIO_RX_EN	2	GPIO Out source control	0b0: This GPIO source disabled for GMSL2 reception 0b1: This GPIO source enabled for GMSL2 reception
GPIO_TX_EN	1	GPIO TX source control	0b0: This GPIO source disabled for GMSL2 transmission 0b1: This GPIO source enabled for GMSL2 transmission
GPIO_OUT_DIS	0	Disable GPIO output driver	0b0: Output driver enabled 0b1: Output driver disabled

GPIO_B (0x2C5)

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Field	PULL_UPDN_SEL[1:0]		OUT_TYPE	GPIO_TX_ID[4:0]				
Reset	00		1	00010				
Access Type	Write, Read		Write, Read	Write, Read				

Bitfield	Bits	Description	Decode
PULL_UPDN_SEL	7:6	Buffer pull up/down configuration	0b00: None 0b01: Pull-up 0b10: Pull-down 0b11: Reserved

Bitfield	Bits	Description	Decode
OUT_TYPE	5	Driver type selection	0b0: Open-drain 0b1: Push-pull
GPIO_TX_ID	4:0	GPIO ID for pin while transmitting	0bXXXXX: This GPIO transmit ID

GPIO_C (0x2C6)

Bit	7	6	5	4	3	2	1	0
Field	OVR_RES_CFG	–	–	GPIO_RX_ID[4:0]				
Reset	0	–	–	00010				
Access Type	Write, Read	–	–	Write, Read				

Bitfield	Bits	Description	Decode
OVR_RES_CFG	7	Override non-GPIO port function IO setting. When set, res_cfg and pull_updn_sel are effective when pin is configured as non-GPIO. When cleared, non-GPIO pin function determines IO type.	0b0: Non-GPIO function determines IO type when alternate function is selected 0b1: RES_CFG and PULL_UPDN_SEL determine IO type for non-GPIO configuration
GPIO_RX_ID	4:0	GPIO ID for pin while receiving	0bXXXXX: This GPIO receive ID

GPIO_A (0x2C7)

Bit	7	6	5	4	3	2	1	0
Field	RES_CFG	TX_PRIO	TX_COMP_EN	GPIO_OUT	GPIO_IN	GPIO_RX_EN	GPIO_TX_EN	GPIO_OUT_DIS
Reset	1	0	0	0	0	0	0	1
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	Read Only	Write, Read	Write, Read	Write, Read

Bitfield	Bits	Description	Decode
RES_CFG	7	Resistor pull-up/pull-down strength	0b0: 40kΩ 0b1: 1MΩ
TX_PRIO	6	Priority for GPIO scheduling	0b0: Low priority 0b1: High priority
TX_COMP_EN	5	Jitter minimization compensation enable	0b0: Jitter compensation disabled 0b1: Jitter compensation enabled
GPIO_OUT	4	GPIO pin output drive value when gpio_rx_en=0	0b0: This GPIO pin output is driven to 0 0b1: This GPIO pin output is driven to 1
GPIO_IN	3	GPIO pin input level	0b0: This GPIO pin value is 0 0b1: This GPIO pin value is 1
GPIO_RX_EN	2	GPIO Out source control	0b0: This GPIO source disabled for GMSL2

Bitfield	Bits	Description	Decode
			reception 0b1: This GPIO source enabled for GMSL2 reception
GPIO_TX_EN	1	GPIO TX source control	0b0: This GPIO source disabled for GMSL2 transmission 0b1: This GPIO source enabled for GMSL2 transmission
GPIO_OUT_DIS	0	Disable GPIO output driver	0b0: Output driver enabled 0b1: Output driver disabled

GPIO_B (0x2C8)

Bit	7	6	5	4	3	2	1	0
Field	PULL_UPDN_SEL[1:0]		OUT_TYPE	GPIO_TX_ID[4:0]				
Reset	00		1	00011				
Access Type	Write, Read		Write, Read	Write, Read				

Bitfield	Bits	Description	Decode
PULL_UPDN_SEL	7:6	Buffer pull up/down configuration	0b00: None 0b01: Pull-up 0b10: Pull-down 0b11: Reserved
OUT_TYPE	5	Driver type selection	0b0: Open-drain 0b1: Push-pull
GPIO_TX_ID	4:0	GPIO ID for pin while transmitting	0bXXXXX: This GPIO transmit ID

GPIO_C (0x2C9)

Bit	7	6	5	4	3	2	1	0
Field	OVR_RES_CFG	–	–	GPIO_RX_ID[4:0]				
Reset	0	–	–	00011				
Access Type	Write, Read	–	–	Write, Read				

Bitfield	Bits	Description	Decode
OVR_RES_CFG	7	Override non-GPIO port function IO setting. When set, res_cfg and pull_updn_sel are effective when pin is configured as non-GPIO. When cleared, non-GPIO pin function determines IO type.	0b0: Non-GPIO function determines IO type when alternate function is selected 0b1: RES_CFG and PULL_UPDN_SEL determine IO type for non-GPIO configuration
GPIO_RX_ID	4:0	GPIO ID for pin while receiving	0bXXXXX: This GPIO receive ID

GPIO_A (0x2CA)

Bit	7	6	5	4	3	2	1	0
Field	RES_CFG	TX_PRIO	TX_COMP_EN	GPIO_OUT	GPIO_IN	GPIO_RX_EN	GPIO_TX_EN	GPIO_OUT_DIS
Reset	1	0	0	1	1	0	0	1
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	Read Only	Write, Read	Write, Read	Write, Read

Bitfield	Bits	Description	Decode
RES_CFG	7	Resistor pull-up/pull-down strength	0b0: 40kΩ 0b1: 1MΩ
TX_PRIO	6	Priority for GPIO scheduling	0b0: Low priority 0b1: High priority
TX_COMP_EN	5	Jitter minimization compensation enable	0b0: Jitter compensation disabled 0b1: Jitter compensation enabled
GPIO_OUT	4	GPIO pin output drive value when gpio_rx_en=0	0b0: This GPIO pin output is driven to 0 0b1: This GPIO pin output is driven to 1
GPIO_IN	3	GPIO pin input level	0b0: This GPIO pin value is 0 0b1: This GPIO pin value is 1
GPIO_RX_EN	2	GPIO Out source control	0b0: This GPIO source disabled for GMSL2 reception 0b1: This GPIO source enabled for GMSL2 reception
GPIO_TX_EN	1	GPIO TX source control	0b0: This GPIO source disabled for GMSL2 transmission 0b1: This GPIO source enabled for GMSL2 transmission
GPIO_OUT_DIS	0	Disable GPIO output driver	0b0: Output driver enabled 0b1: Output driver disabled

GPIO_B (0x2CB)

Bit	7	6	5	4	3	2	1	0
Field	PULL_UPDN_SEL[1:0]		OUT_TYPE	GPIO_TX_ID[4:0]				
Reset	00		1	00100				
Access Type	Write, Read		Write, Read	Write, Read				

Bitfield	Bits	Description	Decode
PULL_UPDN_SEL	7:6	Buffer pull up/down configuration	0b00: None 0b01: Pull-up 0b10: Pull-down 0b11: Reserved

Bitfield	Bits	Description	Decode
OUT_TYPE	5	Driver type selection	0b0: Open-drain 0b1: Push-pull
GPIO_TX_ID	4:0	GPIO ID for pin while transmitting	0bXXXXX: This GPIO transmit ID

GPIO_C (0x2CC)

Bit	7	6	5	4	3	2	1	0
Field	OVR_RES_CFG	–	–	GPIO_RX_ID[4:0]				
Reset	0	–	–	00100				
Access Type	Write, Read	–	–	Write, Read				

Bitfield	Bits	Description	Decode
OVR_RES_CFG	7	Override non-GPIO port function IO setting. When set, res_cfg and pull_updn_sel are effective when pin is configured as non-GPIO. When cleared, non-GPIO pin function determines IO type.	0b0: Non-GPIO function determines IO type when alternate function is selected 0b1: RES_CFG and PULL_UPDN_SEL determine IO type for non-GPIO configuration
GPIO_RX_ID	4:0	GPIO ID for pin while receiving	

GPIO_A (0x2CD)

Bit	7	6	5	4	3	2	1	0
Field	RES_CFG	TX_PRIO	TX_COMP_EN	GPIO_OUT	GPIO_IN	GPIO_RX_EN	GPIO_TX_EN	GPIO_OUT_DIS
Reset	1	0	0	0	0	0	0	1
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	Read Only	Write, Read	Write, Read	Write, Read

Bitfield	Bits	Description	Decode
RES_CFG	7	Resistor pull-up/pull-down strength	0b0: 40kΩ 0b1: 1MΩ
TX_PRIO	6	Priority for GPIO scheduling	0b0: Low priority 0b1: High priority
TX_COMP_EN	5	Jitter minimization compensation enable	0b0: Jitter compensation disabled 0b1: Jitter compensation enabled
GPIO_OUT	4	GPIO pin output drive value when gpio_rx_en=0	0b0: This GPIO pin output is driven to 0 0b1: This GPIO pin output is driven to 1
GPIO_IN	3	GPIO pin input level	0b0: This GPIO pin value is 0 0b1: This GPIO pin value is 1
GPIO_RX_EN	2	GPIO Out source control	0b0: This GPIO source disabled for GMSL2 reception

Bitfield	Bits	Description	Decode
			0b1: This GPIO source enabled for GMSL2 reception
GPIO_TX_EN	1	GPIO TX source control	0b0: This GPIO source disabled for GMSL2 transmission 0b1: This GPIO source enabled for GMSL2 transmission
GPIO_OUT_DIS	0	Disable GPIO output driver	0b0: Output driver enabled 0b1: Output driver disabled

GPIO_B (0x2CE)

Bit	7	6	5	4	3	2	1	0
Field	PULL_UPDN_SEL[1:0]		OUT_TYPE	GPIO_TX_ID[4:0]				
Reset	00		1	00101				
Access Type	Write, Read		Write, Read	Write, Read				

Bitfield	Bits	Description	Decode
PULL_UPDN_SEL	7:6	Buffer pull up/down configuration	0b00: None 0b01: Pull-up 0b10: Pull-down 0b11: Reserved
OUT_TYPE	5	Driver type selection	0b0: Open-drain 0b1: Push-pull
GPIO_TX_ID	4:0	GPIO ID for pin while transmitting	0bXXXXX: This GPIO transmit ID

GPIO_C (0x2CF)

Bit	7	6	5	4	3	2	1	0
Field	OVR_RES_CFG	–	–	GPIO_RX_ID[4:0]				
Reset	0	–	–	00101				
Access Type	Write, Read	–	–	Write, Read				

Bitfield	Bits	Description	Decode
OVR_RES_CFG	7	Override non-GPIO port function IO setting. When set, res_cfg and pull_updn_sel are effective when pin is configured as non-GPIO. When cleared, non-GPIO pin function determines IO type.	0b0: Non-GPIO function determines IO type when alternate function is selected 0b1: RES_CFG and PULL_UPDN_SEL determine IO type for non-GPIO configuration
GPIO_RX_ID	4:0	GPIO ID for pin while receiving	0bXXXXX: This GPIO receive ID

GPIO_A (0x2D0)

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Field	RES_CFG	TX_PRIO	TX_COMP_EN	GPIO_OUT	GPIO_IN	GPIO_RX_EN	GPIO_TX_EN	GPIO_OUT_DIS
Reset	1	0	0	1	1	0	0	1
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	Read Only	Write, Read	Write, Read	Write, Read

Bitfield	Bits	Description	Decode
RES_CFG	7	Resistor pull-up/pull-down strength	0b0: 40kΩ 0b1: 1MΩ
TX_PRIO	6	Priority for GPIO scheduling	0b0: Low priority 0b1: High priority
TX_COMP_EN	5	Jitter minimization compensation enable	0b0: Jitter compensation disabled 0b1: Jitter compensation enabled
GPIO_OUT	4	GPIO pin output drive value when gpio_rx_en=0	0b0: This GPIO pin output is driven to 0 0b1: This GPIO pin output is driven to 1
GPIO_IN	3	GPIO pin input level	0b0: This GPIO pin value is 0 0b1: This GPIO pin value is 1
GPIO_RX_EN	2	GPIO Out source control	0b0: This GPIO source disabled for GMSL2 reception 0b1: This GPIO source enabled for GMSL2 reception
GPIO_TX_EN	1	GPIO TX source control	0b0: This GPIO source disabled for GMSL2 transmission 0b1: This GPIO source enabled for GMSL2 transmission
GPIO_OUT_DIS	0	Disable GPIO output driver	0b0: Output driver enabled 0b1: Output driver disabled

GPIO_B (0x2D1)

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Field	PULL_UPDN_SEL[1:0]		OUT_TYPE	GPIO_TX_ID[4:0]				
Reset	00		1	00110				
Access Type	Write, Read		Write, Read	Write, Read				

Bitfield	Bits	Description	Decode
PULL_UPDN_SEL	7:6	Buffer pull up/down configuration	0b00: None 0b01: Pull-up 0b10: Pull-down 0b11: Reserved

Bitfield	Bits	Description	Decode
OUT_TYPE	5	Driver type selection	0b0: Open-drain 0b1: Push-pull
GPIO_TX_ID	4:0	GPIO ID for pin while transmitting	0bXXXXX: This GPIO transmit ID

GPIO_C (0x2D2)

Bit	7	6	5	4	3	2	1	0
Field	OVR_RES_CFG	–	–	GPIO_RX_ID[4:0]				
Reset	0	–	–	00110				
Access Type	Write, Read	–	–	Write, Read				

Bitfield	Bits	Description	Decode
OVR_RES_CFG	7	Override non-GPIO port function IO setting. When set, res_cfg and pull_updn_sel are effective when pin is configured as non-GPIO. When cleared, non-GPIO pin function determines IO type.	0b0: Non-GPIO function determines IO type when alternate function is selected 0b1: RES_CFG and PULL_UPDN_SEL determine IO type for non-GPIO configuration
GPIO_RX_ID	4:0	GPIO ID for pin while receiving	0bXXXXX: This GPIO receive ID

GPIO_A (0x2D3)

Bit	7	6	5	4	3	2	1	0
Field	RES_CFG	TX_PRIO	TX_COMP_EN	GPIO_OUT	GPIO_IN	GPIO_RX_EN	GPIO_TX_EN	GPIO_OUT_DIS
Reset	1	0	0	0	0	0	1	1
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	Read Only	Write, Read	Write, Read	Write, Read

Bitfield	Bits	Description	Decode
RES_CFG	7	Resistor pull-up/pull-down strength	0b0: 40kΩ 0b1: 1MΩ
TX_PRIO	6	Priority for GPIO scheduling	0b0: Low priority 0b1: High priority
TX_COMP_EN	5	Jitter minimization compensation enable	0b0: Jitter compensation disabled 0b1: Jitter compensation enabled
GPIO_OUT	4	GPIO pin output drive value when gpio_rx_en=0	0b0: This GPIO pin output is driven to 0 0b1: This GPIO pin output is driven to 1
GPIO_IN	3	GPIO pin input level	0b0: This GPIO pin value is 0 0b1: This GPIO pin value is 1
GPIO_RX_EN	2	GPIO Out source control	0b0: This GPIO source disabled for GMSL2

Bitfield	Bits	Description	Decode
			reception 0b1: This GPIO source enabled for GMSL2 reception
GPIO_TX_EN	1	GPIO TX source control	0b0: This GPIO source disabled for GMSL2 transmission 0b1: This GPIO source enabled for GMSL2 transmission
GPIO_OUT_DIS	0	Disable GPIO output driver	0b0: Output driver enabled 0b1: Output driver disabled

GPIO_B (0x2D4)

Bit	7	6	5	4	3	2	1	0
Field	PULL_UPDN_SEL[1:0]		OUT_TYPE	GPIO_TX_ID[4:0]				
Reset	10		1	00111				
Access Type	Write, Read		Write, Read	Write, Read				

Bitfield	Bits	Description	Decode
PULL_UPDN_SEL	7:6	Buffer pull up/down configuration	0b00: None 0b01: Pull-up 0b10: Pull-down 0b11: Reserved
OUT_TYPE	5	Driver type selection	0b0: Open-drain 0b1: Push-pull
GPIO_TX_ID	4:0	GPIO ID for pin while transmitting	0bXXXXX: This GPIO transmit ID

GPIO_C (0x2D5)

Bit	7	6	5	4	3	2	1	0
Field	OVR_RES_CFG	–	–	GPIO_RX_ID[4:0]				
Reset	0	–	–	00111				
Access Type	Write, Read	–	–	Write, Read				

Bitfield	Bits	Description	Decode
OVR_RES_CFG	7	Override non-GPIO port function IO setting. When set, res_cfg and pull_updn_sel are effective when pin is configured as non-GPIO. When cleared, non-GPIO pin function determines IO type.	0b0: Non-GPIO function determines IO type when alternate function is selected 0b1: RES_CFG and PULL_UPDN_SEL determine IO type for non-GPIO configuration
GPIO_RX_ID	4:0	GPIO ID for pin while receiving	0bXXXXX: This GPIO receive ID

GPIO_A (0x2D6)

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Field	RES_CFG	TX_PRIO	TX_COMP_EN	GPIO_OUT	GPIO_IN	GPIO_RX_EN	GPIO_TX_EN	GPIO_OUT_DIS
Reset	1	0	0	1	1	1	0	0
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	Read Only	Write, Read	Write, Read	Write, Read

Bitfield	Bits	Description	Decode
RES_CFG	7	Resistor pull-up/pull-down strength	0b0: 40kΩ 0b1: 1MΩ
TX_PRIO	6	Priority for GPIO scheduling	0b0: Low priority 0b1: High priority
TX_COMP_EN	5	Jitter minimization compensation enable	0b0: Jitter compensation disabled 0b1: Jitter compensation enabled
GPIO_OUT	4	GPIO pin output drive value when gpio_rx_en=0	0b0: This GPIO pin output is driven to 0 0b1: This GPIO pin output is driven to 1
GPIO_IN	3	GPIO pin input level	0b0: This GPIO pin value is 0 0b1: This GPIO pin value is 1
GPIO_RX_EN	2	GPIO Out source control	0b0: This GPIO source disabled for GMSL2 reception 0b1: This GPIO source enabled for GMSL2 reception
GPIO_TX_EN	1	GPIO TX source control	0b0: This GPIO source disabled for GMSL2 transmission 0b1: This GPIO source enabled for GMSL2 transmission
GPIO_OUT_DIS	0	Disable GPIO output driver	0b0: Output driver enabled 0b1: Output driver disabled

GPIO_B (0x2D7)

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Field	PULL_UPDN_SEL[1:0]		OUT_TYPE	GPIO_TX_ID[4:0]				
Reset	10		1	01000				
Access Type	Write, Read		Write, Read	Write, Read				

Bitfield	Bits	Description	Decode
PULL_UPDN_SEL	7:6	Buffer pull up/down configuration	0b00: None 0b01: Pull-up 0b10: Pull-down 0b11: Reserved

Bitfield	Bits	Description	Decode
OUT_TYPE	5	Driver type selection	0b0: Open-drain 0b1: Push-pull
GPIO_TX_ID	4:0	GPIO ID for pin while transmitting	0bXXXXX: This GPIO transmit ID

GPIO_C (0x2D8)

Bit	7	6	5	4	3	2	1	0
Field	OVR_RES_CFG	–	–	GPIO_RX_ID[4:0]				
Reset	0	–	–	01000				
Access Type	Write, Read	–	–	Write, Read				

Bitfield	Bits	Description	Decode
OVR_RES_CFG	7	Override non-GPIO port function IO setting. When set, res_cfg and pull_updn_sel are effective when pin is configured as non-GPIO. When cleared, non-GPIO pin function determines IO type.	0b0: Non-GPIO function determines IO type when alternate function is selected 0b1: RES_CFG and PULL_UPDN_SEL determine IO type for non-GPIO configuration
GPIO_RX_ID	4:0	GPIO ID for pin while receiving	0bXXXXX: This GPIO receive ID

GPIO_A (0x2D9)

Bit	7	6	5	4	3	2	1	0
Field	RES_CFG	TX_PRIO	TX_COMP_EN	GPIO_OUT	GPIO_IN	GPIO_RX_EN	GPIO_TX_EN	GPIO_OUT_DIS
Reset	1	0	0	0	0	0	0	1
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	Read Only	Write, Read	Write, Read	Write, Read

Bitfield	Bits	Description	Decode
RES_CFG	7	Resistor pull-up/pull-down strength	0b0: 40kΩ 0b1: 1MΩ
TX_PRIO	6	Priority for GPIO scheduling	0b0: Low priority 0b1: High priority
TX_COMP_EN	5	Jitter minimization compensation enable	0b0: Jitter compensation disabled 0b1: Jitter compensation enabled
GPIO_OUT	4	GPIO pin output drive value when gpio_rx_en=0	0b0: This GPIO pin output is driven to 0 0b1: This GPIO pin output is driven to 1
GPIO_IN	3	GPIO pin input level	0b0: This GPIO pin value is 0 0b1: This GPIO pin value is 1
GPIO_RX_EN	2	GPIO Out source control	0b0: This GPIO source disabled for GMSL2

Bitfield	Bits	Description	Decode
			reception 0b1: This GPIO source enabled for GMSL2 reception
GPIO_TX_EN	1	GPIO TX source control	0b0: This GPIO source disabled for GMSL2 transmission 0b1: This GPIO source enabled for GMSL2 transmission
GPIO_OUT_DIS	0	Disable GPIO output driver	0b0: Output driver enabled 0b1: Output driver disabled

GPIO_B (0x2DA)

Bit	7	6	5	4	3	2	1	0
Field	PULL_UPDN_SEL[1:0]		OUT_TYPE	GPIO_TX_ID[4:0]				
Reset	00		1	01001				
Access Type	Write, Read		Write, Read	Write, Read				

Bitfield	Bits	Description	Decode
PULL_UPDN_SEL	7:6	Buffer pull up/down configuration	0b00: None 0b01: Pull-up 0b10: Pull-down 0b11: Reserved
OUT_TYPE	5	Driver type selection	0b0: Open-drain 0b1: Push-pull
GPIO_TX_ID	4:0	GPIO ID for pin while transmitting	0bXXXXX: This GPIO transmit ID

GPIO_C (0x2DB)

Bit	7	6	5	4	3	2	1	0
Field	OVR_RES_CFG	–	–	GPIO_RX_ID[4:0]				
Reset	0	–	–	01001				
Access Type	Write, Read	–	–	Write, Read				

Bitfield	Bits	Description	Decode
OVR_RES_CFG	7	Override non-GPIO port function IO setting. When set, res_cfg and pull_updn_sel are effective when pin is configured as non-GPIO. When cleared, non-GPIO pin function determines IO type.	0b0: Non-GPIO function determines IO type when alternate function is selected 0b1: RES_CFG and PULL_UPDN_SEL determine IO type for non-GPIO configuration
GPIO_RX_ID	4:0	GPIO ID for pin while receiving	0bXXXXX: This GPIO receive ID

GPIO_A (0x2DC)

Bit	7	6	5	4	3	2	1	0
Field	RES_CFG	TX_PRIO	TX_COMP_EN	GPIO_OUT	GPIO_IN	GPIO_RX_EN	GPIO_TX_EN	GPIO_OUT_DIS
Reset	1	0	0	1	1	0	0	1
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	Read Only	Write, Read	Write, Read	Write, Read

Bitfield	Bits	Description	Decode
RES_CFG	7	Resistor pull-up/pull-down strength	0b0: 40kΩ 0b1: 1MΩ
TX_PRIO	6	Priority for GPIO scheduling	0b0: Low priority 0b1: High priority
TX_COMP_EN	5	Jitter minimization compensation enable	0b0: Jitter compensation disabled 0b1: Jitter compensation enabled
GPIO_OUT	4	GPIO pin output drive value when gpio_rx_en=0	0b0: This GPIO pin output is driven to 0 0b1: This GPIO pin output is driven to 1
GPIO_IN	3	GPIO pin input level	0b0: This GPIO pin value is 0 0b1: This GPIO pin value is 1
GPIO_RX_EN	2	GPIO Out source control	0b0: This GPIO source disabled for GMSL2 reception 0b1: This GPIO source enabled for GMSL2 reception
GPIO_TX_EN	1	GPIO TX source control	0b0: This GPIO source disabled for GMSL2 transmission 0b1: This GPIO source enabled for GMSL2 transmission
GPIO_OUT_DIS	0	Disable GPIO output driver	0b0: Output driver enabled 0b1: Output driver disabled

GPIO_B (0x2DD)

Bit	7	6	5	4	3	2	1	0
Field	PULL_UPDN_SEL[1:0]		OUT_TYPE	GPIO_TX_ID[4:0]				
Reset	00		1	0xA				
Access Type	Write, Read		Write, Read	Write, Read				

Bitfield	Bits	Description	Decode
PULL_UPDN_SEL	7:6	Buffer pull up/down configuration	0b00: None 0b01: Pull-up 0b10: Pull-down 0b11: Reserved

Bitfield	Bits	Description	Decode
OUT_TYPE	5	Driver type selection	0b0: Open-drain 0b1: Push-pull
GPIO_TX_ID	4:0	GPIO ID for pin while transmitting	0bXXXXX: This GPIO transmit ID

GPIO_C (0x2DE)

Bit	7	6	5	4	3	2	1	0
Field	OVR_RES_CFG	–	–	GPIO_RX_ID[4:0]				
Reset	0	–	–	0xA				
Access Type	Write, Read	–	–	Write, Read				

Bitfield	Bits	Description	Decode
OVR_RES_CFG	7	Override non-GPIO port function IO setting. When set, res_cfg and pull_updn_sel are effective when pin is configured as non-GPIO. When cleared, non-GPIO pin function determines IO type.	0b0: Non-GPIO function determines IO type when alternate function is selected 0b1: RES_CFG and PULL_UPDN_SEL determine IO type for non-GPIO configuration
GPIO_RX_ID	4:0	GPIO ID for pin while receiving	0bXXXXX: This GPIO receive ID

FRONTTOP_0 (0x308)

Bit	7	6	5	4	3	2	1	0
Field	–	enable_line_info	START_PORTB	START_PORTA	CLK_SELU	CLK_SELZ	CLK_SELY	CLK_SELX
Reset	–	1	1	1	1	1	0	0
Access Type	–	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read

Bitfield	Bits	Description	Decode
enable_line_info	6	Enable sending line start info frames	0b0: Line start info frames disabled 0x1: Line start info frames enabled
START_PORTB	5	Enable CSI Port B	0b0: CSI on Port B disabled 0x1: CSI on Port B enabled
START_PORTA	4	Enable CSI port A	0b0: CSI on Port A disabled 0x1: CSI on Port A enabled
CLK_SELU	3	CSI port selection for video pipeline U	0b0: Port A 0b1: Port B
CLK_SELZ	2	CSI port selection for video pipeline Z	0b0: Port A 0b1: Port B
CLK_SELY	1	CSI port selection for video pipeline Y	0b0: Port A

Bitfield	Bits	Description	Decode
			0b1: Port B
CLK_SELX	0	CSI port selection for video pipeline X	0b0: Port A 0b1: Port B

FRONTTOP_1 (0x309)

Bit	7	6	5	4	3	2	1	0
Field	VC_SELX_L[7:0]							
Reset	0xFF							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
VC_SELX_L	7:0	Virtual channel filter for VID PIPE X - low byte	0xXX: Virtual channel filter low byte

FRONTTOP_2 (0x30A)

Bit	7	6	5	4	3	2	1	0
Field	VC_SELX_H[7:0]							
Reset	0xFF							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
VC_SELX_H	7:0	Virtual channel filter for VID PIPE X - high byte	0xXX: Virtual channel filter high byte

FRONTTOP_3 (0x30B)

Bit	7	6	5	4	3	2	1	0
Field	VC_SELY_L[7:0]							
Reset	0xFF							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
VC_SELY_L	7:0	Virtual channel filter for VID PIPE Y - low byte	0xXX: Virtual channel filter low byte

FRONTTOP_4 (0x30C)

Bit	7	6	5	4	3	2	1	0
Field	VC_SELY_H[7:0]							
Reset	0xFF							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
VC_SELY_H	7:0	Virtual channel filter for VID PIPE Y - high byte	0xXX: Virtual channel filter high byte

FRONTTOP_5 (0x30D)

Bit	7	6	5	4	3	2	1	0
Field	VC_SELZ_L[7:0]							
Reset	0xFF							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
VC_SELZ_L	7:0	Virtual channel filter for VID PIPE Z - low byte	0xXX: Virtual channel filter low byte

FRONTTOP_6 (0x30E)

Bit	7	6	5	4	3	2	1	0
Field	VC_SELZ_H[7:0]							
Reset	0xFF							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
VC_SELZ_H	7:0	Virtual channel filter for VID PIPE Z - high byte	0xXX: Virtual channel filter high byte

FRONTTOP_7 (0x30F)

Bit	7	6	5	4	3	2	1	0
Field	VC_SELU_L[7:0]							
Reset	0xFF							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
VC_SELU_L	7:0	Virtual channel filter for VID PIPE U - low byte	0xXX: Virtual channel filter low byte

FRONTTOP_8 (0x310)

Bit	7	6	5	4	3	2	1	0
Field	VC_SELU_H[7:0]							
Reset	0xFF							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
VC_SELU_H	7:0	Virtual channel filter for VID PIPE U - high byte	0xXX: Virtual channel filter high byte

FRONTTOP_9 (0x311)

Bit	7	6	5	4	3	2	1	0
Field	START_PORTBU	START_PORTBZ	START_PORTBY	START_PORTBX	START_PORTAU	START_PORTAZ	START_PORTAY	START_PORTAX
Reset	0	1	0	1	0	1	0	1
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read

Bitfield	Bits	Description	Decode
START_PORTBU	7	Start video pipe U from CSI port B	0b0: Video not started 0b1: Start video
START_PORTBZ	6	Start video pipe Z from CSI port B	0b0: Video not started 0b1: Start video
START_PORTBY	5	Start video pipe Y from CSI port B	0b0: Video not started 0b1: Start video
START_PORTBX	4	Start video pipe X from CSI port B	0b0: Video not started 0b1: Start video
START_PORTAU	3	Start video pipe U from CSI port A	0b0: Video not started 0b1: Start video
START_PORTAZ	2	Start video pipe Z from CSI port A	0b0: Video not started 0b1: Start video
START_PORTAY	1	Start video pipe Y from CSI port A	0b0: Video not started 0b1: Start video
START_PORTAX	0	Start vidoe pipe X from CSI port A	0b0: Video not started 0b1: Start video

FRONTTOP_10 (0x312)

Bit	7	6	5	4	3	2	1	0
Field	–	–	–	–	bpp8dblu	bpp8dblz	bpp8dbly	bpp8dblx
Reset	–	–	–	–	0	0	0	0
Access Type	–	–	–	–	Write, Read	Write, Read	Write, Read	Write, Read

Bitfield	Bits	Description	Decode
bpp8dblu	3	Send 8-bit pixels as 16-bit on video pipe U	0b0: Send as 8-bit pixels 0b1: Send 8-bit pixels as 16-bit

Bitfield	Bits	Description	Decode
bpp8dblz	2	Send 8-bit pixels as 16-bit on video pipe Z	0b0: Send as 8-bit pixels 0b1: Send 8-bit pixels as 16-bit
bpp8dbly	1	Send 8-bit pixels as 16-bit on video pipe Y	0b0: Send as 8-bit pixels 0b1: Send 8-bit pixels as 16-bit
bpp8dblx	0	Send 8-bit pixels as 16-bit on video pipe X	0b0: Send as 8-bit pixels 0b1: Send 8-bit pixels as 16-bit

FRONTTOP_11 (0x313)

Bit	7	6	5	4	3	2	1	0
Field	bpp12dblu	bpp12dblz	bpp12dbly	bpp12dblx	bpp10dblu	bpp10dblz	bpp10dbly	bpp10dblx
Reset	0	0	0	0	0	0	0	0
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read

Bitfield	Bits	Description	Decode
bpp12dblu	7	Send 12-bit pixels as 24-bit on video pipe U	0b0: Send as 12-bit pixels 0b1: Send 12-bit pixels as 24-bit
bpp12dblz	6	Send 12-bit pixels as 24-bit on video pipe Z	0b0: Send as 12-bit pixels 0b1: Send 12-bit pixels as 24-bit
bpp12dbly	5	Send 12-bit pixels as 24-bit on video pipe Y	0b0: Send as 12-bit pixels 0b1: Send 12-bit pixels as 24-bit
bpp12dblx	4	Send 12-bit pixels as 24-bit on video pipe X	0b0: Send as 12-bit pixels 0b1: Send 12-bit pixels as 24-bit
bpp10dblu	3	Send 10-bit pixels as 20-bit on video pipe U	0b0: Send as 10-bit pixels 0b1: Send 10-bit pixels as 20-bit
bpp10dblz	2	Send 10-bit pixels as 20-bit on video pipe Z	0b0: Send as 10-bit pixels 0b1: Send 10-bit pixels as 20-bit
bpp10dbly	1	Send 10-bit pixels as 20-bit on video pipe Y	0b0: Send as 10-bit pixels 0b1: Send 10-bit pixels as 20-bit
bpp10dblx	0	Send 10-bit pixels as 20-bit on video pipe X	0b0: Send as 10-bit pixels 0b1: Send 10-bit pixels as 20-bit

FRONTTOP_12 (0x314)

Bit	7	6	5	4	3	2	1	0
Field	–	mem_dt1_selx[6:0]						
Reset	–	0000000						
Access Type	–	Write, Read						

Bitfield	Bits	Description	Decode
mem_dt1_selx	6:0	Select designated datatype to route to video pipeline X (MSB is enable)	0xXXXXXXX: Datatype selected to route to video pipeline

FRONTTOP_13 (0x315)

Bit	7	6	5	4	3	2	1	0
Field	–	mem_dt2_selx[6:0]						
Reset	–	0000000						
Access Type	–	Write, Read						

Bitfield	Bits	Description	Decode
mem_dt2_selx	6:0	Select designated datatype to route to video pipeline X (MSB is enable)	0xXXXXXXX: Datatype selected to route to video pipeline

FRONTTOP_14 (0x316)

Bit	7	6	5	4	3	2	1	0
Field	–	mem_dt1_sely[6:0]						
Reset	–	0000000						
Access Type	–	Write, Read						

Bitfield	Bits	Description	Decode
mem_dt1_sely	6:0	Select designated datatype to route to video pipeline Y (MSB is enable)	0xXXXXXXX: Datatype selected to route to video pipeline

FRONTTOP_15 (0x317)

Bit	7	6	5	4	3	2	1	0
Field	–	mem_dt2_sely[6:0]						
Reset	–	0000000						
Access Type	–	Write, Read						

Bitfield	Bits	Description	Decode
mem_dt2_sely	6:0	Select designated datatype to route to video pipeline Y (MSB is enable)	0xXXXXXXX: Datatype selected to route to video pipeline

FRONTTOP_16 (0x318)

Bit	7	6	5	4	3	2	1	0
Field	–	mem_dt1_selz[6:0]						
Reset	–	0000000						
Access Type	–	Write, Read						

Bitfield	Bits	Description	Decode
mem_dt1_selz	6:0	Select designated datatype to route to video pipeline Z (MSB is enable)	0xXXXXXXX: Datatype selected to route to video pipeline

FRONTTOP_17 (0x319)

Bit	7	6	5	4	3	2	1	0
Field	–	mem_dt2_selz[6:0]						
Reset	–	0000000						
Access Type	–	Write, Read						

Bitfield	Bits	Description	Decode
mem_dt2_selz	6:0	Select designated datatype to route to video pipeline Z (MSB is enable)	0xXXXXXXX: Datatype selected to route to video pipeline

FRONTTOP_18 (0x31A)

Bit	7	6	5	4	3	2	1	0
Field	–	mem_dt1_selu[6:0]						
Reset	–	0000000						
Access Type	–	Write, Read						

Bitfield	Bits	Description	Decode
mem_dt1_selu	6:0	Select designated datatype to route to video pipeline U (MSB is enable)	0xXXXXXXX: Datatype selected to route to video pipeline

FRONTTOP_19 (0x31B)

Bit	7	6	5	4	3	2	1	0
Field	–	mem_dt2_selu[6:0]						
Reset	–	0000000						
Access Type	–	Write, Read						

Bitfield	Bits	Description	Decode
mem_dt2_selu	6:0	Select designated datatype to route to video pipeline U (MSB is enable)	0xXXXXXXX: Datatype selected to route to video pipeline

FRONTTOP_20 (0x31C)

Bit	7	6	5	4	3	2	1	0
Field	soft_dtx_en	soft_vcx_en	soft_bppx_en	soft_bppx[4:0]				
Reset	0	0	0	11000				
Access Type	Write, Read	Write, Read	Write, Read	Write, Read				

Bitfield	Bits	Description	Decode
soft_dtx_en	7	Datatype software override enable for video pipeline X	0b0: Software override disabled 0b1: Software override enabled
soft_vcx_en	6	Virtual channel software override enable for video pipeline X	0b0: Software override disabled 0b1: Software override enabled
soft_bppx_en	5	BPP software override enable for video pipeline X	0b0: Software override disabled 0b1: Software override enabled
soft_bppx	4:0	Software override of BPP on video pipeline X	0bXXXXX: Software override value

FRONTTOP_21 (0x31D)

Bit	7	6	5	4	3	2	1	0
Field	soft_dty_en	soft_vcy_en	soft_bppy_en	soft_bppy[4:0]				
Reset	0	0	0	11000				
Access Type	Write, Read	Write, Read	Write, Read	Write, Read				

Bitfield	Bits	Description	Decode
soft_dty_en	7	Datatype software override enable for video pipeline Y	0b0: Software override disabled 0b1: Software override enabled
soft_vcy_en	6	Virtual channel software override enable for video pipeline Y	0b0: Software override disabled 0b1: Software override enabled
soft_bppy_en	5	BPP software override enable for video pipeline Y	0b0: Software override disabled 0b1: Software override enabled
soft_bppy	4:0	Software override of BPP on video pipeline Y	0bXXXXX: Software override value

FRONTTOP_22 (0x31E)

Bit	7	6	5	4	3	2	1	0
Field	soft_dtz_en	soft_vcz_en	soft_bppz_en	soft_bppz[4:0]				
Reset	0	0	0	11000				
Access Type	Write, Read	Write, Read	Write, Read	Write, Read				

Bitfield	Bits	Description	Decode
soft_dtz_en	7	Datatype software override enable for video pipeline Z	0b0: Software override disabled 0b1: Software override enabled
soft_vcz_en	6	Virtual channel software override enable for video pipeline Z	0b0: Software override disabled 0b1: Software override enabled
soft_bppz_en	5	BPP software override enable for video pipeline Z	0b0: Software override disabled 0b1: Software override enabled
soft_bppz	4:0	Software override of BPP on video pipeline Z	0bXXXXX: Software override value

FRONTTOP_23 (0x31F)

Bit	7	6	5	4	3	2	1	0
Field	soft_dtu_en	soft_vcu_en	soft_bppu_en	soft_bppu[4:0]				
Reset	0	0	0	11000				
Access Type	Write, Read	Write, Read	Write, Read	Write, Read				

Bitfield	Bits	Description	Decode
soft_dtu_en	7	Datatype software override enable for video pipeline U	0b0: Software override disabled 0b1: Software override enabled
soft_vcu_en	6	Virtual channel software override enable for video pipeline U	0b0: Software override disabled 0b1: Software override enabled
soft_bppu_en	5	BPP software override enable for video pipeline U	0b0: Software override disabled 0b1: Software override enabled
soft_bppu	4:0	Software override of BPP on video pipeline U	0bXXXXX: Software override value

FRONTTOP_24 (0x320)

Bit	7	6	5	4	3	2	1	0
Field	soft_vcu[1:0]		soft_vcz[1:0]		soft_vcy[1:0]		soft_vcx[1:0]	
Reset	00		00		00		00	
Access Type	Write, Read		Write, Read		Write, Read		Write, Read	

Bitfield	Bits	Description	Decode
soft_vcu	7:6	Virtual channel software override for video pipeline U	0bXX: Software override value
soft_vcz	5:4	Virtual channel software override for video pipeline Z	0bXX: Software override value
soft_vcy	3:2	Virtual channel software override for video pipeline Y	0bXX: Software override value
soft_vcx	1:0	Virtual channel software override for video pipeline X	0bXX: Software override value

FRONTTOP_25 (0x321)

Bit	7	6	5	4	3	2	1	0
Field	–	–	soft_dtx[5:0]					
Reset	–	–	110000					
Access Type	–	–	Write, Read					

Bitfield	Bits	Description	Decode
soft_dtx	5:0	Datatype software override for video channel X	0bXXXXXX: Software override value

FRONTTOP_26 (0x322)

Bit	7	6	5	4	3	2	1	0
Field	–	–	soft_dty[5:0]					
Reset	–	–	110000					
Access Type	–	–	Write, Read					

Bitfield	Bits	Description	Decode
soft_dty	5:0	Datatype software override for video channel Y	0bXXXXXX: Software override value

FRONTTOP_27 (0x323)

Bit	7	6	5	4	3	2	1	0
Field	–	–	soft_dtz[5:0]					
Reset	–	–	110000					
Access Type	–	–	Write, Read					

Bitfield	Bits	Description	Decode
soft_dtz	5:0	Datatype software override for video channel Z	0bXXXXXX: Software override value

FRONTTOP_28 (0x324)

Bit	7	6	5	4	3	2	1	0
Field	–	–	soft_dtu[5:0]					
Reset	–	–	110000					
Access Type	–	–	Write, Read					

Bitfield	Bits	Description	Decode
soft_dtu	5:0	Datatype software override for video channel U	0bXXXXXX: Software override value

DV0 (0x32A)

Bit	7	6	5	4	3	2	1	0
Field	RSVD	DV_SWP_AB	LINE_ALT	–	–	DV_CONV	DV_SPL	DV_EN
Reset	0	0	0	–	–	1	0	0
Access Type	Read Only	Write, Read	Write, Read	–	–	Write, Read	Write, Read	Write, Read

Bitfield	Bits	Description	Decode
RSVD	7	Dualview pixel per line count stable, processor can start	
DV_SWP_AB	6	Swap video data going to video transmit pipelines	0b0: Normal video split between SIOA and SIOB 0b1: Inverts video split between SIOA and SIOB
LINE_ALT	5	Enable line alternating mode	0b0: Line alternating mode disabled 0b1: Line alternating mode enabled
DV_CONV	2	Dual View Conversion. When enabled, side-by-side input is converted to pixel interleaved.	0b0: Bypass 0b1: Convert
DV_SPL	1	Enables Dual View split mode	0b0: Dual view split disabled 0b1: Dual view split enabled
DV_EN	0	Dualview processor enable	0b0: Dual view processing disabled 0b1: Dual view processing enabled

MIPI_RX0 (0x330)

Bit	7	6	5	4	3	2	1	0
Field	–	–	ctrl1_vc_map_en	ctrl0_vc_map_en	mipi_rx_reset	phy_config[2:0]		
Reset	–	–	0	0	0	000		
Access Type	–	–	Write, Read	Write, Read	Write, Read	Write, Read		

Bitfield	Bits	Description	Decode
ctrl1_vc_map_en	5	Virtual channel mapping enable	0x0: Disable virtual channel mapping 0x1: Enable virtual channel mapping
ctrl0_vc_map_en	4	Virtual channel mapping enable	0x0: Disable virtual channel mapping 0x1: Enable virtual channel mapping
mipi_rx_reset	3	Reset MIPI RX receiver	0b0: Do not reset MIPI RX 0b1: Reset MIPI RX
phy_config	2:0	MIPI Rx PHY configuration. Configure CSI port as 1x4, 2x4, or 2x2.	0b000: 1x4. One Port with four data lanes 0b001: 2x2, only A (1x2). Port A with two data lanes enabled. 0b010: 2x2, only B (1x2). Port B with two data lanes enabled. 0b011: 2x2, A and B. Both Port A and Port B with

Bitfield	Bits	Description	Decode
			two data lanes each port enabled 0b100: 2x4, only A (1x4). Two Ports with four data lanes each. Only Port A enabled. 0b101: 2x4, only B (1x4).Two Ports with four data lanes each. Only Port B enabled. 0b110: 2x4, A and B. Two Ports with four data lanes each. Both Port A and Port B enabled. 0b111: Reserved

MIPI_RX1 (0x331)

Bit	7	6	5	4	3	2	1	0
Field	ctrl1_vcx_en	ctrl1_deskew en	ctrl1_num_lanes[1:0]		ctrl0_vcx_en	ctrl0_deskew en	ctrl0_num_lanes[1:0]	
Reset	0	0	11		0	0	11	
Access Type	Write, Read	Write, Read	Write, Read		Write, Read	Write, Read	Write, Read	

Bitfield	Bits	Description	Decode
ctrl1_vcx_en	7	Enable the extended Virtual Channels feature	0b0: Extended VC disabled 0b1: Extended VC enabled
ctrl1_deskewen	6	Enable the deskew calibration for 1.5Gbps and above for Port B	0x0: Deskew calibration disabled 0x1: Deskew calibration enabled
ctrl1_num_lanes	5:4	Select number of data lanes for Port B	0b00: one data lane 0b01: two data lanes 0b10: three data lanes 0b11: four data lanes
ctrl0_vcx_en	3	Enable the extended Virtual Channels feature	0b0: Extended VC disabled 0b1: Extended VC enabled
ctrl0_deskewen	2	Enable the deskew calibration for 1.5Gbps and above for Port A	0b0: Deskew calibration disabled 0b1: Deskew calibration enabled
ctrl0_num_lanes	1:0	Select number of data lanes for Port A	0b00: one data lane 0b01: two data lanes 0b10: three data lanes 0b11: four data lanes

MIPI_RX2 (0x332)

Bit	7	6	5	4	3	2	1	0
Field	phy1_lane_map[3:0]				phy0_lane_map[3:0]			
Reset	0xE				0xE			
Access Type	Write, Read				Write, Read			

Bitfield	Bits	Description	Decode
phy1_lane_map	7:4	PHY1 lane mapping For 2X4 and 2X2 configuration, Bit[5:4] map PHY1 data lane 0 to Port A. Bit[7:6] map PHY1 data lane 1 to Port A. Works together with phy0_lane_map register in 2X4 configuration. PHY0 and PHY1 lane maps should be exclusive For 1X4 configuration, Bit[5:4] map PHY1 data lane 0 to single port DSI. Bit[7:6] map PHY1 data lane 1 to single port DSI. Works together with phy2_lane_map register. PHY1 and PHY2 lane maps should be exclusive.	0bXX00: 2X4 and 2X2: Map PHY1 D0 to Port A D0; 1X4: Map PHY1 D0 to DSI 0 0bXX01: 2X4 and 2X2: Map PHY1 D0 to Port A D1; 1X4: Map PHY1 D0 to DSI 1 0bXX10: 2x4: Map PHY1 D0 to Port A D2; 1X4: Map PHY1 DO to DSI 2 0bXX11: 2x4: Map PHY1 D0 to Port A D3; 1X4: Map PHY1D0 to DSI 3 0b00XX: 2X4 and 2X2: Map PHY1 1 to Port A D0; 1X4: Map PHY1 D1 to DSI 0 0b01XX: 2x4 and 2X2: Map PHY1 D1 to Port A D1 1X4: Map PHY1 D1 to DSI 1 0b10XX: 2x4: Map PHY1 D1 to Port A D2; 1X4: Map PHY1 D1 to DSI 2 0b11XX: 2x4: Map PHY1 D1 to Port A D3; 1X4: Map PHY1 D1 to DSI 3
phy0_lane_map	3:0	PHY0 to Port A lane mapping in 2X4 configuration. Works together with phy1_lane_map register. PHY0 and PHY1 lane maps should be exclusive. Bit[1:0] map PHY0 data lane 0. Bit[3:2] map PHY0 data lane 1.	0bXX00: Map PHY0 D0 to Port A D0 0bXX01: Map PHY0 D0 to Port A D1 0bXX10: Map PHY0 D0 to Port A D2 0bXX11: Map PHY0 D0 to Port A D3 0b00XX: Map PHY0 D1 to Port A D0 0b01XX: Map PHY0 D1 to Port A D1 0b10XX: Map PHY0 D1 to Port A D2 0b11XX: Map PHY0 D1 to Port A D3

MIPI_RX3 (0x333)

Bit	7	6	5	4	3	2	1	0
Field	phy3_lane_map[3:0]				phy2_lane_map[3:0]			
Reset	0xE				0100			
Access Type	Write, Read				Write, Read			

Bitfield	Bits	Description	Decode
phy3_lane_map	7:4	PHY3 to Port B lane mapping in 2X4 configuration. Works together with phy2_lane_map register. PHY2 and PHY3 lane maps should be exclusive. Bit[5:4] map PHY3 data lane 0. Bit[7:6] map PHY3 data lane 1	0bXX00: Map PHY3 D0 to Port B D0 0bXX01: Map PHY3 D0 to Port B D1 0bXX10: Map PHY3 D0 to Port B D2 0bXX11: Map PHY3 D0 to Port B D3 0bXX00: Map PHY3 D1 to Port B D0 0bXX00: Map PHY3 D1 to Port B D1 0bXX00: Map PHY3 D1 to Port B D2 0bXX00: Map PHY3 D1 to Port B D3
phy2_lane_map	3:0	PHY2 lane mapping For 2X4 and 2X2 configuration, Bit[1:0] map PHY2 data lane 0 to Port B. Bit[3:2] map PHY2 data lane 1 to PORT B. For 2X4 configuration these bits work together with phy3_lane_map register. PHY2 and PHY3	0bXX00: 2x4 and 2X2: Map PHY2 D0 to Port B D0; 1X4: Map PHY1 D0 to DSI 0 0bXX01: 2x4 and 2X2: Map PHY2 D0 to Port B D1; 1x4: Map PHY1 D0 to DSI 1 0bXX10: 2x4 and 2X2: Map PHY2 D0 to Port B D2; 1X4 and 2X2: Map PHY1 DO to DSI 2

Bitfield	Bits	Description	Decode
		lane maps should be exclusive. For 1X4 configuration, Bit[1:0] map PHY2 data lane 0 to single port DSI. Bit[3:2] map PHY2 data lane 1 to single port DSI. Works together with phy1_lane_map register. PHY1 and PH2 lane maps should be exclusive.	0bXX11: 2X4 and 2X2: Map PHY2 D0 to Port B D3; 1X4: Map PHY1D0 to DSI 3 0b00XX: 2x4 and 2X2: Map PHY2 1 to Port B D0; 1x4: Map PHY1 D1 to DSI 0 0b01XX: 2x4 and 2X2: Map PHY2 D1 to Port B D1 1X4: Map PHY1 D1 to DSI 1 0b10XX: 2x4 and 2X2: Map PHY2 D1 to Port B D2; 1X4: Map PHY1 D1 to DSI 2 0b11XX: 2x4 and 2X2: Map PHY2 D1 to Port B D3; 1X4: Map PHY1 D1 to DSI 3

MIPI_RX4 (0x334)

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Field	–	phy1_pol_map[2:0]			–	phy0_pol_map[2:0]		
Reset	–	000			–	000		
Access Type	–	Write, Read			–	Write, Read		

Bitfield	Bits	Description	Decode
phy1_pol_map	6:4	PHY1 Lane Polarity Setting	0bXX0: Normal Polarity for data lane 0 0bXX1: Inverse Polarity for data lane 0 0bX0X: Normal Polarity for data lane 1 0bX1X: Inverse Polarity for data lane 1 0b0XX: Normal Polarity for clock lane 0b1XX: Inverse Polarity for clock lane
phy0_pol_map	2:0	PHY0 Lane Polarity Setting	0bXX0: Normal Polarity for data lane 0 0bXX1: Inverse Polarity for data lane 0 0bX0X: Normal Polarity for data lane 1 0bX1X: Inverse Polarity for data lane 1 0b0XX: Normal Polarity for clock lane 0b1XX: Inverse Polarity for clock lane

MIPI_RX5 (0x335)

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Field	–	phy3_pol_map[2:0]			–	phy2_pol_map[2:0]		
Reset	–	000			–	000		
Access Type	–	Write, Read			–	Write, Read		

Bitfield	Bits	Description	Decode
phy3_pol_map	6:4	PHY3 Lane Polarity Setting	0bXX0: Normal Polarity for data lane 0 0bXX1: Inverse Polarity for data lane 0 0bX0X: Normal Polarity for data lane 1 0bX1X: Inverse Polarity for data lane 1 0b0XX: Normal Polarity for clock lane

Bitfield	Bits	Description	Decode
			0b1XX: Inverse Polarity for clock lane
phy2_pol_map	2:0	PHY2 Lane Polarity Setting	0bXX0: Normal Polarity for data lane 0 0bXX1: Inverse Polarity for data lane 0 0bX0X: Normal Polarity for data lane 1 0bX1X: Inverse Polarity for data lane 1 0b0XX: Normal Polarity for clock lane 0b1XX: Inverse Polarity for clock lane

MIPI_RX8 (0x338)

Bit	7	6	5	4	3	2	1	0
Field	t_term_en[1:0]		t_hs_settle[1:0]		t_clk_miss[1:0]		t_clk_settle[1:0]	
Reset	01		01		01		01	
Access Type	Write, Read		Write, Read		Write, Read		Write, Read	

Bitfield	Bits	Description	Decode
t_term_en	7:6	Set typical DPHY Tclk_term_en and Ths_term_en timing in ns	0b00: 13 0b01: 20 0b10: 26 0b11: 33
t_hs_settle	5:4	Set typical DPHY hs_settle timing in ns	0b00: 86 0b01: 93 0b10: 120 0b11: 146
t_clk_miss	3:2	Set typical DPHY Tclk_miss timing in ns	0b00: disabled 0b01: 26 0b10: 40 0b11: 66
t_clk_settle	1:0	Set typical DPHY Tclk_settle timing in ns	0b00: 100 0b01: 160 0b10: 226 0b11: 293

MIPI_RX9 (0x339)

Bit	7	6	5	4	3	2	1	0
Field	–	–	–	phy0_lp_err[4:0]				
Reset	–	–	–	00000				
Access Type	–	–	–	Read Clears All				

Bitfield	Bits	Description	Decode
phy0_lp_err	4:0	Phy0 LP status	0bXXXX1: Unrecognized Escape command

Bitfield	Bits	Description	Decode
			received from data lane D0 0bXXX1X: Unrecognized Escape command received from CLK lane 0bXX1XX: Invalid line sequence detected from data lane D0 0bX1XXX: Invalid line sequence detected from data lane D1 0b1XXXX: Invalid line sequence detected from CLK lane

MIPI_RX10 (0x33A)

Bit	7	6	5	4	3	2	1	0
Field	phy0_hs_err[7:0]							
Reset	00000000							
Access Type	Read Clears All							

Bitfield	Bits	Description	Decode
phy0_hs_err	7:0	PHY0 high-speed status	0bXXXXXXX1: HS sync pattern with 2 or more bit errors detected on data lane D0 0bXXXXXX1X: HS sync pattern with 2 or more bit errors detected on data lane D1 0bXXXXX1XX: HS sync pattern with 1 bit error detected on data lane D0 0bXXXX1XXX: HS sync pattern with 1 bit error detected on data lane D1 0bXXX1XXXX: High speed receiver skew calibration failed on data lane D1 0bXX1XXXXX: High speed receiver skew calibration failed on data lane D0 0bX1XXXXXX: High speed receiver skew calibration run on data lane D1 0b1XXXXXXX: High speed receiver skew calibration run on data lane D0

MIPI_RX11 (0x33B)

Bit	7	6	5	4	3	2	1	0
Field	–	–	–	phy1_lp_err[4:0]				
Reset	–	–	–	00000				
Access Type	–	–	–	Read Clears All				

Bitfield	Bits	Description	Decode
phy1_lp_err	4:0	Phy1 LP status	0bXXXX1: Unrecognized Escape command received from data lane D0 0bXXX1X: Unrecognized Escape command

Bitfield	Bits	Description	Decode
			received from CLK lane 0bXX1XX: Invalid line sequence detected from data lane D0 0bX1XXX: Invalid line sequence detected from data lane D1 0b1XXXX: Invalid line sequence detected from CLK lane

MIPI_RX12 (0x33C)

Bit	7	6	5	4	3	2	1	0
Field	phy1_hs_err[7:0]							
Reset	00000000							
Access Type	Read Clears All							

Bitfield	Bits	Description	Decode
phy1_hs_err	7:0	PHY1 high-speed status	0bXXXXXXX1: HS sync pattern with 2 or more bit errors detected on data lane D0 0bXXXXXX1X: HS sync pattern with 2 or more bit errors detected on data lane D1 0bXXXXX1XX: HS sync pattern with 1 bit error detected on data lane D0 0bXXXX1XXX: HS sync pattern with 1 bit error detected on data lane D1 0bXXX1XXXX: High speed receiver skew calibration failed on data lane D1 0bXX1XXXXX: High speed receiver skew calibration failed on data lane D0 0bX1XXXXXX: High speed receiver skew calibration run on data lane D1 0b1XXXXXXX: High speed receiver skew calibration run on data lane D0

MIPI_RX13 (0x33D)

Bit	7	6	5	4	3	2	1	0
Field	–	–	–	phy2_lp_err[4:0]				
Reset	–	–	–	00000				
Access Type	–	–	–	Read Clears All				

Bitfield	Bits	Description	Decode
phy2_lp_err	4:0	Phy2 LP status	0bXXXX1: Unrecognized Escape command received from data lane D0 0bXXX1X: Unrecognized Escape command received from CLK lane 0bXX1XX: Invalid line sequence detected from data

Bitfield	Bits	Description	Decode
			lane D0 0bX1XXX: Invalid line sequence detected from data lane D1 0b1XXXX: Invalid line sequence detected from CLK lane

MIPI_RX14 (0x33E)

Bit	7	6	5	4	3	2	1	0
Field	phy2_hs_err[7:0]							
Reset	00000000							
Access Type	Read Clears All							

Bitfield	Bits	Description	Decode
phy2_hs_err	7:0	PHY2 high-speed status	0bXXXXXXX1: HS sync pattern with 2 or more bit errors detected on data lane D0 0bXXXXXX1X: HS sync pattern with 2 or more bit errors detected on data lane D1 0bXXXXX1XX: HS sync pattern with 1 bit error detected on data lane D0 0bXXXX1XXX: HS sync pattern with 1 bit error detected on data lane D1 0bXXX1XXXX: High speed receiver skew calibration failed on data lane D1 0bXX1XXXXX: High speed receiver skew calibration failed on data lane D0 0bX1XXXXXX: High speed receiver skew calibration run on data lane D1 0b1XXXXXXX: High speed receiver skew calibration run on data lane D0

MIPI_RX15 (0x33F)

Bit	7	6	5	4	3	2	1	0
Field	–	–	–	phy3_lp_err[4:0]				
Reset	–	–	–	00000				
Access Type	–	–	–	Read Clears All				

Bitfield	Bits	Description	Decode
phy3_lp_err	4:0	Phy3 LP status	0bXXXX1: Unrecognized Escape command received from data lane D0 0bXXX1X: Unrecognized Escape command received from CLK lane 0bXX1XX: Invalid line sequence detected from data lane D0 0bX1XXX: Invalid line sequence detected from data

Bitfield	Bits	Description	Decode
			lane D1 0b1XXXX: Invalid line sequence detected from CLK lane

MIPI_RX16 (0x340)

Bit	7	6	5	4	3	2	1	0
Field	phy3_hs_err[7:0]							
Reset	00000000							
Access Type	Read Clears All							

Bitfield	Bits	Description	Decode
phy3_hs_err	7:0	PHY3 high-speed status	0bXXXXXXX1: HS sync pattern with 2 or more bit errors detected on data lane D0 0bXXXXXX1X: HS sync pattern with 2 or more bit errors detected on data lane D1 0bXXXXX1XX: HS sync pattern with 1 bit error detected on data lane D0 0bXXXX1XXX: HS sync pattern with 1 bit error detected on data lane D1 0bXXX1XXXX: High speed receiver skew calibration failed on data lane D1 0bXX1XXXXX: High speed receiver skew calibration failed on data lane D0 0bX1XXXXXX: High speed receiver skew calibration run on data lane D1 0b1XXXXXXX: High speed receiver skew calibration run on data lane D0

MIPI_RX17 (0x341)

Bit	7	6	5	4	3	2	1	0
Field	ctrl0_csi_err_l[7:0]							
Reset	00000000							
Access Type	Read Clears All							

Bitfield	Bits	Description	Decode
ctrl0_csi_err_l	7:0	CSI-2 Controller 0 Status, low byte	0bXXXXXXX1: 1-bit ECC error detected 0bXXXXXX1X: 2-bit ECC error detected 0bXYYYYYXX: YYYYY = bit position of the 1-bit error 0b1XXXXXXX: CRC error detected

MIPI_RX18 (0x342)

Bit	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	ctrl0_csi_err_h[2:0]		
Reset	–	–	–	–	–	000		
Access Type	–	–	–	–	–	Read Clears All		

Bitfield	Bits	Description	Decode
ctrl0_csi_err_h	2:0	CSI-2 Controller 0 Status, high bits	0bX1: Packets terminated early 0b1X: Frame count error detected

MIPI_RX19 (0x343)

Bit	7	6	5	4	3	2	1	0
Field	ctrl1_csi_err_l[7:0]							
Reset	00000000							
Access Type	Read Clears All							

Bitfield	Bits	Description	Decode
ctrl1_csi_err_l	7:0	CSI-2 Controller 1 Status, low byte	0bXXXXXXX1: 1-bit ECC error detected 0bXXXXXX1X: 2-bit ECC error detected 0bXYYYYYXX: YYYYY = bit position of the 1-bit error 0b1XXXXXXX: CRC error detected

MIPI_RX20 (0x344)

Bit	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	ctrl1_csi_err_h[2:0]		
Reset	–	–	–	–	–	000		
Access Type	–	–	–	–	–	Read Clears All		

Bitfield	Bits	Description	Decode
ctrl1_csi_err_h	2:0	CSI-2 Controller 1 Status, high bits	0bX1: Packets terminated early 0b1X: Frame count error detected

MIPI_RX21 (0x345)

Bit	7	6	5	4	3	2	1	0
Field	ctrl1_vc_map0[3:0]				ctrl0_vc_map0[3:0]			
Reset	0000				0000			
Access Type	Write, Read				Write, Read			

Bitfield	Bits	Description
ctrl1_vc_map0	7:4	New virtual channel for VC=0
ctrl0_vc_map0	3:0	New virtual channel for VC=0

MIPI_RX22 (0x346)

Bit	7	6	5	4	3	2	1	0
Field	ctrl1_vc_map1[3:0]				ctrl0_vc_map1[3:0]			
Reset	0000				0000			
Access Type	Write, Read				Write, Read			

Bitfield	Bits	Description
ctrl1_vc_map1	7:4	New virtual channel for VC=1
ctrl0_vc_map1	3:0	New virtual channel for VC=1

MIPI_LPB0 (0x370)

Bit	7	6	5	4	3	2	1	0
Field	–	LPB_MIPI_RX_PRBSEN_1	LPB_MIPI_TX_PRBSEN_1	LPB_PATTERN_SEL[2:0]			LPB_MIPI_RX_PRBSEN_0	LPB_MIPI_TX_PRBSEN_0
Reset	–	0	0	000			0	0
Access Type	–	Write, Read	Write, Read	Write, Read			Write, Read	Write, Read

Bitfield	Bits	Description	Decode
LPB_MIPI_RX_PRBSEN_1	6	PRBS pattern checker enable for MIPI loop-back test	0b0: PRBS pattern checker for MIPI loopback disable 0b1: PRBS pattern checker for MIPI loopback enable
LPB_MIPI_TX_PRBSEN_1	5	PRBS pattern generator enable for MIPI loop-back test	0b0: PRBS pattern generator for MIPI loopback disable 0b1: PRBS pattern generator for MIPI loopback enable
LPB_PATTERN_SEL	4:2	PRBS pattern selection for MIPI loop-back test	0b000: Output 0x55, 0xAA 0b001: Output 0xAA, 0x55 0b010: PRBS pattern 0b011: PRBS pattern with error injected 0b100: Reserved 0b101: Reserved 0b110: Reserved 0b111: Reserved
LPB_MIPI_RX_PRBSEN_0	1	PRBS pattern checker enable for MIPI loop-back test	0b0: PRBS pattern checker for MIPI loopback disable 0b1: PRBS pattern checker for MIPI loopback

Bitfield	Bits	Description	Decode
			enable
LPB_MIPI_TX_PRBSEN_0	0	PRBS pattern generator enable for MIPI loop-back test	0b0: PRBS pattern generator for MIPI loopback disable 0b1: PRBS pattern generator for MIPI loopback enable

MIPI_LPB1 (0x371)

Bit	7	6	5	4	3	2	1	0
Field	LPB_LINE_PF[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
LPB_LINE_PF	7:0	Lines per frame for MIPI loop-back test	0xXX: Number of lines per frame

MIPI_LPB2 (0x372)

Bit	7	6	5	4	3	2	1	0
Field	LPB_PIXEL_COUNT_L[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
LPB_PIXEL_COUNT_L	7:0	Pixel count least significant byte for MIPI loop-back test	0xXX: Pixel count LSB

MIPI_LPB3 (0x373)

Bit	7	6	5	4	3	2	1	0
Field	LPB_PIXEL_COUNT_H[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
LPB_PIXEL_COUNT_H	7:0	Pixel count least significant byte for MIPI loop-back test	0xXX: Pixel count MSB

MIPI_LPB4 (0x374)

Bit	7	6	5	4	3	2	1	0
Field	LPB_DATA_INVALID1[3:0]				LPB_DATA_INVALID0[3:0]			
Reset	0000				0000			
Access Type	Read Only				Read Only			

Bitfield	Bits	Description	Decode
LPB_DATA_INVALID1	7:4	Controller 1 data checker status for MIPI Loop-Back Test	0bXXX0: Data lane D0 data valid 0bXXX1: Data lane D0 data invalid 0bXX0X: Data lane D1 data valid 0bXX1X: Data lane D1 data invalid 0bX0XX: Data lane D2 data valid 0bX1XX: Data lane D2 data invalid 0b0XXX: Data lane D3 data valid 0b1XXX: Data lane D3 data invalid
LPB_DATA_INVALID0	3:0	Controller 0 data checker status for MIPI Loop-Back Test	0bXXX0: Data lane D0 data valid 0bXXX1: Data lane D0 data invalid 0bXX0X: Data lane D1 data valid 0bXX1X: Data lane D1 data invalid 0bX0XX: Data lane D2 data valid 0bX1XX: Data lane D2 data invalid 0b0XXX: Data lane D3 data valid 0b1XXX: Data lane D3 data invalid

FRONTTOP_EXT0 (0x3C0)

Bit	7	6	5	4	3	2	1	0
Field	mem_dt3_selx[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
mem_dt3_selx	7:0	Select a designated datatype to route to video pipeline X for VS	0xXX: Designated datatype

FRONTTOP_EXT1 (0x3C1)

Bit	7	6	5	4	3	2	1	0
Field	mem_dt4_selx[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
mem_dt4_selx	7:0	Select a designated datatype to route to video pipeline X for VS	0xXX: Designated datatype

FRONTTOP_EXT2 (0x3C2)

Bit	7	6	5	4	3	2	1	0
Field	mem_dt5_selx[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
mem_dt5_selx	7:0	Select a designated datatype to route to video pipeline X for HS	0xXX: Designated datatype

FRONTTOP_EXT3 (0x3C3)

Bit	7	6	5	4	3	2	1	0
Field	mem_dt6_selx[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
mem_dt6_selx	7:0	Select a designated datatype to route to video pipeline X for HS	0xXX: Designated datatype

FRONTTOP_EXT4 (0x3C4)

Bit	7	6	5	4	3	2	1	0
Field	mem_dt3_sely[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
mem_dt3_sely	7:0	Select a designated datatype to route to video pipeline Y for VS	0xXX: Designated datatype

FRONTTOP_EXT5 (0x3C5)

Bit	7	6	5	4	3	2	1	0
Field	mem_dt4_sely[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
mem_dt4_sely	7:0	Select a designated datatype to route to video pipeline Y for VS	0xXX: Designated datatype

FRONTTOP_EXT6 (0x3C6)

Bit	7	6	5	4	3	2	1	0
Field	mem_dt5_sely[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
mem_dt5_sely	7:0	Select a designated datatype to route to video pipeline Y for HS	0xXX: Designated datatype

FRONTTOP_EXT7 (0x3C7)

Bit	7	6	5	4	3	2	1	0
Field	mem_dt6_sely[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
mem_dt6_sely	7:0	Select a designated datatype to route to video pipeline Y for HS	0xXX: Designated datatype

FRONTTOP_EXT8 (0x3C8)

Bit	7	6	5	4	3	2	1	0
Field	mem_dt3_selz[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
mem_dt3_selz	7:0	Select a designated datatype to route to video pipeline Z for VS	0xXX: Designated datatype

FRONTTOP_EXT9 (0x3C9)

Bit	7	6	5	4	3	2	1	0
Field	mem_dt4_selz[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
mem_dt4_selz	7:0	Select a designated datatype to route to video pipeline Z for VS	0xXX: Designated datatype

FRONTTOP_EXT10 (0x3CA)

Bit	7	6	5	4	3	2	1	0
Field	mem_dt5_selz[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
mem_dt5_selz	7:0	Select a designated datatype to route to video pipeline Z for HS	0xXX: Designated datatype

FRONTTOP_EXT11 (0x3CB)

Bit	7	6	5	4	3	2	1	0
Field	mem_dt6_selz[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
mem_dt6_selz	7:0	Select a designated datatype to route to video pipeline Z for HS	0xXX: Designated datatype

FRONTTOP_EXT12 (0x3CC)

Bit	7	6	5	4	3	2	1	0
Field	mem_dt3_selu[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
mem_dt3_selu	7:0	Select a designated datatype to route to video pipeline U for VS	0xXX: Designated datatype

FRONTTOP_EXT13 (0x3CD)

Bit	7	6	5	4	3	2	1	0
Field	mem_dt4_selu[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
mem_dt4_selu	7:0	Select a designated datatype to route to video pipeline U for VS	0xXX: Designated datatype

FRONTTOP_EXT14 (0x3CE)

Bit	7	6	5	4	3	2	1	0
Field	mem_dt5_selu[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
mem_dt5_selu	7:0	Select a designated datatype to route to video pipeline U for HS	0xXX: Designated datatype

FRONTTOP_EXT15 (0x3CF)

Bit	7	6	5	4	3	2	1	0
Field	mem_dt6_selu[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
mem_dt6_selu	7:0	Select a designated datatype to route to video pipeline U for HS	0xXX: Designated datatype

FRONTTOP_EXT16 (0x3D0)

Bit	7	6	5	4	3	2	1	0
Field	mem_dt6_sely_en	mem_dt5_sely_en	mem_dt4_sely_en	mem_dt3_sely_en	mem_dt6_selx_en	mem_dt5_selx_en	mem_dt4_selx_en	mem_dt3_selx_en
Reset	0	0	0	0	0	0	0	0
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read

Bitfield	Bits	Description	Decode
mem_dt6_sely_en	7	Enable datatype designated in mem_dt6_sely to route to video pipeline Y	0b0: Disable datatype routing 0b1: Enable datatype routing
mem_dt5_sely_en	6	Enable datatype designated in mem_dt5_sely to route to video pipeline Y	0b0: Disable datatype routing 0b1: Enable datatype routing
mem_dt4_sely_en	5	Enable datatype designated in mem_dt4_sely to route to video pipeline Y	0b0: Disable datatype routing 0b1: Enable datatype routing
mem_dt3_sely_en	4	Enable datatype designated in mem_dt3_sely to route to video pipeline Y	0b0: Disable datatype routing 0b1: Enable datatype routing
mem_dt6_selx_en	3	Enable datatype designated in mem_dt6_selx to route to video pipeline X	0b0: Disable datatype routing 0b1: Enable datatype routing
mem_dt5_selx_en	2	Enable datatype designated in mem_dt5_selx to route to video pipeline X	0b0: Disable datatype routing

Bitfield	Bits	Description	Decode
			0b1: Enable datatype routing
mem_dt4_selx_en	1	Enable datatype designated in mem_dt4_selx to route to video pipeline X	0b0: Disable datatype routing 0b1: Enable datatype routing
mem_dt3_selx_en	0	Enable datatype designated in mem_dt3_selx to route to video pipeline X	0b0: Disable datatype routing 0b1: Enable datatype routing

FRONTTOP_EXT17 (0x3D1)

Bit	7	6	5	4	3	2	1	0
Field	mem_dt6_selu_en	mem_dt5_selu_en	mem_dt4_selu_en	mem_dt3_selu_en	mem_dt6_selz_en	mem_dt5_selz_en	mem_dt4_selz_en	mem_dt3_selz_en
Reset	0	0	0	0	0	0	0	0
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read

Bitfield	Bits	Description	Decode
mem_dt6_selu_en	7	Enable datatype designated in mem_dt6_selu to route to video pipeline U	0b0: Disable datatype routing 0b1: Enable datatype routing
mem_dt5_selu_en	6	Enable datatype designated in mem_dt5_selu to route to video pipeline U	0b0: Disable datatype routing 0b1: Enable datatype routing
mem_dt4_selu_en	5	Enable datatype designated in mem_dt4_selu to route to video pipeline U	0b0: Disable datatype routing 0b1: Enable datatype routing
mem_dt3_selu_en	4	Enable datatype designated in mem_dt3_selu to route to video pipeline U	0b0: Disable datatype routing 0b1: Enable datatype routing
mem_dt6_selz_en	3	Enable datatype designated in mem_dt6_selz to route to video pipeline Z	0b0: Disable datatype routing 0b1: Enable datatype routing
mem_dt5_selz_en	2	Enable datatype designated in mem_dt5_selz to route to video pipeline Z	0b0: Disable datatype routing 0b1: Enable datatype routing
mem_dt4_selz_en	1	Enable datatype designated in mem_dt4_selz to route to video pipeline Z	0b0: Disable datatype routing 0b1: Enable datatype routing
mem_dt3_selz_en	0	Enable datatype designated in mem_dt3_selz to route to video pipeline Z	0b0: Disable datatype routing 0b1: Enable datatype routing

VTX0 (0x3E0)

Bit	7	6	5	4	3	2	1	0
Field	–	VS_TRIG	REF_VTG_MODE[1:0]		HS_INV	GEN_HS	VS_INV	GEN_VS
Reset	–	1	11		0	0	0	0
Access Type	–	Write, Read	Write, Read		Write, Read	Write, Read	Write, Read	Write, Read

Bitfield	Bits	Description	Decode
VS_TRIG	6	Select VS trigger edge	0b0: Falling edge is VS trigger 0b1: Rising edge is VS trigger
REF_VTG_MODE	5:4	Selects one of the following modes for video interface timing generation. VS tracking mode. VS input's period (VS_HIGH+VS_LOW) is tracked. After VS tracking is locked, any VS input edge (glitches) not in the expected pclk cycle is ignored. VS tracking is locked with 3 consecutive matches and unlocked by 3 consecutive mismatches. When unlocked or power up, the next VS input edge is assumed to be the right VS edge. VS one trigger mode. One VS input edge triggers the generation of one frame of VSO/HSO/DEO. If the next VS input edge comes earlier or later than expected by VS period, the newly generated frame will be correct. The current VSO/HSO/DEO is cut or extended at the point of the rising edge of the newly generated VSO/HSO/DEO. Auto repeat mode. VS input edge triggers the generation of continuous frames of VSO/HSO/DEO even if there are no more VS input edges. If next VS input edge comes earlier or later than expected by VS period, the newly generated frame will be correct. The current VSO/HSO/DEO is cut or extended at the point of the rising edge of the newly generated VSO/HSO/DEO.	0b00: VS tracking mode 0b01: VS one trigger mode 0b10: Auto repeat mode 0b11: VS tracking mode
HS_INV	3	Invert HS output of video timing generator	0b0: Do not invert HS 0b1: Invert HS
GEN_HS	2	Enable generation of HS output	0b0: Disable generation of HS 0b1: Enable generation of HS
VS_INV	1	Invert VS output of video timing generator	0b0: Do not invert VS 0b1: Invert VS
GEN_VS	0	Enable generation of VS output	0b0: Disable generation of VS 0b1: Enable generation of VS

VTX1 (0x3E1)

Bit	7	6	5	4	3	2	1	0
Field	VS_HIGH_2[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
VS_HIGH_2	7:0	VS High Period in terms of PCLK cycles (Bits [23:16])	0xXX: VS high period high byte

VTX2 (0x3E2)

Bit	7	6	5	4	3	2	1	0
Field	VS_HIGH_1[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
VS_HIGH_1	7:0	VS High Period in terms of PCLK cycles (Bits [15:8])	0xXX: VS high period middle byte

VTX3 (0x3E3)

Bit	7	6	5	4	3	2	1	0
Field	VS_HIGH_0[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
VS_HIGH_0	7:0	VS High Period in terms of PCLK cycles (Bits [7:0])	0xXX: VS high period low byte

VTX4 (0x3E4)

Bit	7	6	5	4	3	2	1	0
Field	VS_LOW_2[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
VS_LOW_2	7:0	VS Low Period in terms of PCLK cycles (Bits [23:16])	0xXX: VS low period high byte

VTX5 (0x3E5)

Bit	7	6	5	4	3	2	1	0
Field	VS_LOW_1[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
VS_LOW_1	7:0	VS Low Period in terms of PCLK cycles (Bits [15:8])	0xXX: VS low period middle byte

VTX6 (0x3E6)

Bit	7	6	5	4	3	2	1	0
Field	VS_LOW_0[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
VS_LOW_0	7:0	VS Low Period in terms of PCLK cycles (Bits [7:0])	0xXX: VS low period low byte

VTX7 (0x3E7)

Bit	7	6	5	4	3	2	1	0
Field	V2H_2[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
V2H_2	7:0	Horizontal sync delay. VS edge to the rising edge of the first HS in terms of PCLK cycles (Bits [23:16])	0xXX: HS delay high byte

VTX8 (0x3E8)

Bit	7	6	5	4	3	2	1	0
Field	V2H_1[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
V2H_1	7:0	Horizontal sync delay. VS edge to the rising edge of the first HS in terms of PCLK cycles (Bits [15:8])	0xXX: HS delay middle byte

VTX9 (0x3E9)

Bit	7	6	5	4	3	2	1	0
Field	V2H_0[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
V2H_0	7:0	Horizontal sync delay. VS edge to the rising edge of the first HS in terms of PCLK cycles (Bits [7:0])	0xXX: HS delay low byte

VTX10 (0x3EA)

Bit	7	6	5	4	3	2	1	0
Field	HS_HIGH_1[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
HS_HIGH_1	7:0	HS High Period in terms of PCLK cycles (Bits [15:8])	0xXX: HS high period high byte

VTX11 (0x3EB)

Bit	7	6	5	4	3	2	1	0
Field	HS_HIGH_0[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
HS_HIGH_0	7:0	HS High Period in terms of PCLK cycles (Bits [7:0])	0xXX: HS high period low byte

VTX12 (0x3EC)

Bit	7	6	5	4	3	2	1	0
Field	HS_LOW_1[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
HS_LOW_1	7:0	HS Low Period in terms of PCLK cycles (Bits [15:8])	0xXX: HS low period high byte

VTX13 (0x3ED)

Bit	7	6	5	4	3	2	1	0
Field	HS_LOW_0[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
HS_LOW_0	7:0	HS Low Period in terms of PCLK cycles (Bits [7:0])	0xXX: HS low period low byte

VTX14 (0x3EE)

Bit	7	6	5	4	3	2	1	0
Field	HS_CNT_1[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
HS_CNT_1	7:0	Number of HS pulses per frame (Bits [15:8])	0xXX: HS pulses per frame high byte

VTX15 (0x3EF)

Bit	7	6	5	4	3	2	1	0
Field	HS_CNT_0[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
HS_CNT_0	7:0	Number of HS pulses per frame (Bits [7:0])	0xXX: HS pulses per frame low byte

REF_VTG0 (0x3F0)

Bit	7	6	5	4	3	2	1	0
Field	REFGEN_LOCKED	REFGEN_PREDEF_EN	REFGEN_PREDEF_FREQ[1:0]		–	–	REFGEN_RST	REFGEN_EN
Reset	0	1	01		–	–	0	0
Access Type	Read Only	Write, Read	Write, Read		–	–	Write, Read	Write, Read

Bitfield	Bits	Description	Decode
REFGEN_LOCKED	7	Reference generation PLL is locked	0b0: Reference generation PLL not locked 0b1: Reference generation PLL locked
REFGEN_PREDEF_EN	6	Enable pre-defined clock settings for reference generation PLL	0b0: Disable pre-defined clock settings for

Bitfield	Bits	Description	Decode
			reference generation PLL 0b1: Enable pre-defined clock settings for reference generation PLL
REFGEN_PREDEF_FREQ	5:4	Pre-defined reference generation PLL frequency setting	0b00: 19.2 MHz 0b01: 27 MHz 0b10: 37.125 MHz 0b11: 74.25 MHz
REFGEN_RST	1	Reset reference generation PLL	0b0: Do not reset reference generation PLL 0b1: Reset reference generation PLL
REFGEN_EN	0	Enable reference generation PLL	0b0: Disable reference generation PLL 0b1: Enable reference generation PLL

REF_VTG1 (0x3F1)

Bit	7	6	5	4	3	2	1	0
Field	RCLKEN_Y	–	PCLK_GPIO[4:0]					PCLKEN
Reset	0	–	00000					0
Access Type	Write, Read	–	Write, Read					Write, Read

Bitfield	Bits	Description	Decode
RCLKEN_Y	7	Output RCLK from PCLKOUT pin	0b0: Disable RCLK output on PCLKOUT pin 0b1: Enable RCLK output on PCLKOUT pin
PCLK_GPIO	5:1	Output PCLK from selected GPIO	0bXXXXX: GIPO pin selected for PCLK output
PCLKEN	0	Output PCLK on GPIO	0b0: Disable PCLK output on GPIO 0b1: Enable PCLK output on GPIO selected by PCLK_GPIO

REF_VTG2 (0x3F2)

Bit	7	6	5	4	3	2	1	0
Field	–	–	HS_GPIO[4:0]					HSEN
Reset	–	–	00000					0
Access Type	–	–	Write, Read					Write, Read

Bitfield	Bits	Description	Decode
HS_GPIO	5:1	Output HS from selected GPIO	0bXXXXX: GIPO pin selected for HS output
HSEN	0	Output HS on GPIO	0b0: Disable HS output on GPIO 0b1: Enable HS output on GPIO selected by HS_GPIO

REF_VTG3 (0x3F3)

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Field	–	–	VS_GPIO[4:0]					VSEN
Reset	–	–	00000					0
Access Type	–	–	Write, Read					Write, Read

Bitfield	Bits	Description	Decode
VS_GPIO	5:1	Output VS from selected GPIO	0bXXXXX: GIPO pin selected for VS output
VSEN	0	Output VS on GPIO	0b0: Disable VS output on GPIO 0b1: Enable VS output on GPIO selected by VS_GPIO

REF_VTG4 (0x3F4)

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Field	REFGEN_FB_FRACT_L[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
REFGEN_FB_FRACT_L	7:0	Reference generator PLL feedback divider fraction value when pre-defined mode is disabled (REFGEN_PREDEF_EN=0)	0xXX: Low byte of reference generator PLL feedback divider fraction value

REF_VTG5 (0x3F5)

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Field	–	–	–	–	REFGEN_FB_FRACT_H[3:0]			
Reset	–	–	–	–	0000			
Access Type	–	–	–	–	Write, Read			

Bitfield	Bits	Description	Decode
REFGEN_FB_FRACT_H	3:0	Reference generator PLL feedback divider fraction value when pre-defined mode is disabled (REFGEN_PREDEF_EN=0)	0xX: High nibble of reference generator PLL feedback divider fraction value

REF_VTG6 (0x3F6)

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Field	VS_DLY_2[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
VS_DLY_2	7:0	VS Delay in terms of PCLK cycles. The output VS is delayed by VS_DELAY cycles from the input VS. (Bits [23:16])	0xXX: VS delay high byte

REF_VTG7 (0x3F7)

Bit	7	6	5	4	3	2	1	0
Field	VS_DLY_1[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
VS_DLY_1	7:0	VS Delay in terms of PCLK cycles. The output VS is delayed by VS_DELAY cycles from the input VS. (Bits [15:8])	0xXX: VS delay middle byte

REF_VTG8 (0x3F8)

Bit	7	6	5	4	3	2	1	0
Field	VS_DLY_0[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
VS_DLY_0	7:0	VS Delay in terms of PCLK cycles. The output VS is delayed by VS_DELAY cycles from the input VS. (Bits [7:0])	0xXX: VS delay low byte

REF_VTG9 (0x3F9)

Bit	7	6	5	4	3	2	1	0
Field	REF_VTG_TRIG_EN	–	–	REF_VTG_TRIG_ID[4:0]				
Reset	0	–	–	11110				
Access Type	Write, Read	–	–	Write, Read				

Bitfield	Bits	Description	Decode
REF_VTG_TRIG_EN	7	Enable receiving REF VTG trigger signal	0b0: Disable reception of REF VTG input signal 0b1: Enable reception of REF VTG input signal
REF_VTG_TRIG_ID	4:0	GPIO ID used for receiving REF_VTG_TRIG	0bXXXXX: GIPO ID selected for receiving REF_VTG_TRIG

GMSL1_2 (0x402)

Bit	7	6	5	4	3	2	1	0
Field	–	–	SSEN	–	–	–	–	–
Reset	–	–	0	–	–	–	–	–
Access Type	–	–	Write, Read	–	–	–	–	–

Bitfield	Bits	Description	Decode
SSEN	5	Spread spectrum enable	0b0: Spread spectrum disabled 0b1: Spread spectrum enabled

GMSL1_4 (0x404)

Bit	7	6	5	4	3	2	1	0
Field	SEREN	CLINKEN	PRBSEN	–	–	–	REVCCEN	FWDCCEN
Reset	1	0	0	–	–	–	1	1
Access Type	Write, Read	Write, Read	Write, Read	–	–	–	Write, Read	Write, Read

Bitfield	Bits	Description	Decode
SEREN	7	Serialization enable	0b0: Disable serialization 0b1: Enable serialization
CLINKEN	6	Configuration link enable	0b0: Disable configuration link 0b1: Enable configuration link
PRBSEN	5	PRBS test enable (In HIBW mode set PRBS_TYPE=0)	0b0: Set device normal operation 0b1: Enable PRBS test
REVCCEN	1	Enable reverse control channel from deserializer	0b0: Disable reverse control channel receiver 0b1: Enable reverse control channel receiver
FWDCCEN	0	Enable forward control channel to deserializer	0b0: Disable forward control channel transmitter 0b1: Enable forward control channel transmitter

GMSL1_5 (0x405)

Bit	7	6	5	4	3	2	1	0
Field	–	–	PRBS_LEN[1:0]		–	–	–	–
Reset	–	–	00		–	–	–	–
Access Type	–	–	Write, Read		–	–	–	–

Bitfield	Bits	Description	Decode
PRBS_LEN	5:4	PRBS test length	0b00: Continuous 0b01: 167.1 Mbits 0b10: 9.83 Mbits 0b11: 1341.5 Mbits

GMSL1_7 (0x407)

Bit	7	6	5	4	3	2	1	0
Field	DBL	HIBW	BWS	–	DRS	HVEN	–	PXL_CRC
Reset	0	0	0	–	0	0	–	0
Access Type	Write, Read	Write, Read	Write, Read	–	Write, Read	Write, Read	–	Write, Read

Bitfield	Bits	Description	Decode
DBL	7	Double-output mode	0b0: Use single-rate output 0b1: Use double-rate output (2x word rate at 1/2x width)
HIBW	6	High-bandwidth mode enable	0b0: Disable high-bandwidth mode 0b1: Enable high-bandwidth mode (when BWS = 0)
BWS	5	Bus width select	0b0: Set bus width for 22-/24-bit bus, 24-/27-bit mode (depending on HIBW setting) 0b1: Set bus width for 30-bit bus (32-bit mode)
DRS	3	Data rate select	0b0: Use normal data rate output 0b1: Use 1/2 rate data output (for use with low data rates)
HVEN	2	HS/VS encoding enable	0b0: Disable HS/VS encoding 0b1: Enable HS/VS encoding
PXL_CRC	0	Pixel CRC type	0b0: Use 1-bit parity (compatible with all devices) 0b1: Use 6-bit CRC

GMSL1_D (0x40D)

Bit	7	6	5	4	3	2	1	0
Field	I2C_LOC_ACK	–	–	–	–	–	–	–
Reset	0	–	–	–	–	–	–	–
Access Type	Write, Read	–	–	–	–	–	–	–

Bitfield	Bits	Description	Decode
I2C_LOC_ACK	7	I2C-to-I2C slave generates local ack when fwd ch is not available	0b0: Disable local acknowledge when forward channel is not available 0b1: Enable local acknowledge when forward channel is not available

GMSL1_F (0x40F)

Bit	7	6	5	4	3	2	1	0
Field	CNTL_IN_EN[4:0]					GPO_RX_EN	GPO_OUT_SEL	SET_GPO
Reset	01111					1	0	0
Access Type	Write, Read					Write, Read	Write, Read	Write, Read

Bitfield	Bits	Description	Decode
CNTL_IN_EN	7:3	CNTL input enable for CNTL 0,1,2,3,4	0xXXXX0: CNTL 0 input disabled 0xXXXX1: CNTL 0 input enabled 0xXXX0X: CNTL 1 input disabled 0xXXX1X: CNTL 1 input enabled 0xXX0XX: CNTL 2 input disabled 0xXX1XX: CNTL 2 input enabled 0xX0XXX: CNTL 3 input disabled 0xX1XXX: CNTL 3 input enabled 0x0XXXX: CNTL 4 input disabled 0x1XXXX: CNTL 4 input enabled
GPO_RX_EN	2	Enable receiving GPO signal from reverse channel	0x0: Disable receiving of GPO signal 0x1: Enable receiving of GPO signal
GPO_OUT_SEL	1	Enable alternative GPO output	0b0: Disable alternative GPO output 0b1: Enable alternative GPO output
SET_GPO	0	Set GPO output high or low	0b0: Set GPO output low 0b1: Set GPO output high

GMSL1_11 (0x411)

Bit	7	6	5	4	3	2	1	0
Field	ERRG_RATE2[1:0]		ERRG_TYPE2[1:0]		ERRG_CNT2[1:0]		ERRG_PER2	ERRG_EN
Reset	00		00		00		0	0
Access Type	Write, Read		Write, Read		Write, Read		Write, Read	Write, Read

Bitfield	Bits	Description	Decode
ERRG_RATE2	7:6	Error generation rate, on average	0b00: Every 2560 bits 0b01: Every 40960 bits 0b10: Every 655360 bit 0b11: Every 10485760 bits
ERRG_TYPE2	5:4	Type of generated errors	0b00: Single bit 0b01: Two 8b/10b symbols 0b10: Three 8b/10b symbols 0b11: Four 8b/10b symbols
ERRG_CNT2	3:2	Number of generated errors	0b00: Continuous 0b01: 16

Bitfield	Bits	Description	Decode
			0b10: 128 0b11: 1024
ERRG_PER2	1	Periodic error generation enable	0b0: Periodic error generator disabled 0b1: Periodic error generator enabled
ERRG_EN	0	Enable Error generator, auto clears when done	0b0: Disable error generator 0b1: Enable error generator

GMSL1_14 (0x414)

Bit	7	6	5	4	3	2	1	0
Field	–	–	–	I2C_TIMED_OUT2	CC_WBLOCK_LOST	REV_BIT_LOCK	CC_WBLOCK	REM_CCLOCK
Reset	–	–	–	0	0	0	0	0
Access Type	–	–	–	Read Only	Read Only	Read Only	Read Only	Read Only

Bitfield	Bits	Description	Decode
I2C_TIMED_OUT2	4	I2C channel timed out	0b0: I2C channel not timed out 0b1: I2C channel timed out
CC_WBLOCK_LOST	3	Word boundary lock lost	0b0: Word boundary lock not lost 0b1: Word boundary lock lost
REV_BIT_LOCK	2	Reverse channel bit locked	0b0: Reverse channel bit not locked 0b1: Reverse channel bit locked
CC_WBLOCK	1	Control channel word boundary lock	0b0: Control channel word boundary not locked 0b1: Control channel word boundary locked
REM_CCLOCK	0	Remote control channel locked	0b0: Remote control channel not locked 0b1: Remote control channel locked

GMSL1_15 (0x415)

Bit	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	–	SEL_VESA	SEL_RGB888
Reset	–	–	–	–	–	–	1	1
Access Type	–	–	–	–	–	–	Write, Read	Write, Read

Bitfield	Bits	Description	Decode
SEL_VESA	1	VESA or OLDI mapping selection	0b0: OLDI mapping 0b1: VESA mapping
SEL_RGB888	0	Select RGB888 bit mapping	0b0: Non-RGB888

Bitfield	Bits	Description	Decode
			0b1: RGB888

GMSL1_16 (0x416)

Bit	7	6	5	4	3	2	1	0
Field	–	MAX_RT_ERR	–	–	–	–	–	–
Reset	–	0	–	–	–	–	–	–
Access Type	–	Read Only	–	–	–	–	–	–

Bitfield	Bits	Description	Decode
MAX_RT_ERR	6	Maximum retransmission error flag. Cleared when read.	0b0: No control-channel retransmission error 0b1: Control-channel retransmission maximum limit reached

GMSL1_18 (0x41C)

Bit	7	6	5	4	3	2	1	0
Field	CC_I2C_RETR_CNT[7:0]							
Reset	00000000							
Access Type	Read Only							

Bitfield	Bits	Description	Decode
CC_I2C_RETR_CNT	7:0	I2C packet retransmit count	0xXX: Number of I2C packets retransmitted

GMSL1_19 (0x41D)

Bit	7	6	5	4	3	2	1	0
Field	CC_CRC_ERRCNT[7:0]							
Reset	00000000							
Access Type	Read Only							

Bitfield	Bits	Description	Decode
CC_CRC_ERRCNT	7:0	Packet based control channel CRC error counter	0xXX: Number of control channel CRC errors

GMSL1_39 (0x438)

Bit	7	6	5	4	3	2	1	0
Field	–	CROSS24_I	CROSS24_F	CROSS24[4:0]				
Reset	–	0	0	11000				
Access Type	–	Write, Read	Write, Read	Write, Read				

Bitfield	Bits	Description	Decode
CROSS24_I	6	Invert CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS24_F	5	Force CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS24	4:0	Maps incoming bit position set by this field to the outgoing bit position 24	0bXXXXX: Incoming bit position

GMSL1_3A (0x439)

Bit	7	6	5	4	3	2	1	0
Field	–	CROSS25_I	CROSS25_F	CROSS25[4:0]				
Reset	–	0	0	11001				
Access Type	–	Write, Read	Write, Read	Write, Read				

Bitfield	Bits	Description	Decode
CROSS25_I	6	Invert CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS25_F	5	Force CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS25	4:0	Maps incoming bit position set by this field to the outgoing bit position 25	0bXXXXX: Incoming bit position

GMSL1_3B (0x43A)

Bit	7	6	5	4	3	2	1	0
Field	–	CROSS26_I	CROSS26_F	CROSS26[4:0]				
Reset	–	0	0	11010				
Access Type	–	Write, Read	Write, Read	Write, Read				

Bitfield	Bits	Description	Decode
CROSS26_I	6	Invert CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS26_F	5	Force CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS26	4:0	Maps incoming bit position set by this field to the outgoing bit position 26	0bXXXXX: Incoming bit position

GMSL1_3C (0x43B)

Bit	7	6	5	4	3	2	1	0
Field	–	CROSS27_I	CROSS27_F	CROSS27[4:0]				
Reset	–	0	0	11011				
Access Type	–	Write, Read	Write, Read	Write, Read				

Bitfield	Bits	Description	Decode
CROSS27_I	6	Invert CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS27_F	5	Force CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS27	4:0	Maps incoming bit position set by this field to the outgoing bit position 27	0bXXXXX: Incoming bit position

GMSL1_3D (0x43C)

Bit	7	6	5	4	3	2	1	0
Field	–	CROSS28_I	CROSS28_F	CROSS28[4:0]				
Reset	–	0	0	11100				
Access Type	–	Write, Read	Write, Read	Write, Read				

Bitfield	Bits	Description	Decode
CROSS28_I	6	Invert CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS28_F	5	Force CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS28	4:0	Maps incoming bit position set by this field to the outgoing bit position 28	0bXXXXX: Incoming bit position

GMSL1_3E (0x43D)

Bit	7	6	5	4	3	2	1	0
Field	–	CROSS29_I	CROSS29_F	CROSS29[4:0]				
Reset	–	0	0	11101				
Access Type	–	Write, Read	Write, Read	Write, Read				

Bitfield	Bits	Description	Decode
CROSS29_I	6	Invert CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS29_F	5	Force CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero

Bitfield	Bits	Description	Decode
CROSS29	4:0	Maps incoming bit position set by this field to the outgoing bit position 29	0bXXXXX: Incoming bit position

GMSL1_3F (0x43E)

Bit	7	6	5	4	3	2	1	0
Field	–	CROSS30_I	CROSS30_F	CROSS30[4:0]				
Reset	–	0	0	11110				
Access Type	–	Write, Read	Write, Read	Write, Read				

Bitfield	Bits	Description	Decode
CROSS30_I	6	Invert CrossX	0b0: Do not invert bit 0b1: Invert bit
CROSS30_F	5	Force CrossX to 0 before inversion	0b0: Do not force bit to zero 0b1: Force bit to zero
CROSS30	4:0	Maps incoming bit position set by this field to the outgoing bit position 30	0bXXXXX: Incoming bit position

GMSL1_42 (0x442)

Bit	7	6	5	4	3	2	1	0
Field	–	–	–	MAX_RT_EN	I2C_RT_EN	GPI_COMP_EN	GPI_RT_EN	GPO_EN
Reset	–	–	–	1	1	0	1	1
Access Type	–	–	–	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read

Bitfield	Bits	Description	Decode
MAX_RT_EN	4	Maximum retransmission limit enable	0b0: Max retransmission limit disabled 0b1: Max retransmission limit enabled
I2C_RT_EN	3	I2C retransmission enable	0b0: I2C retransmission disabled 0b1: I2C retransmission enabled
GPI_COMP_EN	2	General purpose input compensation enable	0b0: GPI compensation disabled 0b1: GPI compensation enabled
GPI_RT_EN	1	Enable general purpose input pin	0b0: GPI pin disabled 0b1: GPI pin enabled
GPO_EN	0	Enable general purpose output pin	0b0: GPO pin disabled 0b1: GPO pin enabled

GMSL1_4D (0x44D)

Bit	7	6	5	4	3	2	1	0
Field	HIGHIMM	–	–	–	–	–	–	–
Reset	0	–	–	–	–	–	–	–
Access Type	Write, Read	–	–	–	–	–	–	–

Bitfield	Bits	Description	Decode
HIGHIMM	7	Reverse channel high immunity mode	0b0: Reverse channel high immunity mode disabled 0b1: Reverse channel high immunity mode enabled

GMSL1_66 (0x466)

Bit	7	6	5	4	3	2	1	0
Field	–	–	PRBS_TYPE	REV_FAST	DIS_DE	–	–	–
Reset	–	–	1	0	0	–	–	–
Access Type	–	–	Write, Read	Write, Read	Write, Read	–	–	–

Bitfield	Bits	Description	Decode
PRBS_TYPE	5	PRBS type select (In HIBW mode set PRBS_TYPE=0	0b0: GMSL legacy style PRBS test 0b1: MAX9272 style PRBS test
REV_FAST	4	Reverse channel fast mode	0b0: Disable reverse channel fast mode 0b1: Enable reverse channel fast mode
DIS_DE	3	Disable processing separate HS and DE signals	0b0: Enable processing separate HS and DE signals 0b1: Disable processing separate HS and DE signals

GMSL1_67 (0x467)

Bit	7	6	5	4	3	2	1	0
Field	CNTL_TRIG[1:0]		AUTO_CLINK	DE_RGB888_DBL	–	DBL_ALIGN_TO[2:0]		
Reset	00		0	0	–	111		
Access Type	Write, Read		Write, Read	Write, Read	–	Write, Read		

Bitfield	Bits	Description	Decode
CNTL_TRIG	7:6	Select CNTL trigger of encoded packets in high bandwidth mode	0b00: no trigger 0b01: always trigger 0b10: trigger when DE is low 0b11: trigger when HS is low
AUTO_CLINK	5	Automatic control of configuration link	0b0: Enable configuration link only when CLINKEN=1 and SEREN=0

Bitfield	Bits	Description	Decode
			0b1: Automatically enable configuration link when SEREN=1 and PCLKDET=0
DE_RGB888_DBL	4	DBL DE RGB888 processing mode select	0b0: Regular DBL DE processing while mapping input 0b1: Process DBL DE for RGB888 mode while mapping input
DBL_ALIGN_TO	2:0	Double alignment mode	0b000: Align at each rising edge of HS and turn off the alignment aftger HS is low (MAX9286) 0b001: Reserved 0b010: Force align 0b011: Force align 0b100: Align at each rising edge of HS 0b101: Align at each rising edge of DE 0b110: Reserved 0b111: DBL-DBL mode with no alignment

GMSL1_68 (0x468)

Bit	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	–	CC_CRC_LENGTH[1:0]	
Reset	–	–	–	–	–	–	01	
Access Type	–	–	–	–	–	–	Write, Read	

Bitfield	Bits	Description	Decode
CC_CRC_LENGTH	1:0	Control channel CRC length	0x0: 1-bit 0x1: 5-bit 0x2: 8-bit 0x3: Reserved

GMSL1_96 (0x496)

Bit	7	6	5	4	3	2	1	0
Field	–	–	–	–	BYPASS_HVD_ALIGN	–	–	–
Reset	–	–	–	–	0	–	–	–
Access Type	–	–	–	–	Write, Read	–	–	–

Bitfield	Bits	Description	Decode
BYPASS_HVD_ALIGN	3	Bypass HS, VS, DE alignment	0b0: align HS, VS and DE 0b1: bypass HS, VS and DE through alignment block

GMSL1_99 (0x499)

Bit	7	6	5	4	3	2	1	0
Field	–	–	–	REV_FILT_EN	ALIGN_VS_MODE	ALIGN_INFO	–	–
Reset	–	–	–	0	1	1	–	–
Access Type	–	–	–	Write, Read	Write, Read	Write, Read	–	–

Bitfield	Bits	Description	Decode
REV_FILT_EN	4	Reverse channel digital filter enable	0b0: Disable filter 0b1: Enable filter
ALIGN_VS_MODE	3	Processing mode while aligning VS in HIBW DBL mode	0b0: Send VSL 0b1: Send VSL or VSH
ALIGN_INFO	2	Send alignment information through CNTL signals in HBIW mode	0b0: Do not send alignment information 0b1: Send alignment information

GMSL1_9A (0x49A)

Bit	7	6	5	4	3	2	1	0
Field	–	–	–	–	PKTCC_EN	–	–	–
Reset	–	–	–	–	0	–	–	–
Access Type	–	–	–	–	Write, Read	–	–	–

Bitfield	Bits	Description	Decode
PKTCC_EN	3	Packet based control channel mode enable, when enabled and the user enters sleep mode, it is required to reduce MST_TO.	0b0: Disable packet-based control-channel mode 0b1: Enable packet-based control-channel mode

GMSL1_C8 (0x4C8)

Bit	7	6	5	4	3	2	1	0
Field	–	ALIGNING	I2C_ACK_RECVED	I2C_ACK_ACKED	–	–	–	–
Reset	–	0	0	0	–	–	–	–
Access Type	–	Read Only	Read Only	Read Only	–	–	–	–

Bitfield	Bits	Description	Decode
ALIGNING	6	Status of DBL alignment	0b0: Not aligning 0b1: Aligning
I2C_ACK_RECVED	5	Remote ACK received	0b0: Remote ACK not received 0b1: Remote ACK received
I2C_ACK_ACKED	4	Remote ACK acknowledged	0b0: Remote ACK not acknowledged 0b1: Remote ACK acknowledged

ADC_CTRL_0 (0x500)

Bit	7	6	5	4	3	2	1	0
Field	buf_bypass	–	–	adc_chgpump_pu	adc_refbuf_pu	buf_pu	adc_pu	cpu_adc_start
Reset	0	–	–	0	0	0	0	0
Access Type	Write, Read	–	–	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read

Bitfield	Bits	Description	Decode
buf_bypass	7	Bypass input buffer	0b0: Use input buffer 0b1: Bypass input buffer stage
adc_chgpump_pu	4	ADC charge pump power up	0b0: ADC charge pump powered off 0b1: ADC charge pump powered on
adc_refbuf_pu	3	ADC reference buffer power up	0b0: ADC reference buffer powered off 0b1: ADC reference buffer powered on
buf_pu	2	ADC input buffer power up	0b0: ADC input buffer powered off 0b1: ADC input buffer powered on
adc_pu	1	ADC power up	0b0: ADC powered off 0b1: ADC powered on
cpu_adc_start	0	Start ADC conversion. Bit is automatically cleared to zero after completion of the conversion	0b0: Conversion complete 0b1: Start ADC conversion

ADC_CTRL_1 (0x501)

Bit	7	6	5	4	3	2	1	0
Field	adc_chsel[3:0]				adc_clk_en	adc_refsel	adc_scale	adc_refscl
Reset	0000				0	0	0	0
Access Type	Write, Read				Write, Read	Write, Read	Write, Read	Write, Read

Bitfield	Bits	Description	Decode
adc_chsel	7:4	ADC channel select - GPIO channels available vary by part number	0x0: GPIO0 0x1: GPIO1 0x2: GPIO2 0x3: GPIO3 0x4: GPIO4 0x5: GPIO5 0x6: GPIO6 0x7: GPIO7 0x8: VSUP0 / 4 0x9: VSUP1 / 2 0xA: VSUP2 /2 0xB: TMON 0xC: ABUS<0> 0xD: ABUS<1>

Bitfield	Bits	Description	Decode
			0xE: ABUS<2> 0xF: ABUS<3>
adc_clk_en	3	ADC clock enable	0b0: ADC clock disable 0b1: ADC clock enable
adc_refsel	2	ADC reference select	0b0: Select internal bandgap voltage as ADC reference 0b1: Select VDD18 as ADC reference
adc_scale	1	ADC scale	0b0: Normal operation 0b1: Scale ADC input down by 1/2
adc_refscl	0	ADC reference stage	0b0: Normal operation 0b1: Scale ADC reference down by 1/2

ADC_CTRL_2 (0x502)

Bit	7	6	5	4	3	2	1	0
Field	–	–	–	–	adc_div[1:0]		adc_xref	lnmux_en
Reset	–	–	–	–	00		0	0
Access Type	–	–	–	–	Write, Read		Write, Read	Write, Read

Bitfield	Bits	Description	Decode
adc_div	3:2	GPIO internal divider	0b00: Divide by 1 0b01: Divide by 2 0b10: Divide by 3 0b11: Divide by 4
adc_xref	1	Enable use of ADC external reference	0b0: Use internal reference for ADC 0b1: Use external reference for ADC
lnmux_en	0	Enable the input mux to the ADC to allow for a break before make connection sequence	0b0: Mux is opened 0b1: Mux selected by adc_chsel field is closed

ADC_CTRL_3 (0x503)

Bit	7	6	5	4	3	2	1	0
Field	AfePwrUpDly[7:0]							
Reset	00001010							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
AfePwrUpDly	7:0	AFE power up delay select (in 2.8 μs increments when using a 2.5MHz ADC clock.)	0xXX: Number of 2.8 μs delays

ADC_STATUS0 (0x504)

Bit	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	–	–	adc_active
Reset	–	–	–	–	–	–	–	0
Access Type	–	–	–	–	–	–	–	Read Only

Bitfield	Bits	Description	Decode
adc_active	0	ADC conversion in progress	0b0: ADC is idle 0b1: ADC conversion currently in progress

ADC_DATA0 (0x508)

Bit	7	6	5	4	3	2	1	0
Field	adc_data_l[7:0]							
Reset	00000000							
Access Type	Read Only							

Bitfield	Bits	Description	Decode
adc_data_l	7:0	Lower byte of 10-bit ADC converted sample data output	0xX: Lower byte of 10-bit ADC converted value

ADC_DATA1 (0x509)

Bit	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	–	adc_data_h[1:0]	
Reset	–	–	–	–	–	–	00	
Access Type	–	–	–	–	–	–	Read Only	

Bitfield	Bits	Description	Decode
adc_data_h	1:0	Upper 2-bits of 10-bit ADC converted sample data output	0xX: Upper 2-bits of 10-bit ADC converted value

ADC_INTRIE0 (0x50C)

Bit	7	6	5	4	3	2	1	0
Field	–	–	–	adc_overflow_ie	adc_lo_limit_ie	adc_hi_limit_ie	adc_ref_ready_ie	adc_done_ie
Reset	–	–	–	0	0	0	0	0
Access Type	–	–	–	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read

Bitfield	Bits	Description	Decode
adc_overflow_ie	4	ADC overflow interrupt enable	0b0: ADC interrupt source disabled 0b1: Enable assertion of ADC interrupt when ADC overflow interrupt flag is set

Bitfield	Bits	Description	Decode
adc_lo_limit_ie	3	ADC lo limit monitor interrupt enable	0b0: ADC interrupt source disabled 0b1: Enable assertion of ADC interrupt when ADC lo limit monitor interrupt flag is set
adc_hi_limit_ie	2	ADC Hi limit monitor interrupt enable	0b0: ADC interrupt source disabled 0b1: Enable assertion of ADC interrupt when ADC hi limit monitor interrupt flag is set
adc_ref_ready_ie	1	ADC reference ready interrupt enable	0b0: ADC interrupt source disabled 0b1: Enable assertion of ADC interrupt when ADC reference ready interrupt flag is set
adc_done_ie	0	ADC done interrupt enable	0b0: ADC interrupt source disabled 0b1: Enable assertion of ADC interrupt when ADC done interrupt flag is set

ADC_INTRIE1 (0x50D)

Bit	7	6	5	4	3	2	1	0
Field	ch7_hi_limit_ie	ch6_hi_limit_ie	ch5_hi_limit_ie	ch4_hi_limit_ie	ch3_hi_limit_ie	ch2_hi_limit_ie	ch1_hi_limit_ie	ch0_hi_limit_ie
Reset	0	0	0	0	0	0	0	0
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read

Bitfield	Bits	Description	Decode
ch7_hi_limit_ie	7	Channel 7 hi limit monitor interrupt enable	0b0: ADC interrupt source disabled 0b1: Enable assertion of ADC interrupt when channel 7 hi limit monitor interrupt flag is set
ch6_hi_limit_ie	6	Channel 6 hi limit monitor interrupt enable	0b0: ADC interrupt source disabled 0b1: Enable assertion of ADC interrupt when channel 6 hi limit monitor interrupt flag is set
ch5_hi_limit_ie	5	Channel 5 hi limit monitor interrupt enable	0b0: ADC interrupt source disabled 0b1: Enable assertion of ADC interrupt when channel 5 hi limit monitor interrupt flag is set
ch4_hi_limit_ie	4	Channel 4 hi limit monitor interrupt enable	0b0: ADC interrupt source disabled 0b1: Enable assertion of ADC interrupt when channel 4 hi limit monitor interrupt flag is set
ch3_hi_limit_ie	3	Channel 3 hi limit monitor interrupt enable	0b0: ADC interrupt source disabled 0b1: Enable assertion of ADC interrupt when channel 3 hi limit monitor interrupt flag is set
ch2_hi_limit_ie	2	Channel 2 hi limit monitor interrupt enable	0b0: ADC interrupt source disabled 0b1: Enable assertion of ADC interrupt when channel 2 hi limit monitor interrupt flag is set

Bitfield	Bits	Description	Decode
ch1_hi_limit_ie	1	Channel 1 hi limit monitor interrupt enable	0b0: ADC interrupt source disabled 0b1: Enable assertion of ADC interrupt when channel 1 hi limit monitor interrupt flag is set
ch0_hi_limit_ie	0	Channel 0 hi limit monitor interrupt enable	0b0: ADC interrupt source disabled 0b1: Enable assertion of ADC interrupt when channel 0 hi limit monitor interrupt flag is set

ADC_INTRIE2 (0x50E)

Bit	7	6	5	4	3	2	1	0
Field	ch7_lo_limit_ie	ch6_lo_limit_ie	ch5_lo_limit_ie	ch4_lo_limit_ie	ch3_lo_limit_ie	ch2_lo_limit_ie	ch1_lo_limit_ie	ch0_lo_limit_ie
Reset	0	0	0	0	0	0	0	0
Access Type	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read	Write, Read

Bitfield	Bits	Description	Decode
ch7_lo_limit_ie	7	Channel 7 lo limit monitor interrupt enable	0b0: ADC interrupt source disabled 0b1: Enable assertion of ADC interrupt when channel 7 lo limit monitor interrupt flag is set
ch6_lo_limit_ie	6	Channel 6 lo limit monitor interrupt enable	0b0: ADC interrupt source disabled 0b1: Enable assertion of ADC interrupt when channel 6 lo limit monitor interrupt flag is set
ch5_lo_limit_ie	5	Channel 5 lo limit monitor interrupt enable	0b0: ADC interrupt source disabled 0b1: Enable assertion of ADC interrupt when channel 5 lo limit monitor interrupt flag is set
ch4_lo_limit_ie	4	Channel 4 lo limit monitor interrupt enable	0b0: ADC interrupt source disabled 0b1: Enable assertion of ADC interrupt when channel 4 lo limit monitor interrupt flag is set
ch3_lo_limit_ie	3	Channel 3 lo limit monitor interrupt enable	0b0: ADC interrupt source disabled 0b1: Enable assertion of ADC interrupt when channel 3 lo limit monitor interrupt flag is set
ch2_lo_limit_ie	2	Channel 2 lo limit monitor interrupt enable	0b0: ADC interrupt source disabled 0b1: Enable assertion of ADC interrupt when channel 2 lo limit monitor interrupt flag is set
ch1_lo_limit_ie	1	Channel 1 lo limit monitor interrupt enable	0b0: ADC interrupt source disabled 0b1: Enable assertion of ADC interrupt when channel 1 lo limit monitor interrupt flag is set
ch0_lo_limit_ie	0	Channel 0 lo limit monitor interrupt enable	0b0: ADC interrupt source disabled 0b1: Enable assertion of ADC interrupt when channel 0 lo limit monitor interrupt flag is set

ADC_INTRIE3 (0x50F)

Bit	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	–	–	invalid_ch_sel_ie
Reset	–	–	–	–	–	–	–	0
Access Type	–	–	–	–	–	–	–	Write, Read

Bitfield	Bits	Description	Decode
invalid_ch_sel_ie	0	Enable the ADC channel selection invalid function	0b0: Disable ADC channel selection invalid function 0b1: Enable ADC channel selection invalid function

ADC_INTR0 (0x510)

Bit	7	6	5	4	3	2	1	0
Field	–	–	–	adc_overflow_if	adc_lo_limit_if	adc_hi_limit_if	adc_ref_ready_if	adc_done_if
Reset	–	–	–	0	0	0	0	0
Access Type	–	–	–	Read Only	Read Only	Read Only	Read Only	Read Only

Bitfield	Bits	Description	Decode
adc_overflow_if	4	ADC overflow interrupt flag. Cleared when read.	0b0: Flag cleared 0b1: Flag set
adc_lo_limit_if	3	ADC lo limit monitor interrupt flag. Cleared when read.	0b0: Flag cleared 0b1: Flag set
adc_hi_limit_if	2	ADC Hi limit monitor interrupt flag. Cleared when read.	0b0: Flag cleared 0b1: Flag set
adc_ref_ready_if	1	ADC reference ready interrupt flag. Cleared when read.	0b0: Flag cleared 0b1: Flag set
adc_done_if	0	ADC done interrupt flag. Cleared when read.	0b0: Flag cleared 0b1: Flag set

ADC_INTR1 (0x511)

Bit	7	6	5	4	3	2	1	0
Field	ch7_hi_limit_if	ch6_hi_limit_if	ch5_hi_limit_if	ch4_hi_limit_if	ch3_hi_limit_if	ch2_hi_limit_if	ch1_hi_limit_if	ch0_hi_limit_if
Reset	0	0	0	0	0	0	0	0
Access Type	Read Only	Read Only	Read Only	Read Only	Read Only	Read Only	Read Only	Read Only

Bitfield	Bits	Description	Decode
ch7_hi_limit_if	7	Channel 7 hi limit monitor interrupt flag. Cleared when read.	0b0: Flag cleared 0b1: Flag set

Bitfield	Bits	Description	Decode
ch6_hi_limit_if	6	Channel 6 hi limit monitor interrupt flag. Cleared when read.	0b0: Flag cleared 0b1: Flag set
ch5_hi_limit_if	5	Channel 5 hi limit monitor interrupt flag. Cleared when read.	0b0: Flag cleared 0b1: Flag set
ch4_hi_limit_if	4	Channel 4 hi limit monitor interrupt flag. Cleared when read.	0b0: Flag cleared 0b1: Flag set
ch3_hi_limit_if	3	Channel 3 hi limit monitor interrupt flag. Cleared when read.	0b0: Flag cleared 0b1: Flag set
ch2_hi_limit_if	2	Channel 2 hi limit monitor interrupt flag. Cleared when read.	0b0: Flag cleared 0b1: Flag set
ch1_hi_limit_if	1	Channel 1 hi limit monitor interrupt flag. Cleared when read.	0b0: Flag cleared 0b1: Flag set
ch0_hi_limit_if	0	Channel 0 hi limit monitor interrupt flag. Cleared when read.	0b0: Flag cleared 0b1: Flag set

ADC_INTR2 (0x512)

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Field	ch7_lo_limit_if	ch6_lo_limit_if	ch5_lo_limit_if	ch4_lo_limit_if	ch3_lo_limit_if	ch2_lo_limit_if	ch1_lo_limit_if	ch0_lo_limit_if
Reset	0	0	0	0	0	0	0	0
Access Type	Read Only	Read Only	Read Only	Read Only	Read Only	Read Only	Read Only	Read Only

Bitfield	Bits	Description	Decode
ch7_lo_limit_if	7	Channel 7 lo limit monitor interrupt flag. Cleared when read.	0b0: Flag cleared 0b1: Flag set
ch6_lo_limit_if	6	Channel 6 lo limit monitor interrupt flag. Cleared when read.	0b0: Flag cleared 0b1: Flag set
ch5_lo_limit_if	5	Channel 5 lo limit monitor interrupt flag. Cleared when read.	0b0: Flag cleared 0b1: Flag set
ch4_lo_limit_if	4	Channel 4 lo limit monitor interrupt flag. Cleared when read.	0b0: Flag cleared 0b1: Flag set
ch3_lo_limit_if	3	Channel 3 lo limit monitor interrupt flag. Cleared when read.	0b0: Flag cleared 0b1: Flag set
ch2_lo_limit_if	2	Channel 2 lo limit monitor interrupt flag. Cleared when read.	0b0: Flag cleared 0b1: Flag set
ch1_lo_limit_if	1	Channel 1 lo limit monitor interrupt flag. Cleared when read.	0b0: Flag cleared

Bitfield	Bits	Description	Decode
			0b1: Flag set
ch0_lo_limit_if	0	Channel 0 lo limit monitor interrupt flag. Cleared when read.	0b0: Flag cleared 0b1: Flag set

ADC_INTR3 (0x513)

Bit	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	–	–	invalid_ch_sel_if
Reset	–	–	–	–	–	–	–	0
Access Type	–	–	–	–	–	–	–	Read Only

Bitfield	Bits	Description	Decode
invalid_ch_sel_if	0	Invalid channel flag. Indicates that an invalid GPIO channel was attempted to be connected to the ADC input. This can be a result of the package selection or GPIO port configuration. Cleared when read.	0b0: Flag cleared 0b1: Flag set

ADC_LIMIT0_0 (0x514)

Bit	7	6	5	4	3	2	1	0
Field	chLoLimit_l0[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
chLoLimit_l0	7:0	Register set 0 - lo limit threshold value, low bits	0xXX: Lo limit low byte

ADC_LIMIT0_1 (0x515)

Bit	7	6	5	4	3	2	1	0
Field	chHiLimit_l0[3:0]				–	–	chLoLimit_h0[1:0]	
Reset	0xF				–	–	00	
Access Type	Write, Read				–	–	Write, Read	

Bitfield	Bits	Description	Decode
chHiLimit_l0	7:4	Register set 0 - hi limit threshold value, low bits	0bX: Hi limit low nibble
chLoLimit_h0	1:0	Register set 0 - lo limit threshold value, high bits	0bXX: Lo limit high bits

ADC_LIMIT0_2 (0x516)

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Field	–	–	chHiLimit_h0[5:0]					
Reset	–	–	111111					
Access Type	–	–	Write, Read					

Bitfield	Bits	Description	Decode
chHiLimit_h0	5:0	Register set 0 - hi limit threshold value, high bits	0bXXXXXX: Hi limit high bits

ADC_LIMIT0_3 (0x517)

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Field	–	–	div_sel0[1:0]		ch_sel0[3:0]			
Reset	–	–	00		0011			
Access Type	–	–	Write, Read		Write, Read			

Bitfield	Bits	Description	Decode
div_sel0	5:4	ADC channel 0 divider setting	0b00: Divide by 1 0b01: Divide by 2 0b10: Divide by 3 0b11: Divide by 4
ch_sel0	3:0	ADC Channel Select for Register set 0	0x0: GPIO0 0x1: GPIO1 0x2: GPIO2 0x3: GPIO3 0x4: GPIO4 0x5: GPIO5 0x6: GPIO6 0x7: GPIO7 0x8: VSUP0 / 4 0x9: VSUP1 / 2 0xA: VSUP2 /2 0xB: TMON 0xC: ABUS<0> 0xD: ABUS<1> 0xE: ABUS<2> 0xF: ABUS<3>

ADC_LIMIT1_0 (0x518)

Bit	**7**	**6**	**5**	**4**	**3**	**2**	**1**	**0**
Field	chLoLimit_l1[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
chLoLimit_l1	7:0	Register set 1 - lo limit threshold value, low bits	0xXX: Lo limit low byte

ADC_LIMIT1_1 (0x519)

Bit	7	6	5	4	3	2	1	0
Field	chHiLimit_l1[3:0]				–	–	chLoLimit_h1[1:0]	
Reset	0xF				–	–	00	
Access Type	Write, Read				–	–	Write, Read	

Bitfield	Bits	Description	Decode
chHiLimit_l1	7:4	Register set 1 - hi limit threshold value, low bits	0bX: Hi limit low nibble
chLoLimit_h1	1:0	Register set 1 - lo limit threshold value, high bits	0bXX: Lo limit high bits

ADC_LIMIT1_2 (0x51A)

Bit	7	6	5	4	3	2	1	0
Field	–	–	chHiLimit_h1[5:0]					
Reset	–	–	111111					
Access Type	–	–	Write, Read					

Bitfield	Bits	Description	Decode
chHiLimit_h1	5:0	Register set 1 - hi limit threshold value, high bits	0bXXXXXX: Hi limit high bits

ADC_LIMIT1_3 (0x51B)

Bit	7	6	5	4	3	2	1	0
Field	–	–	div_sel1[1:0]		ch_sel1[3:0]			
Reset	–	–	00		0011			
Access Type	–	–	Write, Read		Write, Read			

Bitfield	Bits	Description	Decode
div_sel1	5:4	ADC channel 1 divider setting	0b00: Divide by 1 0b01: Divide by 2 0b10: Divide by 3 0b11: Divide by 4
ch_sel1	3:0	ADC Channel Select for Register set 1	0x0: GPIO0 0x1: GPIO1 0x2: GPIO2 0x3: GPIO3 0x4: GPIO4

Bitfield	Bits	Description	Decode
			0x5: GPIO5 0x6: GPIO6 0x7: GPIO7 0x8: VSUP0 / 4 0x9: VSUP1 / 2 0xA: VSUP2 /2 0xB: TMON 0xC: ABUS<0> 0xD: ABUS<1> 0xE: ABUS<2> 0xF: ABUS<3>

ADC_LIMIT2_0 (0x51C)

Bit	7	6	5	4	3	2	1	0
Field	chLoLimit_l2[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
chLoLimit_l2	7:0	Register set 2 - lo limit threshold value, low bits	0xXX: Lo limit low byte

ADC_LIMIT2_1 (0x51D)

Bit	7	6	5	4	3	2	1	0
Field	chHiLimit_l2[3:0]				–	–	chLoLimit_h2[1:0]	
Reset	0xF				–	–	00	
Access Type	Write, Read				–	–	Write, Read	

Bitfield	Bits	Description	Decode
chHiLimit_l2	7:4	Register set 2 - hi limit threshold value, low bits	0bX: Hi limit low nibble
chLoLimit_h2	1:0	Register set 2 - lo limit threshold value, high bits	0bXX: Lo limit high bits

ADC_LIMIT2_2 (0x51E)

Bit	7	6	5	4	3	2	1	0
Field	–	–	chHiLimit_h2[5:0]					
Reset	–	–	111111					
Access Type	–	–	Write, Read					

Bitfield	Bits	Description	Decode
chHiLimit_h2	5:0	Register set 2 - hi limit threshold value, high bits	0bXXXXXX: Hi limit high bits

ADC_LIMIT2_3 (0x51F)

Bit	7	6	5	4	3	2	1	0
Field	–	–	div_sel2[1:0]		ch_sel2[3:0]			
Reset	–	–	00		0011			
Access Type	–	–	Write, Read		Write, Read			

Bitfield	Bits	Description	Decode
div_sel2	5:4	ADC channel 2 divider setting	0b00: Divide by 1 0b01: Divide by 2 0b10: Divide by 3 0b11: Divide by 4
ch_sel2	3:0	ADC Channel Select for Register set 2	0x0: GPIO0 0x1: GPIO1 0x2: GPIO2 0x3: GPIO3 0x4: GPIO4 0x5: GPIO5 0x6: GPIO6 0x7: GPIO7 0x8: VSUP0 / 4 0x9: VSUP1 / 2 0xA: VSUP2 /2 0xB: TMON 0xC: ABUS<0> 0xD: ABUS<1> 0xE: ABUS<2> 0xF: ABUS<3>

ADC_LIMIT3_0 (0x520)

Bit	7	6	5	4	3	2	1	0
Field	chLoLimit_l3[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
chLoLimit_l3	7:0	Register set 3 - lo limit threshold value, low bits	0xXX: Lo limit low byte

ADC_LIMIT3_1 (0x521)

Bit	7	6	5	4	3	2	1	0
Field	chHiLimit_l3[3:0]				–	–	chLoLimit_h3[1:0]	
Reset	0xF				–	–	00	
Access Type	Write, Read				–	–	Write, Read	

Bitfield	Bits	Description	Decode
chHiLimit_l3	7:4	Register set 3 - hi limit threshold value, low bits	0bX: Hi limit low nibble
chLoLimit_h3	1:0	Register set 3 - lo limit threshold value, high bits	0bXX: Lo limit high bits

ADC_LIMIT3_2 (0x522)

Bit	7	6	5	4	3	2	1	0
Field	–	–	chHiLimit_h3[5:0]					
Reset	–	–	111111					
Access Type	–	–	Write, Read					

Bitfield	Bits	Description	Decode
chHiLimit_h3	5:0	Register set 3 - hi limit threshold value, high bits	0bXXXXXX: Hi limit high bits

ADC_LIMIT3_3 (0x523)

Bit	7	6	5	4	3	2	1	0
Field	–	–	div_sel3[1:0]		ch_sel3[3:0]			
Reset	–	–	00		0011			
Access Type	–	–	Write, Read		Write, Read			

Bitfield	Bits	Description	Decode
div_sel3	5:4	ADC channel 3 divider setting	0b00: Divide by 1 0b01: Divide by 2 0b10: Divide by 3 0b11: Divide by 4
ch_sel3	3:0	ADC Channel Select for Register set 3	0x0: GPIO0 0x1: GPIO1 0x2: GPIO2 0x3: GPIO3 0x4: GPIO4 0x5: GPIO5 0x6: GPIO6 0x7: GPIO7 0x8: VSUP0 / 4 0x9: VSUP1 / 2

Bitfield	Bits	Description	Decode
			0xA: VSUP2 /2 0xB: TMON 0xC: ABUS<0> 0xD: ABUS<1> 0xE: ABUS<2> 0xF: ABUS<3>

ADC_LIMIT4_0 (0x524)

Bit	7	6	5	4	3	2	1	0
Field	chLoLimit_l4[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
chLoLimit_l4	7:0	Register set 4 - lo limit threshold value, low bits	0xXX: Lo limit low byte

ADC_LIMIT4_1 (0x525)

Bit	7	6	5	4	3	2	1	0
Field	chHiLimit_l4[3:0]				–	–	chLoLimit_h4[1:0]	
Reset	0xF				–	–	00	
Access Type	Write, Read				–	–	Write, Read	

Bitfield	Bits	Description	Decode
chHiLimit_l4	7:4	Register set 4 - hi limit threshold value, low bits	0bX: Hi limit low nibble
chLoLimit_h4	1:0	Register set 4 - lo limit threshold value, high bits	0bXX: Lo limit high bits

ADC_LIMIT4_2 (0x526)

Bit	7	6	5	4	3	2	1	0
Field	–	–	chHiLimit_h4[5:0]					
Reset	–	–	111111					
Access Type	–	–	Write, Read					

Bitfield	Bits	Description	Decode
chHiLimit_h4	5:0	Register set 4 - hi limit threshold value, high bits	0bXXXXXX: Hi limit high bits

ADC_LIMIT4_3 (0x527)

Bit	7	6	5	4	3	2	1	0
Field	–	–	div_sel4[1:0]		ch_sel4[3:0]			
Reset	–	–	00		0011			
Access Type	–	–	Write, Read		Write, Read			

Bitfield	Bits	Description	Decode
div_sel4	5:4	ADC channel 4 divider setting	0b00: Divide by 1 0b01: Divide by 2 0b10: Divide by 3 0b11: Divide by 4
ch_sel4	3:0	ADC Channel Select for Register set 4	0x0: GPIO0 0x1: GPIO1 0x2: GPIO2 0x3: GPIO3 0x4: GPIO4 0x5: GPIO5 0x6: GPIO6 0x7: GPIO7 0x8: VSUP0 / 4 0x9: VSUP1 / 2 0xA: VSUP2 /2 0xB: TMON 0xC: ABUS<0> 0xD: ABUS<1> 0xE: ABUS<2> 0xF: ABUS<3>

ADC_LIMIT5_0 (0x528)

Bit	7	6	5	4	3	2	1	0
Field	chLoLimit_l5[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
chLoLimit_l5	7:0	Register set 5 - lo limit threshold value, low bits	0xXX: Lo limit low byte

ADC_LIMIT5_1 (0x529)

Bit	7	6	5	4	3	2	1	0
Field	chHiLimit_l5[3:0]				–	–	chLoLimit_h5[1:0]	
Reset	0xF				–	–	00	
Access Type	Write, Read				–	–	Write, Read	

Bitfield	Bits	Description	Decode
chHiLimit_l5	7:4	Register set 5 - hi limit threshold value, low bits	0bX: Hi limit low nibble
chLoLimit_h5	1:0	Register set 5 - lo limit threshold value, high bits	0bXX: Lo limit high bits

ADC_LIMIT5_2 (0x52A)

Bit	7	6	5	4	3	2	1	0
Field	–	–	chHiLimit_h5[5:0]					
Reset	–	–	111111					
Access Type	–	–	Write, Read					

Bitfield	Bits	Description	Decode
chHiLimit_h5	5:0	Register set 5 - hi limit threshold value, high bits	0bXXXXXX: Hi limit high bits

ADC_LIMIT5_3 (0x52B)

Bit	7	6	5	4	3	2	1	0
Field	–	–	div_sel5[1:0]		ch_sel5[3:0]			
Reset	–	–	00		0011			
Access Type	–	–	Write, Read		Write, Read			

Bitfield	Bits	Description	Decode
div_sel5	5:4	ADC channel 5 divider setting	0b00: Divide by 1 0b01: Divide by 2 0b10: Divide by 3 0b11: Divide by 4
ch_sel5	3:0	ADC Channel Select for Register set 5	0x0: GPIO0 0x1: GPIO1 0x2: GPIO2 0x3: GPIO3 0x4: GPIO4 0x5: GPIO5 0x6: GPIO6 0x7: GPIO7 0x8: VSUP0 / 4 0x9: VSUP1 / 2 0xA: VSUP2 /2 0xB: TMON 0xC: ABUS<0> 0xD: ABUS<1> 0xE: ABUS<2> 0xF: ABUS<3>

ADC_LIMIT6_0 (0x52C)

Bit	7	6	5	4	3	2	1	0
Field	chLoLimit_l6[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
chLoLimit_l6	7:0	Register set 6 - lo limit threshold value, low bits	0xXX: Lo limit low byte

ADC_LIMIT6_1 (0x52D)

Bit	7	6	5	4	3	2	1	0
Field	chHiLimit_l6[3:0]				–	–	chLoLimit_h6[1:0]	
Reset	0xF				–	–	00	
Access Type	Write, Read				–	–	Write, Read	

Bitfield	Bits	Description	Decode
chHiLimit_l6	7:4	Register set 6 - hi limit threshold value, low bits	0bX: Hi limit low nibble
chLoLimit_h6	1:0	Register set 6 - lo limit threshold value, high bits	0bXX: Lo limit high bits

ADC_LIMIT6_2 (0x52E)

Bit	7	6	5	4	3	2	1	0
Field	–	–	chHiLimit_h6[5:0]					
Reset	–	–	111111					
Access Type	–	–	Write, Read					

Bitfield	Bits	Description	Decode
chHiLimit_h6	5:0	Register set 5 - hi limit threshold value, high bits	0bXXXXXX: Hi limit high bits

ADC_LIMIT6_3 (0x52F)

Bit	7	6	5	4	3	2	1	0
Field	–	–	div_sel6[1:0]		ch_sel6[3:0]			
Reset	–	–	00		0011			
Access Type	–	–	Write, Read		Write, Read			

Bitfield	Bits	Description	Decode
div_sel6	5:4	ADC channel 6 divider setting	0b00: Divide by 1 0b01: Divide by 2

Bitfield	Bits	Description	Decode
			0b10: Divide by 3 0b11: Divide by 4
ch_sel6	3:0	ADC Channel Select for Register set 6	0x0: GPIO0 0x1: GPIO1 0x2: GPIO2 0x3: GPIO3 0x4: GPIO4 0x5: GPIO5 0x6: GPIO6 0x7: GPIO7 0x8: VSUP0 / 4 0x9: VSUP1 / 2 0xA: VSUP2 /2 0xB: TMON 0xC: ABUS<0> 0xD: ABUS<1> 0xE: ABUS<2> 0xF: ABUS<3>

ADC_LIMIT7_0 (0x530)

Bit	7	6	5	4	3	2	1	0
Field	chLoLimit_l7[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
chLoLimit_l7	7:0	Register set 7 - lo limit threshold value, low bits	0xXX: Lo limit low byte

ADC_LIMIT7_1 (0x531)

Bit	7	6	5	4	3	2	1	0
Field	chHiLimit_l7[3:0]				–	–	chLoLimit_h7[1:0]	
Reset	0xF				–	–	00	
Access Type	Write, Read				–	–	Write, Read	

Bitfield	Bits	Description	Decode
chHiLimit_l7	7:4	Register set 7 - hi limit threshold value, low bits	0bX: Hi limit low nibble
chLoLimit_h7	1:0	Register set 7- lo limit threshold value, high bits	0bXX: Lo limit high bits

ADC_LIMIT7_2 (0x532)

Bit	7	6	5	4	3	2	1	0
Field	–	–	chHiLimit_h7[5:0]					
Reset	–	–	111111					
Access Type	–	–	Write, Read					

Bitfield	Bits	Description	Decode
chHiLimit_h7	5:0	Register set 7 - hi limit threshold value, high bits	0bXXXXXX: Hi limit high bits

ADC_LIMIT7_3 (0x533)

Bit	7	6	5	4	3	2	1	0
Field	–	–	div_sel7[1:0]		ch_sel7[3:0]			
Reset	–	–	00		0011			
Access Type	–	–	Write, Read		Write, Read			

Bitfield	Bits	Description	Decode
div_sel7	5:4	ADC channel 7 divider setting	0b00: Divide by 1 0b01: Divide by 2 0b10: Divide by 3 0b11: Divide by 4
ch_sel7	3:0	ADC Channel Select for Register set 7	0x0: GPIO0 0x1: GPIO1 0x2: GPIO2 0x3: GPIO3 0x4: GPIO4 0x5: GPIO5 0x6: GPIO6 0x7: GPIO7 0x8: VSUP0 / 4 0x9: VSUP1 / 2 0xA: VSUP2 /2 0xB: TMON 0xC: ABUS<0> 0xD: ABUS<1> 0xE: ABUS<2> 0xF: ABUS<3>

ADC_RR_CTRL0 (0x534)

Bit	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	–	–	adc_rr_run
Reset	–	–	–	–	–	–	–	0
Access Type	–	–	–	–	–	–	–	Write, Read

Bitfield	Bits	Description	Decode
adc_rr_run	0	Enable round robin state machine	0b0: Hold round robin state machine in idle mode, manual ADC control 0b1: Run round robin state machine

ADC_RR_CTRL1 (0x535)

Bit	7	6	5	4	3	2	1	0
Field	–	–	–	abus_adcsel	–	–	–	–
Reset	–	–	–	0	–	–	–	–
Access Type	–	–	–	Write, Read	–	–	–	–

Bitfield	Bits	Description	Decode
abus_adcsel	4	Enables the abus to be an input to the ADC	0b0: Abus not enabled to ADC 0x1: Abus enabled as input to ADC

ADC_RR_CTRL2 (0x536)

Bit	7	6	5	4	3	2	1	0
Field	adc_rr_sleep_l[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
adc_rr_sleep_l	7:0	Low byte of number of ADC conversion cycles the state machine sleeps between cycles.	0xXX: Low byte of number of ADC conversion cycles

ADC_RR_CTRL3 (0x537)

Bit	7	6	5	4	3	2	1	0
Field	adc_rr_sleep_h[7:0]							
Reset	00000000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
adc_rr_sleep_h	7:0	High byte of number of ADC conversion cycles the state machine sleeps between cycles.	0xXX: High byte of number of ADC conversion cycles

ADC_CTRL_4 (0x53E)

Bit	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	adc_pin_en[2:0]		
Reset	–	–	–	–	–	000		
Access Type	–	–	–	–	–	Write, Read		

Bitfield	Bits	Description	Decode
adc_pin_en	2:0	Enable monitoring of selected MFP pins with ADC	0bXX0: Disable monitoring of ADC0 0bXX1: Enable monitoring of ADC0 0bX0X: Disable monitoring of ADC1 0bX1X: Enable monitoring of ADC1 0b0XX: Disable monitoring of ADC2 0b1XX: Enable monitoring of ADC2

RLMS4 (0x1404)

Bit	7	6	5	4	3	2	1	0
Field	EOM_CHK_AMOUNT[3:0]				EOM_CHK_THR[1:0]		EOM_PER_MODE	EOM_EN
Reset	0100				10		1	1
Access Type	Write, Read				Write, Read		Write, Read	Write, Read

Bitfield	Bits	Description	Decode
EOM_CHK_AMOUNT	7:4	A factor (N) used to select the order of number of observations in each eye monitor window. N is used in the equation: Observations =6.29*10^(N+2)	0xX: N factor
EOM_CHK_THR	3:2	Eye opening monitor number of error bits to allow in a measurement window	0b00: Allow no errors 0b01: Allow 1 error 0b10: Allow 2 errors 0b11: Allow 3 errors
EOM_PER_MODE	1	Eye opening monitor periodic mode enable	0b0: Eye opening monitor periodic mode disabled 0b1: Eye opening monitor periodic mode enabled
EOM_EN	0	Eye opening monitor enable	0b0: Eye opening monitor disabled 0b1: Eye opening monitor enabled

RLMS5 (0x1405)

Bit	7	6	5	4	3	2	1	0
Field	EOM_MAN_TRG_REQ	EOM_MIN_THR[6:0]						
Reset	0	0010000						
Access Type	Write Only	Write, Read						

Bitfield	Bits	Description	Decode
EOM_MAN_TRG_REQ	7	Eye opening monitor manual trigger. For use when periodic mode is disabled.	0b0: No action 0b1: EOM manual trigger request
EOM_MIN_THR	6:0	The EOM minimum threshold as defined by the equation: % eye opening = EOM_MIN_THR/64. If the value is zero the EOM is disabled.	0bXXXXXXX: EOM minimum threshold factor

RLMS6 (0x1406)

Bit	7	6	5	4	3	2	1	0
Field	EOM_PV_MODE	EOM_RST_THR[6:0]						
Reset	1	0000000						
Access Type	Write, Read	Write, Read						

Bitfield	Bits	Description	Decode
EOM_PV_MODE	7	Eye opening is measured vertically or horizantally	0b0: Vertical opening mode 0b1: Horizonal opening mode
EOM_RST_THR	6:0	The EOM refresh threshold as defined by the equation: % eye opening = EOM_MIN_THR/64. If the value is zero the EOM is disabled.	0bXXXXXXX: EOM refresh threshold factor

RLMS7 (0x1407)

Bit	7	6	5	4	3	2	1	0
Field	EOM_DONE	EOM[6:0]						
Reset	0	0000000						
Access Type	Read Only	Read Only						

Bitfield	Bits	Description	Decode
EOM_DONE	7	Eye opening monitor measurement done	0b0: EOM not complete 0b1: EOM complete
EOM	6:0	Last completed EOM observation	0bXXXXXXX: EOM measurement result

RLMS64 (0x1464)

Bit	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	–	TxSSCMode[1:0]	
Reset	–	–	–	–	–	–	11	
Access Type	–	–	–	–	–	–	Write, Read	

Bitfield	Bits	Description	Decode
TxSSCMode	1:0	Tx spread spectrum mode	00: No spread spectrum (manual phase) 01: Center spread 10: Reserved 11: Reserved

RLMS70 (0x1470)

Bit	7	6	5	4	3	2	1	0
Field	–	TxSSCFrqCtrl[6:0]						
Reset	–	0000001						
Access Type	–	Write, Read						

Bitfield	Bits	Description	Decode
TxSSCFrqCtrl	6:0	Tx spread spectrum frequency control	0bXXXXXXX: Tx spread spectrum center frequency control

RLMS71 (0x1471)

Bit	7	6	5	4	3	2	1	0
Field	–	TxSSCCenSprSt[5:0]						TxSSCEn
Reset	–	000001						0
Access Type	–	Write, Read						Write, Read

Bitfield	Bits	Description	Decode
TxSSCCenSprSt	6:1	Tx spread spectrum center spread startup control	0bXXXXXX: Tx spread spectrum center spread startup control
TxSSCEn	0	Tx spread spectrum enable	0b0: Tx spread spectrum disabled 0b1: Tx spread spectrum enabled

RLMS72 (0x1472)

Bit	7	6	5	4	3	2	1	0
Field	TxSSCPreSclL[7:0]							
Reset	0xCF							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
TxSSCPreSclL	7:0	Tx spread spectrum frequency pre-scaler low byte	0xXX: Tx spread spectrum frequency pre-scaler low byte

RLMS73 (0x1473)

Bit	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	TxSSCPreSclH[2:0]		
Reset	–	–	–	–	–	000		
Access Type	–	–	–	–	–	Write, Read		

Bitfield	Bits	Description	Decode
TxSSCPreSclH	2:0	Tx spread spectrum frequency pre-scaler high bits	0bXXX: Tx spread spectrum frequency pre-scaler high bits

RLMS74 (0x1474)

Bit	7	6	5	4	3	2	1	0
Field	TxSSCPhL[7:0]							
Reset	00101000							
Access Type	Write, Read							

Bitfield	Bits	Description	Decode
TxSSCPhL	7:0	Tx spread spectrum interpolator phase low byte	0xXX: Tx spread spectrum frequency interpolator phase low byte

RLMS75 (0x1475)

Bit	7	6	5	4	3	2	1	0
Field	–	TxSSCPhH[6:0]						
Reset	–	0010110						
Access Type	–	Write, Read						

Bitfield	Bits	Description	Decode
TxSSCPhH	6:0	Tx spread spectrum interpolator phase	0bXXXXXXX: Tx spread spectrum frequency interpolator phase high bits

RLMS76 (0x1476)

Bit	7	6	5	4	3	2	1	0
Field	–	–	–	–	–	–	TxSSCPhQuad[1:0]	
Reset	–	–	–	–	–	–	00	
Access Type	–	–	–	–	–	–	Write, Read	

Bitfield	Bits	Description	Decode
TxSSCPhQuad	1:0	Tx spread spectrum interpolator phase quadrant	0bXX: Tx spread spectrum interpolator phase quadrant

Applications Information

Documentation Changes – Rev B to Rev C

Changes are differences from the previous version. Fixes correct documentation errors in the previous version. Updates are clarifications or enhanced information from the previous version.

Hex Addr.	Register Block	Reg Name	Bitfield	Change Description
6	DEV	REG6		Added register
19	TCTRL	INTR1	(PKT_CNT_THR)	Deleted bitfield
1E	TCTRL	INTR6	VDDBAD_INT_OEN	Added bitfield
1E	TCTRL	INTR6	PORZ_INT_OEN	Added bitfield
1E	TCTRL	INTR6	VDDCMP_INT_FLAG	Added bitfield
1F	TCTRL	INTR7	VDDBAD_INT_OEN	Added bitfield
1F	TCTRL	INTR7	PORZ_INT_OEN	Added bitfield
1F	TCTRL	INTR7	VDDCMP_INT_FLAG	Added bitfield
1F	TCTRL	INTR7	MIPI_ERR_FLAG	Updated Description
4F	CC	UART_PT_0		Added register
68	CFGA__AUDIO_X	TR0	TX_CRC_EN	Updated Description
6D	CFGA__AUDIO_X	ARQ0	EN (ARQ0_EN)	Changed bitfield name (in customer facing documentation only)
78	CFGI__INFOFR	TR0	TX_ECR_EN	Updated Description
7D	CFGI__INFOFR	ARQ0		Deleted register*
7E	CFGI__INFOFR	ARQ1		Deleted register*
7F	CFGI__INFOFR	ARQ2		Deleted register*
80	CFGL__SPI	TR0	TX_ECR_EN	Updated Description
85	CFGL__SPI	ARQ0	EN (ARQ0_EN)	Changed bitfield name (in customer facing documentation only)
88	CFGC__CC	TR0	TX_CRC_EN	Deleted bitfield
88	CFGC__CC	TR0	RX_CRC_EN	Deleted bitfield
8D	CFGC__CC	ARQ0	EN (ARQ0_EN)	Changed bitfield name (in customer facing documentation only)
90	CFGL__GPIO	TR0	TX_ECR_EN	Updated Description
95	CFGL__GPIO	ARQ0	EN (ARQ0_EN)	Changed bitfield name (in customer facing documentation only)
98	CFGL__AHDCP	TR0	TX_ECR_EN	Updated Description

9D	CFGL__AHDCP	ARQ0	EN (ARQ0_EN)	Changed bitfield name (in customer facing documentation only)
A0	CFGL__IIC_X	TR0	TX_ECR_EN	Updated Description
A5	CFGL__IIC_X	ARQ0	EN (ARQ0_EN)	Changed bitfield name (in customer facing documentation only)
A8	CFGL__IIC_Y	TR0	TX_ECR_EN	Updated Description
AD	CFGL__IIC_Y	ARQ0	EN (ARQ0_EN)	Changed bitfield name (in customer facing documentation only)
100	VID_TX__X	VIDEO_TX0	LINE_CRC_SEL	Added bitfield
100	VID_TX__X	VIDEO_TX0	ENC_MODE	Added bitfield
108	VID_TX__Y	VIDEO_TX0	LINE_CRC_SEL	Added bitfield
108	VID_TX__Y	VIDEO_TX0	ENC_MODE	Added bitfield
110	VID_TX__Z	VIDEO_TX0	LINE_CRC_SEL	Added bitfield
110	VID_TX__Z	VIDEO_TX0	ENC_MODE	Added bitfield
118	VID_TX__U	VIDEO_TX0	LINE_CRC_SEL	Added bitfield
118	VID_TX__U	VIDEO_TX0	ENC_MODE	Added bitfield
122	AUD_TX_AX	AUDIO_TX2		Deleted register
123	AUD_TX_AX	AUDIO_TX3		Deleted register
124	AUD_TX_AX	AUDIO_TX4		Deleted register
132	AUD_TX_AY	AUDIO_TX2		Deleted register
133	AUD_TX_AY	AUDIO_TX3		Deleted register
134	AUD_TX_AY	AUDIO_TX4		Deleted register
182	RGMII	RGMII_2	(REQ_HOLD_OFF_RGMII)	Deleted bitfield (bits 2 – 5)*
18C	RGMII	RGMII_12	TX_CLK_SEL	Added bitfield
18C	RGMII	RGMII_12	MDIO_ALT_DRIVER	Added bitfield
192	WM	WM_2	VsyncPol	Updated Description
192	WM	WM_2	HsyncPol	Updated Description
330	MIPI_RX	MIPI_RX0	ctrol0_vc_map_en	Added register*
330	MIPI_RX	MIPI_RX0	ctrol1_vc_map_en	Added register*
370	MIPI_LPB_PRBS	MIPI_LPB0	LPB_MIPI_TX_PRBSEN_1	Added bitfield*
370	MIPI_LPB_PRBS	MIPI_LPB0	LPB_MIPI_RX_PRBSEN_1	Added bitfield*
40F	GMSL1	GMSL1F	GPO_RX_EN	Added bitfield*

Documentation Changes – Rev A to Rev B

The following bits have been added/removed or edited from Rev A to Rev B silicon

REGISTER BLOCK	REG NAME	BITFIELD	CHANGE DESCRIPTION
DEV	REG3	RCLK_ALT	Added bit
DEV	REG3	UART_1_EN	Added bit
DEV	REG3	UART_2_EN	Added bit
TCTRL	CTRL1	BACK_COMP_SPLTR	Added bit
TCTRL	INTR6	ADC_INT_OEN	Added bit
TCTRL	INTR7	ADC_INT_FLAG	Added bit
CC	UART_PT_0	All	Added register and bitfields
VID_TX__X/Y/Z/U	VIDEO_TX2	LIM_HEART	Added bit
WM	WM_2	All	Added register and bitfields
VTX__Y	ALL	ALL	Register address moved + 0x03
VTX__Z	ALL	ALL	Register address moved + 0x06
VTX__U	ALL	ALL	Register address moved + 0x09
VTX__X/Y/Z/U	VTX40	All	Added register and bitfields
VTX__X/Y/Z/U	VTX41	All	Added register and bitfields
VTX__X/Y/Z/U	VTX42	All	Added register and bitfields
GPIO_	ALL	ALL	Register address moved + 0x0E
FRONTTOP	FRONTTOP_0	Enable_line_infor	Added bit
FRONTTOP	FRONTTOP_10	All	Added register and bitfields
FRONTTOP	FRONTTOP_11	All	Added register and bitfields
DUALVIEW__A/B	ALL	ALL	Register address moved + 0x0A
DUALVIEW__A/B	DV0	LIN_ALT	Added bit
MIPI_DSI	DSI36 – DSI58	All	Added registers and bitfields
FRONTTOP_EXT	All	All	Added register block, registers and bitfields
REF_VTG	VTX0	REF_VTG_MODE	Added bit
REF_VTG	VTX0	VS_TRIG	Added bit
REF_VTG	REF_VTG6	All	Added register and bitfields
REF_VTG	REF_VTG7	All	Added register and bitfields
REF_VTG	REF_VTG8	All	Added register and bitfields
REF_VTG	REF_VTG9	All	Added register and bitfields
GMSL1	All	All	Added register block, registers and bitfields

Register Address Changes from Rev A to Rev B

The addition of registers from Rev A to Rev B has changed the register address of registers in this device. The following table lists the offsets needed to map the Rev A table to the Rev B table. As an example GPIO2's GPIOA Register (0x02B6) is now mapped to 0x02B6 + 0x0E = 0x02C4

REGISTER ADDRESS	OFFSET
0x0000 - 0x01EF	0x00
0x01F0 - 0x022D	0x03
0x0230 - 0x0269	0x06
0x0270 - 0x02A6	0x09
0x02B0 - 0x02F1	0x0E
0x0300 - 0x031F	0x00
0x0320 - 0x0325	0x0A
0x0330 - 0x03F7	0x00
0x03F8 - 0x11B7	0x148
0x1300 - 0xFFFF	0x00